

From the Open Plan to the Cubicle:
The Real and Imagined Transformation of American Office Design
and Office Work, 1945-1999

by

Jennifer Kaufmann-Buhler

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The dissertation is approved by the following members of the Final Oral Committee:

Virginia T. Boyd, Professor Emerita, Design Studies

Jung-hye Shin, Assistant Professor, Design Studies

Roberto Rengel, Professor, Design Studies

Pascale Carayon, Professor, Industrial Systems Engineering

Anna Andrzejewski, Professor, Art History

William Jones, Professor, History

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Abstract

This dissertation examines the history of the open plan from its early development in the context of the post-war period through its implementation and widespread use in the late 20th century. Using a combination of design history and material culture approaches, I argue that although initially developed as a progressive office design concept in the late 1960s, by the 1990s the open plan had become a negative symbol of white-collar work in popular culture. Tracing this transformation, this study begins with an overview of the history of office design in the post-war period in the United States. In the second chapter, I situate the progressive vision of the open plan promoted by architects and designers in the late 1960s within the context of an intellectual movement called post-capitalism. I argue that the proponents of the open plan envisioned a future of office work characterized by an increased emphasis on social relations, a change in management culture, and an elimination of organizational hierarchy. In the third chapter, I adapted a model from Industrial Systems Engineering to analyze archival material regarding the implementation of the open plan in Wisconsin State Government (1970-1990). I argue that the open plan was just one change among a number of other changes in this period, and that those myriad organizational, economic, technological, and social changes in white collar work influenced the design, meaning and use of the open plan in Wisconsin State Government offices. In the fourth chapter, I analyze the representation of office cubicles in various popular media in order to analyze the negative imagery associated with the contemporary office cubicle and contemporary white-collar work in the United States since the 1990s. In the final chapter, I argue that the original goals of the open plan failed to manifest not because the design was fundamentally flawed, but rather because the culture and structure of work changed the context in which the open plan was implemented, and created pressure on systems furniture and the open plan to rapidly adapt to changing organizational and technological needs.

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Introduction

In the mid-1980s my mother worked in a giant open plan (“cubicle”) office in a Texas State Government Office Building in Austin, Texas. As a little girl, the office was a huge labyrinth to me. I remember wandering through the office getting lost among the identical cubicles—I was too small to see the signage that hung from the ceiling to help guide the lost visitor. I remember accidentally getting off the elevator on the wrong floor and discovering that all of the offices looked exactly the same. Over the years, I spent many hours waiting for my mom in her cubicle. I remember the feeling of enclosure in that space with the high panel walls on all four sides, and people stopping in or “peeking” in the doorway looking for my mom, and finding me with a coloring book instead. When I think of a “cubicle,” my first association is my mom’s old workplace and its sea of grey cubicles. As I grew older “cubicles” only became more commonplace, particularly in popular media. I read the comic strip *Dilbert* in the newspaper each morning over my breakfast cereal and watched popular movies in which unhappy white collar workers escaped their office life in search of personal fulfillment (*Joe Versus the Volcano*, *City Slickers*, *Singles*, *Jerry Maguire*). By the late 1990s, “the cubicle” as a symbol of “dreary white collar work” was a familiar trope in all forms of popular media, even some of my favorite movies of that time (*The Matrix*, *Office Space*, *Clockwatchers*, *Fight Club*, *American Beauty*) used the cubicle as a symbol of the main character’s unhappiness and dissatisfaction with their

job and life. As an adult I have inhabited several cubicles myself, including one in an office furniture manufacturer showroom where I worked as a manager and spent many long hours feeling bored and unhappy.

I mention this personal history, because as I started this research these were the assumptions, memories, and associations that shaped my own image and understanding of office cubicles. Far from a neutral scholar on this subject, I have in fact been swimming in these cultural waters for decades, and it was not until I began this research that I started to unpack these cultural associations and consider their origin and their significance. First of all, my own upbringing and experience are shaped by my background as a white, educated, middle-class American woman. My expectations about what work is and what work should be are themselves tied to the incredible privilege that I carry because of my social identity. In fact, the idea of the cubicle as “oppressive” (as it is sometimes characterized in popular media) is frankly absurd in comparison to the truly miserable workplaces such as sweatshops, factories, and mines in which workers engage in often very boring or repetitive work and are in real danger of serious personal injury or even death. By comparison even the dreariest cubicle is a palace of safety, security, and comfort. This is not to say that office design and white-collar work are not worthy of study, but it does mean that I must acknowledge up front the incredible social, cultural, and economic privilege that white-collar work represents and reflects for me, particularly in my own experience and my own expectations of what work means. The inherent bias my privilege creates is inevitable to some extent, but by acknowledging that bias I hope to recognize its influence on my interpretation of the material. For example, I tend to assume because of my background that “office work” is primarily a white-collar activity, but there are a number of “blue collar” workers

also associated with office work (maintenance, cleaners, food service, etc) whose experiences are also important, but who are too often forgotten.

For me, the cubicle has long been synonymous with a “boring office job,” through both popular culture, and my own experience as well. I was therefore very surprised when I first learned that the open plan had been developed in the 1960s as a very progressive office design concept tied to an image of work that emphasized worker autonomy, motivation, and engagement.¹ By simply eliminating private offices and re-arranging workers the open plan would improve office communication, enable social, technological, and organizational change, and reduce hierarchy. The open plan was promoted by architects and designers of that time as an ideal office design for the modern organization. In fact, the open plan was more than an office design, it was a vision of the future in which collaboration would foster greater creativity, new management theory would encourage worker autonomy, and the tired symbols of hierarchy and bureaucracy would be dissolved in order to create a more "organic" organization. The open plan was promoted as both a reflection of this new vision of work, and a means of creating it.² By the early 1990s, the open plan in the form of "cubicles" or "cube farms" had become associated with the very attributes of work it was designed to eradicate: social isolation, inflexibility for change, and rigid hierarchy. How did something created with such progressive and idealistic intentions become associated with such strongly negative aspects of white collar work?

1. David Franz, “The Moral Life of Cubicles: the Utopian Origins of Dilbert’s Workspace,” *The New Atlantis* (2008): 132-39.

2. “Chaos as a System,” *Progressive Architecture* (1968): 160-69.

At first glance, this might appear to be a fairly simple story of architects, designers and organizations implementing the open plan in opposition to the needs and wishes of workers. In fact, this was my own assumption when I began this research. If that were the case, that would suggest that the design of the open plan was itself fundamentally anti-worker, but in my research on the origins of the open plan I quickly found that to be untrue. As I will argue, the present and future needs of workers were absolutely central to the open plan concept when it was first conceived, developed, and promoted by architects and designers. Even as the open plan evolved in design and became implemented in American offices more broadly, there were significant concerns about ensuring the well-being of workers in the open plan. The then nascent field of environment and behavior cut its teeth on studies of the effectiveness of the open plan office in supporting the needs of workers, and some of that research directly influenced the design and use of the open plan and systems furniture.³

Instead of focusing exclusively on the design of the open plan in isolation of other variables, this study places the evolving design of the open plan and systems furniture within the context of broader changes in the culture and structure of white-collar work. I argue that the evolution of the open plan is closely tied to the transformation of white-collar work in the same period. When it was first promoted, the open plan was described as a means of improving communication in the office, enabling flexibility for change (both for the individual and the organization),

3. For example: Michael Brill, et al., *Using Office Design to Increase Productivity*, vol. 2 (Buffalo, N.Y: Workplace Design and Productivity, 1984); Michael Brill, et al., *Using Office Design to Increase Productivity*, vol. I (Buffalo: Workplace Design and Productivity, 1984); Malcolm Brookes, and Archie Kaplan, "The Office Environment: Space Planning and Affective Behavior," *Human Factors* 14, no. 5 (1972): 373-91; Eric Sundstrom, *Work Places : the Psychology of the Physical Environment in Offices and Factories* (New York: Cambridge University Press, 1986).

and eliminating hierarchy. Through the 1970s and 1980s, numerous economic, social, technological, and organizational changes in white-collar work altered the design, meaning, and use of the open plan such that over time it became associated with social isolation, inflexibility for accommodating change, and the reproduction of organizational hierarchy. In tandem with the changing context of its use, the popular image of the "cubicle" also became increasingly negative so that by the late 1990s it had become a familiar theme in many forms of popular media often connected to a negative image of white-collar work. This project traces this evolution of the open plan from how it was first promoted in the late 1960s, through its widespread implementation and use in American offices, and ending with an examination of its negative representation in contemporary popular media. It traces how the optimistic goals of improving communication, enabling flexibility, and eliminating hierarchy did not come to fruition as hoped.

The first chapter describes the history and evolution of office design in the post-war period including the changes in architecture of the period, the changes in office interiors, the structure of office work in that period and how that informed office design, the origins of the open plan concept, and the development of systems furniture. The second chapter considers the intellectual ideas underpinning the open plan in the late 1960s, specifically connecting the progressive idealism of the architects and designers promoting the open plan to post-capitalist thought. By situating the "vision" of the architects and designers within the context of social research, new management theory, and new organizational theories, I argue that the open plan as an artifact was deeply embedded with these progressive ideas, and was portrayed by architects and designers as a means of achieving social change in American organizations.

The third chapter examines the adoption of the open plan in Wisconsin State Government Offices in the 1970 and 1980s. This section examines how the open plan became a state office standard in this period, and how various changes in work in this period influenced the way in which the open plan was implemented and used. Using archival material from the Wisconsin Historical Society, I illustrate the complex influence of organizational, social, environmental, and technological change in office work in that period, and reveal how those changes informed decisions about the selection and use of furniture and the arrangement of offices. This then sets the stage for the fourth chapter which offers a broader discussion of what "the cubicle" means in contemporary American culture. Analyzing representations of workers and cubicles in various popular media forms (films, comics, books, news stories, commercials) since the 1990s, I argue that the cubicle as it is represented has become a familiar symbol of the problems of contemporary office work.

Despite the progressive vision of so many architects and designers of the late 1960s and early 1970s, the open plan was not the transformative office design it was heralded to become. In the final chapter, I return to the central claims of the early proponents, particularly the belief that the open plan would improve office communication and interaction, increase flexibility, and reduce hierarchy in order to consider what happened and how the contemporary "office cubicle" became associated with social isolation, inflexibility, and the reproduction of hierarchy. This research suggests that the changes in work in the 1970s and 1980s led to a fundamental transformation of the open plan. The progressive vision of the architects and designers who promoted the

open plan was unrealized not because their ideas were fatally flawed, but rather because the enormous changes in work, American society and the economy in the 1970s and 1980s fundamentally changed the context in which the open plan was implemented and used.

Although interdisciplinary in its approach at its most basic level this is a work of design history, by this I mean that it traces a design idea from its early development through its use and evolution over time. As is typical in the field of design history, this project uses the intentions of architects and designers as the starting point for understanding the open plan as an idea. By exploring their intentions and situating those within the context of social, economic and political research in the postwar period, I argue that the open plan was created and promoted in the late 1960s as a means of supporting and encouraging the transformation of office work and organizational culture.

While design history is well equipped for thinking about the role of designers, it is not very well suited to engaging in the experience of users. To address this aspect, I looked to both material culture and cultural studies. First of all, material culture represents a way of "reading" objects, spaces, and texts that encourages the scholar to identify with the experience of the user through the process of analysis and interpretation.⁴ More recent scholarship in the field of material has focused on the role of "things" more broadly defined both as real objects that are used by people and as ideas of objects that circulate in other ways (for example through representation).⁵ This type of scholarship suggests the importance of understanding an object as an idea imbued

4. Jules Prown, "Mind in Matter: an Introduction to Material Culture Theory and Method," *Winterthur Portfolio* 17, no. 1 (1982): 1-19.

5. Judy Attfield, *Wild Things* (New York: Berg, 2000); Bill Brown, "Thing Theory," *Critical Inquiry* 28, no. 1 (2001): 1-22.

with values, beliefs, memories and feelings in the past and the present as well as a physical artifact embedded in the experience of everyday life and influenced by larger social, cultural and political forces.

In contrast to design history which has tended to focus on objects of "high design," material culture was also important to this work in that it invites the analysis of more mundane objects and their everyday use. This is a critical aspect of this history of the open plan, which intentionally features a case study of a very mundane office environment (Wisconsin State Government) rather than a high design example. Even my analysis of cubicles in popular media in Chapter 4 focuses on the ordinary image of offices as represented in popular media rather than the images of high-end offices in architecture and design circles. This was a deliberate choice in that although high design offices exist (and are often celebrated in both the architectural press and popular media), they do not represent the most common work experiences. This is also important because it illustrates the ways in which a high design idea like the open plan filtered down to mainstream use. This study thus suggests how some of the big ideas of the open plan were translated into more mundane office environments.

Early on in my research on the history of the open plan design, I realized there was a significant gap between the idea of the open plan as it was promoted by the architects and designers of the late 1960s and the representation of the cubicle in contemporary media. In other words what the open plan meant in the late 1960s, and what the cubicle means today are very different, and that difference matters. As I recognized this gap, social historian Carolyn Steedman's article "What a Rag Rug Means" served as a critical model for grappling with the assumptions of the

present day scholar when "reading" objects of the past.⁶ To explore what the cubicle means in contemporary media, I used a cultural studies approach of identifying and analyzing patterns of representation across various forms of popular media. Cultural studies offered a valuable model for conceptualizing the role of representation in creating meaning. Through visual and textual analysis, I examined a large number of media artifacts to understand how the cubicle has been represented visually and textually, and what that representation says about cultural attitudes towards contemporary office work in the United States.

Perhaps the biggest methodological challenge of this project was finding a mechanism for understanding and conceptualizing change in the culture and structure of white collar work. While a "material culture" or design history approach might have focused specifically on the change of the design through a formal analysis of systems furniture, my project was focused on situating the design within the numerous changes in work in the 1970s and 1980s in America. Neither design history nor material culture offered a means of conceptualizing change beyond an analysis of the artifact, and for this reason I turned to a model from Industrial Systems Engineering, called the Work System Model. This powerful methodological tool was created by Pascale Carayon and Michael Smith as a means of conceptualizing the influence of change in the context of job design in five distinct areas: organization, people, environment, technology, and tasks. At its core, the work system model emphasizes the interaction of change across all aspects of work, such that a change in one of these areas will inevitably create change in other areas of the work

6. Carolyn Steedman, "What a Rag Rug Means," *Journal of Material Culture* 3, no. 3 (1998): 259-81.

system.⁷ By adapting this model for my study, I intended to capture a more complex vision of change in the context of work, and trace the interaction of change through the different aspects of work over time.

The Work System Model influenced my analysis, but it also directly influenced my process of identifying and interpreting relevant material for that study. As I went through all 94 boxes of the Department of Administration (DOA) Secretary Subject Files Series I in the Wisconsin Historical Society, the Work System Model changed my approach to collecting data, and led me to look more broadly for relevant materials. For example, if I had not used this approach I might have focused exclusively on folders labeled "space" and "buildings" and likely ignored altogether folders about technology, word processing, data processing, agency reorganizations, affirmative action, clerical studies, specific agency folders and so on. Although it took several months of close reading to get through all of those boxes, the result of this broader reading is that I found references to the open plan, systems furniture, and the experiences and voices of workers peppered in all different parts of the DOA archive, and those diverse documents informed my analysis.

This project is a significant contribution to the field of design history and material culture in that it provides a systems-based model for understanding the dynamics of change beyond the study of artifacts. Whereas a traditional material culture or design history analysis would have focused exclusively on the design itself, my study suggests the importance of understanding the broader social, cultural and economic context of use. I argue that while the open plan was a sig-

7. Michael J Smith, and Pascale C. Sainfort, "A Balance Theory of Job Design for Stress Reduction," *International Journal of Industrial Ergonomics* 4 (1989): 67-79.

nificant intervention in contemporary white collar work, it was just one change among many in the culture and structure of white collar work in the United States in this period. My research offers a more nuanced reading of the myriad forces of change in this period which informed the design, use, and meaning of the open plan.

As office work has changed, our image of the ideal office remains constant. Even today, the big ideas about the open plan--particularly allowing flexibility for change, an emphasis on supporting collaboration, and the goal of reducing spatial hierarchy--continue to be promoted in each new iteration of systems furniture. Thus, underpinning the promotion of each new open plan design is the on-going quest of contemporary architects, designers, and office furniture manufacturers to achieve an "ideal" workplace in which white-collar work and white-collar workers are optimally supported. In this way, the vision of the architects and designers who promoted the open plan in the late 1960s continues to be relevant, and is in fact deeply embedded in the fabric-covered partitions of even the dreariest of office cubicles.

Chapter 1

From the Conventional Plan to the Open Plan: A History of American Office Design, 1945-1990

While the open plan was a change in thinking about the interior design and use of office space, understanding its significance requires an understanding of how it was both similar and different from the ideas, theories, and practices of American office design that preceded its development and use. Throughout the twentieth century, office design has been used as a means of facilitating work processes, representing organizational hierarchy, reflecting organizational values and culture, and controlling or structuring worker activities and behaviors. In fact office architecture and design can be conceived as a kind of technology that has developed in tandem with the rise of the corporate organizational structure, the evolution of office work, and the growth of the white collar work-force from the 19th century through the 20th century. Historians of economic history, business history, and labor history have painted a complex and shifting portrait of white collar work in this period that was at various times influenced by: the role of management; the

changing social composition of the workforce (gender, race and class); the implementation and use of new office technologies; the shifting economic and political climate; and changing ideas about work, corporate structure, organizational culture, and management theory.¹ In these histories, office design (when it is addressed) is typically treated as a means of ordering and controlling social relations and work processes in the office.² Meanwhile there is a separate body of literature in architecture and design history that examines: the development of new building

1. Alfred Chandler, *The Visible Hand: the Managerial Revolution in American Business* (Cambridge: Harvard University Press, 1977); Oliver Zunz, *Making America Corporate, 1870-1920* (Chicago: The University of Chicago Press, 1990); Ileen DeVault, *Sons and Daughters of Labor: Class and Clerical Work in Turn of the Century Pittsburgh* (Ithaca: Cornell University Press, 1990); Angel Kwolek-Folland, *Engendering Business: Men and Women in the Corporate Office, 1870-1930* (Baltimore: The Johns Hopkins University Press, 1994); Venus Green, *Race on the Line: Gender, Labor, and Technology in the Bell System, 1880-1980* (Durham: Duke University Press, 2001); Margery Davies, *Woman's Place is At the Typewriter: Office Work and Office Workers, 1870-1930* (Philadelphia: Temple University Press, 1982); Sharon Hartman Strom, *Beyond the Typewriter: Gender, Class, and the Origins of Modern American Office Work, 1900-1930* (Chicago: University of Illinois Press, 1992); Nelson Lichtenstein, *State of the Union: a Century of American Labor* (Princeton: Princeton University Press, 2002); Alice Kessler-Harris, *In Pursuit of Equity: Women, Men, and the Quest for Economic Citizenship in 20th Century America* (New York: Oxford University Press, 2001); C. Wright Mills, *White Collar: the American Middle Classes* (New York: Oxford University Press, 1953); Art Kleiner, *The Age of Heretics: Heroes, Outlaws, and the Forerunners of Corporate Change* (New York: Currency, 1996).

2. Zunz, *Making America Corporate, 1870-1920*; Kwolek-Folland, *Engendering Business: Men and Women in the Corporate Office, 1870-1930*.

techniques and technologies; the evolution of office furniture and layout; the aesthetic, functional and symbolic aspects of office architecture and layout; and the relationship of office location to the landscape.³

This chapter situates the history and development of the open plan within the broader history of work and post-war office design. Drawing on the research of a number of different scholars and historians, I describe the influence of new architectural technologies and the increase of suburban campuses, as well as the widespread popularity of the “conventional plan” and its relationship to the culture and structure of white-collar work in the post-war period. I then describe the development of the open plan office in the late 1960s particularly in the form of two significant office design concepts: office landscaping and Action Office. The last part of the chapter examines the rapid development and popularity of the open plan, the growth of the systems furniture industry, and the conventions of design for the open plan (typical layouts, furniture types) that emerged in this period.

3. Carol Willis, *Form Follows Finance: Skyscrapers and Skylines in New York and Chicago* (New York: Princeton Architectural Press, 1995); Inaki Abalos, and Juan Herreros, *Tower and Office: From Modernist Theory to Cotemporary Practice* (Cambridge, MA: MIT Press, 2003); Linda Stewart Gatter, “The Office: an Analysis of the Evolution of a Workplace” (M.I.T., 1982); Olga Pantelidou, “Designing for the Workflow,” in *Kevin Roche: Architecture as Environment*, ed. Eeva-Liisa Pelkonen (New Haven: Yale University Press, 2011); James Russell, “Form Follows Fad: the Troubled Love Affair of Architectural Style and Management Ideal,” in *On the Job: Design and the American Office*, ed. Donald Albrecht, and Chrysanthé Broikos (New York: Princeton Architectural Press, 2000); John F Pile, *Interiors 3rd Book of Offices* (New York: Whitney Library of Design, 1976); Reinhold Martin, *The Organizational Complex: Architecture, Media, and Corporate Space* (Cambridge, MA: MIT Press, 2003); Carolyn Torma, “The Spatial Order of Work,” *Perspectives in Vernacular Architecture* 7 (1997): 183-95; Louise Mozingo, *Pastoral Capitalism: a History of Corporate Landscapes* (Cambridge, MA: MIT Press, 2011).

Post-War Corporate Architecture: New Technologies; New Locations

The post-war period is sometimes considered the “golden age” of corporate architecture.⁴

According to James Russell, American corporations of that era were experiencing enormous growth (in terms of their market, their products, and in their workforce), and had the financial capital to invest in new corporate offices. After years of making do with older buildings, American corporations could finally build new corporate homes from the ground up that were designed for their particular needs and culture. Russell describes how these new offices were designed to be the pinnacle of contemporary office design featuring the latest styles in architecture and interior design, the newest forms of building technology, and a spatial arrangement that reflected the organizational structure of the corporation.⁵

Architectural historians have argued that there were two major new technologies that directly influenced the form and design of post-war American corporate architecture: fluorescent lighting and air conditioning. In her book *Form Follows Finance*, architectural historian Carol Willis describes how the use of cool fluorescent lighting allowed architects to create deeper spaces without access to windows, and allowed companies to use all of the interior space more effectively. Air conditioning allowed architects to create buildings that were entirely sealed from the outside and featured dramatic uses of glass in the form of the glass curtain wall.⁶ While these

4.Russell, “Form Follows Fad: the Troubled Love Affair of Architectural Style and Management Ideal,” 56.

5.Russell, “Form Follows Fad: the Troubled Love Affair of Architectural Style and Management Ideal,” 56-59.

6.Willis, *Form Follows Finance: Skyscrapers and Skylines in New York and Chicago*, 132-33.

impressive uses of glass in office buildings became very fashionable, it was also extremely expensive. In addition, having full walls of windows that could not be opened by users required significant air conditioning to maintain comfortable ambient temperatures in the warmer months. One study estimated that a full-height glass curtain wall could increase air conditioning usage in an office building by as much as 50%.⁷

In urban centers, the glass and steel facade of the modern skyscraper became familiar symbols of the contemporary corporation. Such buildings represented a feat of engineering and architectural ingenuity, and even their exorbitant use of glass became a symbol of economic success and corporate power. Carol Willis argues that the International Style skyscrapers of the post-war period were designed to “project an image of affluence and modernity.”⁸ According to architectural historian Reinhold Martin, these modern skyscrapers also served as “corporate advertisements,” often representing the values or materials of the corporation itself in the construction. For example, the Inland Steel building featured a “predominance of stainless steel,” the Alcoa Corporation building featured an aluminum exterior, and the Corning Glass corporation building featured large amounts of glass.⁹

Many of these same building techniques that were first implemented in soaring skyscrapers were also used in the sprawling suburban corporate campuses that became popular in the post-war period. In *Pastoral Capitalism*, Louise Mozingo describes how corporations began their

7. Gail Cooper, *Air-conditioning America: Engineers and the Controlled Environment, 1900-1960* (Baltimore: Johns Hopkins University Press, 1998), 160-62.

8. Willis, *Form Follows Finance: Skyscrapers and Skylines in New York and Chicago*, 141.

9. Martin, *The Organizational Complex: Architecture, Media, and Corporate Space*, 102-03.

exodus from the central business district to the suburbs to accommodate the space and technical demands of research divisions, to reduce the opportunity for unionization among office workers, and to increase the safety and security of the corporate office and its workers in the event of a nuclear attack. These new suburban campuses were often set on large tracts of land with very deep landscaped or wooded areas, and typically featured a sprawling low-rise building set back from the road. Although there was ample parking, architects often took pains to conceal or minimize parking areas so as not to detract from the visual qualities of the landscape. Like the skyscraper, these suburban campuses often became the public image of the corporation itself representing its values, corporate culture, and product.¹⁰

Designing Corporate Interiors: Hierarchy, Efficiency, and Control

Whether situated inside of an urban skyscraper or a suburban corporate campus, the interiors of post-war offices were intentionally planned around the organizational structure and corporate hierarchy, and designed to achieve optimum efficiency and productivity. In fact, the interior arrangement of offices in this period was in many respects a culmination of ideas that had been practiced in corporate architecture since the late 19th century.

The first skyscrapers that were completed in the late 19th and early 20th century featured a combination of private offices and wide open spaces that eventually came to be known as "bullpens." At that time there was a growing interest in applying the principles of Scientific

10. Mozingo, *Pastoral Capitalism: a History of Corporate Landscapes*, 22-43.

Management to the organization and management of clerical work. This had a significant impact on the organization of work (particularly the increasing division of labor in American offices) as well as the physical arrangement of American offices. By the early 20th century it was common practice to arrange the interior space according to departmental function, and workers according to internal work processes. According to historian Angel Kwolek-Folland, by controlling the arrangement of furniture and even the placement of amenities like water coolers, “systemic managers” of the 1920s could use the layout of the office to manage worker behavior and interactions. Kwolek-Folland argues that the office served as a “physical counterpart to the permanent bureaucratic structures of the corporation” and further that the “arrangement of fixtures in the office environment organized the workplace itself.”¹¹ The physical design of the office thus served as a mechanism for order, surveillance, and control within the workplace.

In post-war offices, the physical arrangement of the interior was similarly connected to the organizational structure of the corporation both in terms of the overall bureaucracy and in terms of internal hierarchy. Indeed, post-war offices were in many respects an architectural representation of the official organizational chart mapping the relationships and status of divisions, departments, and individual workers. The planning process for the interior design and arrangement of office work typically began at the level of the individual worker and their specific space needs. Companies created elaborate space standards that dictated the exact amount of space and furniture required (both functionally and symbolically) for each worker to be

11. Kwolek-Folland, *Engendering Business: Men and Women in the Corporate Office, 1870-1930*, 108-110).

effective. In books on office planning, the office space administrator was encouraged to use the individual worker and their specific space needs as the basis for planning the office as a whole.¹²

New construction also gave companies the opportunity to rethink the overall layout of their buildings and the logical relationships between different departments simplifying the flow of paperwork through the entire office horizontally through an individual department as well as across multiple departments and even vertically across multiple floors.¹³ Management manuals such as George Terry's *Office Management and Control* and handbooks such as *How to Plan an Office Layout* offered advice on methods of identifying the flow of paperwork and systematically rearranging the desks in order to minimize the need for workers to walk, which in turn would save time and money.¹⁴ Though these ideas had been in use since the early 20th century, the process of building an entirely new office gave organizations the opportunity to systematically apply these ideas producing a more fully rationalized use of office space.

In constructing a new office, it was common to begin with a survey of interior space needs which were then used to plan the whole building. An excellent study of this process is offered by the book *The Fourth Dimension in Architecture* by Edward and Mildred Hall which describes the design and construction process of the John Deere building in Moline, Illinois in the early 1960s. According to Hall and Hall, the architect (Eero Saarinen) and building committee first

12.Kenneth H Ripnen, *Office Space Administration* (New York: McGraw-Hill, 1974), 15.

13.Mills, *White Collar: the American Middle Classes*, 197.

14.National Stationery and Office Equipment Association, *How to Plan Your Office Layout: a Guide for Planning a Modern, Efficient Office* (Washington, D.C.: National Stationery and Office Equipment Association, 1953), 20; George Terry, *Office Management and Control*, Third ed., (Homewood, IL: Richard D. Irwin, 1959).

established several standard “modules” that would become the basis for the overall plan. In the construction process, the size of the desk (which was designed by Saarinen) became the basis for planning, and different types of workers were allotted different types of furniture: “clerks and secretaries were provided with a desk; draftsmen were given a table and storage space. For employees with private offices, each person was given a desk and a credenza.”¹⁵ This kind of rationalized approach to the allocation and design of office space was common in American offices in the post-war period, and in fact, the process of designing a new corporate headquarters allowed companies to establish fresh space standards based on current needs, and to implement those standards uniformly across the organization.

These space standards dictated the size of a particular workspace, and they often specified the quality and type of interior decoration as well. In constructing new corporate offices, architects and designers sought to achieve a cohesive interior design and style. Modern design with its clean lines and functionalist aesthetic was the ideal expression of this highly standardized and rationalized culture, and became the popular style for new corporate offices in the period. Architects and designers together were responsible for creating a comprehensive design that married the architectural details of the structure with the decorative details of the interiors. According to Hall and Hall, Eero Saarinen reportedly planned every detail of the interior design of the John Deere building from the furniture and equipment all the way down to the china, flatware, salt and pepper dispensers and even the sugar packets used in the dining

15. Mildred Reed Hall, and Edward T. Hall, *The Fourth Dimension in Architecture: the Impact of Building on Behavior* (Santa Fe: Sunstone Press, 1975), 18.

room.¹⁶ As one architectural journalist described, this concept of a "Total Design" "turned the office into a Platonic composition of official art, official plants, and official ashtray positions, implying a military academy gone aesthetic."¹⁷ The visual language of modern art and design became a symbol of the successful and efficient modern corporation evoking the highly rationalized and controlled nature of the organization.

Within the framework of the total design, certain architecture and design elements could be used to produce visual distinction and represent the organizational hierarchy. In his book, *Holy Things and Profane*, architectural historian Dell Upton, developed the twin concepts of style and mode. According to Upton, "style" "provides a context, or system of common understanding," while "mode" (the elements of design that stand out as distinctive) "emphasizes and perpetuates old differences" and "works to create new differences."¹⁸ Applied to the post-war organization, the unified "style" of the corporate office served to create a sense of shared purpose and identity, while the elements of distinction or "mode" granted to individual workers emphasized their status and power within the organization. Space manuals of the period would often specify the size of a particular workspace as well as its furniture and accessories based on position, even specifying drapes, carpeting, carafes, telephone stands and bookcases.¹⁹ One article recommends "utilizing in the private office the fine wood version of the more economical metal furniture that

16.Hall, and Hall, *The Fourth Dimension in Architecture: the Impact of Building on Behavior*, 20.

17. "Chaos as a System," 164.

18.Dell Upton, *Holy Things and Profane: Anglican Parish Churches in Colonial Virginia* (New Haven: Yale University Press, 1997), 102.

19.Vance Packard, *The Pyramid Climbers* (New York: McGraw-Hill, 1962), 19.

is used in the general office," thus creating a unified appearance while also signifying status in the organization.²⁰ The design of the office then became a deliberate reflection of the corporate ladder with each rung represented by different furniture and accessories. As workers moved up within the company, they expected parallel changes in the quality and size of their workspace that would visually communicate their new position.

While the open bullpen had always allowed for an easy rearrangement of space, the increasing use of movable partitions (instead of permanent partitions) gave companies more flexibility even in the allocation and arrangement of private offices. These early partitions were generally made of metal or wood with electrical wiring in the baseboards, and were available in a variety of heights from "the railing to the ceiling."²¹ Private offices were an expensive commodity that hampered a company's ability to alter the layout of the office; in specifying and using movable partitions, companies were able to provide limited privacy while minimizing the cost and maximizing flexibility. Yet, these movable partitions of the early post-war period were far from ideal. In their study, Mildred and Edward Hall describe how movable partitions of the era were often criticized because they were difficult to move, and companies in general did not tend to move them.²² Nonetheless, the early use of these movable partitions speaks to a recognition in the period that the physical arrangement of office space must be adaptable to changes in the organization, particularly expansion. Indeed, some manuals even recommended

20. Kenneth Johnson, "Individual Choice Versus Conformity to Standards in Design of Private Offices," in *New Concepts in Office Design*, ed. Carroll Cihlar (Elmhurst, IL: Business Press, 1968), 5.

21. Terry, *Office Management and Control*, 361-62.

22. Hall, and Hall, *The Fourth Dimension in Architecture: the Impact of Building on Behavior*, 27.

setting generous space standards to begin with and later reducing space standards to accommodate an increase in the number of workers.²³

According to sociologist C. Wright Mills, the physical arrangement and design of new offices in the post-war period was a direct manifestation of the larger culture of white collar work in the period. For clerical workers the factory-like environment of the open bullpen both reflected and facilitated the highly simplified and routinized nature of the work they did. In this period, management relied on a scientific management approach to job design in which each worker's job was tightly defined and controlled by management. In creating mindless routinized jobs, management sought to achieve maximum productivity with minimal room for human error. Combined with the enormous growth of office machines, clerical workers had increasingly less autonomy and control over the work process. By arranging workers in an assembly-line style and emphasizing the use of machines, management maintained greater control over the movements and activities of workers.²⁴

Further, the open space of the bullpen allowed for optimum supervision, preventing workers from taking unauthorized breaks or spending too much time talking with neighbors. Some manuals such as Kenneth Ripnen's book, *Office Space Administration* (originally published as *Office Building and Office Layout Planning* in 1960), suggested that clerical workers be placed far enough apart to reduce "the incidence of gossip, of passing the time of day, or exchanging confidences;" if a worker has to get up and move in order to engage in conversation, personal discussions would be made more obvious to the supervisor discouraging

23.Ripnen, *Office Space Administration*, 25-26.

24.Mills, *White Collar: the American Middle Classes*.

the behavior in the first place.²⁵ In the view of these management and office design handbooks, such casual conversation was a deliberate waste of precious time and efficiency, and needed to be curbed in order to achieve maximum worker output. In this way, the design and layout of the office promoted the underlying work system allowing optimum control over workers and enabling greater efficiency and productivity within the larger work process.

This culture of control was also applied to workers' individual use of their workspace. Management handbooks of the period suggested prescribing the interior organization of desks in order to completely standardize each work area.²⁶ Some companies went so far as to establish clean desk policies that prevented workers from displaying personal objects in their work areas, and even mandated clean desks at the end of each work day. For example, when John Deere completed their new corporate headquarters in 1964, they established a clean desk rule requiring workers to maintain a neat work area (restricting any personal decorations), and requiring them to clean their desk completely at the end of each day. Mildred and Edward Hall found that many workers at John Deere were dissatisfied with this restriction on their decoration and use of work space, stating that while the clean desk policy certainly "preserves the integrity of the interior design" it left some workers feeling like "inmates of the edifice," in other words even though workers recognized the value and aesthetic of Modern design and architecture, some workers felt imprisoned in the overly controlled office environment.²⁷ In this way, the needs and working

25. Rippen, *Office Space Administration*, 26.

26. Terry, *Office Management and Control*, 242.

27. Hall, and Hall, *The Fourth Dimension in Architecture: the Impact of Building on Behavior*, 31.

styles of individual workers mattered less than the visual aesthetic that the Modern design and rows of tidy clean desks produced.

For upper-level white-collar workers, the emphasis on the visual and spatial representation of status in the office reflected and reinforced the strongly defined social hierarchy of the office. According to C. Wright Mills, the titles and appurtenances of authority in the post-war corporation served as "crucial signs of status" that communicated the often minute gradations of power and authority.²⁸ The so-called "organization man" of the post-war corporation was expected to follow the established hierarchy, not challenging the authority of anyone above them by questioning orders or ideas.²⁹ In his wry insider's account of post-war corporate life, Alan Harrington describes the rigid hierarchy in which individuality and creativity were suppressed in favor of a systemic conformity. According to Harrington, the most successful workers in the "Crystal Palace" were those who were able to sublimate their own ambitions and initiative in favor of maintaining the status quo of the corporate hierarchy. Even those who entered the organization with a modicum of talent soon found themselves with a "growing lack of impulse, invention, and desire" thus accepting and internalizing the culture of mediocrity and conformity.³⁰ As a glass and steel manifestation of the organizational hierarchy, post-war corporate offices served as the infrastructure of conformity, reifying the power and authority of those at the top of the corporate ladder.

28. Mills, *White Collar: the American Middle Classes*, 211.

29. William H. Whyte, *The Organization Man* (Garden City, NY: Doubleday, 1956).

30. Alan Harrington, *Life in the Crystal Palace* (New York: Alfred A. Knopf, 1959), 30.

The New Open Plan

The wide open spaces that were critical to the design of the conventional plan also enabled the development of the open plan. In other words, although the open plan was in some respects a significant shift away from conventional planning, it can also be viewed as its direct descendent. Further, in some older discussions of office design, the "conventional plan" is sometimes referred to as an "open plan."³¹ According to interior designer John Pile, the term "conventional plan" only came in to more common use in the mid 1970s to contrast with what became known as the "new open plan."³²

There have been a few histories of office design that have explored the development of the new open plan in the late 1960s through the 1970s. John Pile wrote several books in which he described the history and development of the open plan as part of his explanation of its use and value as did architect Francis Duffy.³³ Stewart Brand also described the development of the open plan in his book *How Buildings Learn* which draws directly from both John Pile and Francis Duffy.³⁴ Linda Stewart Gatter included a history of the open plan as part of her Master's thesis on the history of office design, and Thomas Hench wrote a lengthy Ph.D. dissertation in which he

31.Sydney Rodgers, "Modularity Versus Free-form Clustering," in *New Concepts in Business Design* (Elmhurst, IL: The Business Press, 1968), 14; Gatter, "The Office: an Analysis of the Evolution of a Workplace," 70.

32.John F Pile, *Open Office Planning : a Handbook for Interior Designers and Architects* (New York: Whitney Library of Design, 1978), 14; Pile, *Interiors 3rd Book of Offices*, 38.

33.For example: Pile, *Open Office Planning : a Handbook for Interior Designers and Architects*; Pile, *Interiors 3rd Book of Offices*; John F Pile, *Open Office Space* (New York, N.Y: Facts on File, 1984). Francis Duffy, et al., eds. *Planning Office Space* (New York: Nichols Publishing Company, 1976).

34.Stewart Brand, *How Buildings Learn: What Happens After They're Built* (New York: Viking, 1994), 168-69.

described the evolution of systems furniture from an industrial and business perspective.³⁵ More recently architecture critic James Russell explored the development of the open plan in relation to changes in management in his article “Form Follows Fad,” and architectural historian Olga Pantelidou described the development of the open plan in her article about the work of architect Kevin Roche.³⁶ In all of these histories, office landscaping and Action Office are typically cited as the two original “open plan” concepts that became the basis of our modern open plan.

The office landscape first arrived in the United States in 1967 when the DuPont Company invited Germany’s Quickborner Team to create an office landscaping installation in their Freon Division. While the architectural journals had been following the development of office landscaping in Germany for several years, it was not until the first American installation was completed that office landscaping began receiving significant press in the United States.³⁷ Called “burolandschaft” in German, office landscaping developed as a means of improving communication within the office. Visually, office landscaping was characterized by a wide open space with clusters of desks arranged at different angles and the prominent use of curving screens and plants to demarcate the interior space. Yet its visual style was really a side effect of a layout process that used communication patterns as the basis of the arrangement. Office landscape planning began with a complete survey of all office staff and their individual patterns of

35.Gatter, “The Office: an Analysis of the Evolution of a Workplace.”; Thomas Jackson Hench, “An ‘evolutionary History’ of the Office Systems Furniture Industry and the Nature of Strategic Change: Evolution, Entrepreneurship and Strategy as Complex Adaptive Processes: a Study in Self-organizing Systems” (University of South Carolina, 1997).

36.Russell, “Form Follows Fad: the Troubled Love Affair of Architectural Style and Management Ideal.”; Pantelidou, “Designing for the Workflow.”

37.“Office Landscape,” *Progressive Architecture* (1964): 201-03.

communication. The layout was then developed around this matrix placing workers in close proximity to the other staff members they communicate with the most frequently regardless of status or departmental affiliation. In this way, office landscaping tossed out the American convention of relying on the prescriptive lines of communication as depicted by the organizational chart, and instead sought to uncover the real lines of communication.³⁸

The Quickborner Team believed strongly that office partitions only inhibited effective communication among workers, and for this reason eliminated all private offices from their installations. In an article from the *New York Times*, reporter William Robbins describes their logic thus:

Why, they ask, should it take the time of eight people to try to set up a conference between four executives—discussions between the executive calling the conference and his secretary, telephone calls between secretaries and more discussions between secretaries and bosses—only to find, perhaps, that it is not convenient at the moment to meet? It should be possible, they say, for the man who wants the conference to stand up, look across the screen and check with colleagues if they are all available, they simply walk over to a table, sit down, and start talking.³⁹

In this way, the private office was more than just a harmless symbol of status, it was itself a barrier to communication. By simply removing walls, the corporate executive would supposedly no longer be hampered by the constraints of the corporate bureaucracy, and be free to casually call a meeting at any time without the formalities the conventional office perpetuates.

38. John Pile, “The Nature of Office Landscape,” *AIA Journal* (1969): 40-48.

39. William Robbins, “‘Radicals’ Offer New Office Plan,” *New York Times (1923-Current File)*, July 4 1968, 34, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/118437553?accountid=465>.

In its use of the social context of the office as the basis for design, office landscaping fit well with these ideas that were already circulating in the United States. According to Frank Duffy in his short 1966 monograph *Office Landscaping*, the concept of the office landscape evolved from this new research on the “psychology of group work,” and hinged on the notion that the workplace is not a “hierarchical chain of command” but rather a “united whole in which many complex inter-actions” occur.⁴⁰ Indeed, by placing the social needs of workers at the center of the planning process, the resulting plan was often described as one that was so natural the workers themselves might have created it had they been “left to their own devices.”⁴¹

In contrast to the neatly regimented rows of desks so common in the conventional bullpen, office landscapes featured an apparent jumbled mess of desks arranged willy-nilly around the open space. The architectural press seemed to embrace this visual chaos, featuring images of the office landscape layout prominently in their articles, often on the front page surrounding the article’s title.⁴² In describing the new office landscaping concept, architects and designers also celebrated the seeming visual chaos, for example an author describing the DuPont Company’s Freon Division described their new office landscape installation as “an open, unenclosed space with activities swirling to diffusion in a free-form pattern.”⁴³ To aid in legibility, architectural journals often added notations to indicate groupings of workers or circulation patterns around the office. Figure 1 shows such a drawing, with organic blob-like

40. Frank Duffy, *Office Landscaping: a New Approach to Office Planning* (London: Anbar Publications, 1969), 9.

41. “From Grid to Growth,” *Progressive Architecture* (1969), 103.

42. For example: Pile, “The Nature of Office Landscape.”; “Chaos as a System.”

43. “Burolandschaft USA,” *Progressive Architecture* (1968), 175.

shapes encircling various groupings of desks. The amorphous shapes dominate the image and emphasize the visual chaos, even as they are illustrating the “organic order” underpinning the layout. These types of shapes were becoming popular in visual culture of the 1960s, and according to historian Jennifer Roberts, were evocative of the natural world, suggesting blood cells circulating through the body or the blob-like shape of a human embryo which new technologies had recently rendered visible to the human eye. In this way the shapes were both “natural” but also scientific reflecting the power of new technology to see that which is unseen.⁴⁴ The seeming visual chaos of the open plan was thus re-imagined as a natural product of using the worker’s communication patterns as the basis of planning.

Despite its “disorderly” appearance, articles about office landscaping in the architectural press were largely positive about this new development describing its many benefits in glowing terms. According to interior designer John Pile these included:

improved work performance resulting from improved communications and work relationships; improved morale; greatly improved flexibility for organizational changes rendering replanning easy and thus encouraging the actual making of changes; and—quite incidentally but of great interest—markedly reduced costs in initial construction and, over extended time even greater cost reductions because of the ease of making changes.⁴⁵

Reports of early installations at Eastman Kodak, John Hancock Mutual Insurance and DuPont described highly favorable results of increased productivity and enormous cost savings (particularly in rearranging) that seemed to suggest that office landscaping was a practical solution to

44. Jennifer L. Roberts, “Lucubrations on a Lava Lamp: Tehnocracy, Counterculture, and Containment in the Sixties,” in *American Artifacts*, ed. David Prown, and Kenneth Haltman (Lansing, MI: Michigan State University Press, 2000), 174.

45. Pile, “The Nature of Office Landscape,” 43-44.

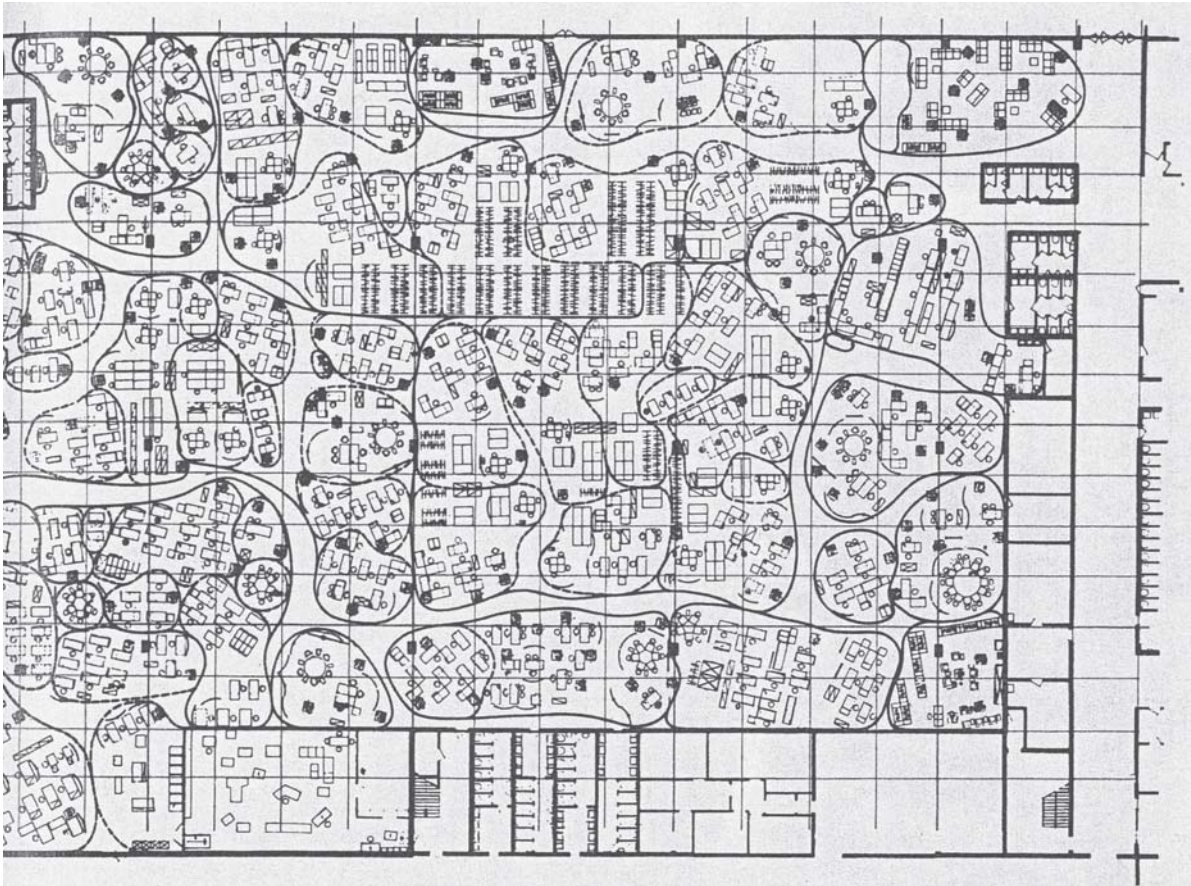


Figure 1: Detail of a "chaotic" office landscaping layout with amoeba-like groupings drawn in to illustrate the "organic order." Reproduced from the article "From Grid to Growth." *Progressive Architecture* (1969): 103. This image is reprinted with the permission of Penton Media, Inc.

real business problems, despite its unconventional appearance. The office landscape offered a less hierarchical and more flexible work environment that could be changed as necessary to suit the changing needs of the organization.

In many ways the office landscape re-imagined the workplace as a more permissive and casual environment than the conventional office of the post-war period. By arranging workers based on patterns of communication, it was expected that workers would talk; a stark contrast to the bullpen in which workers were actively discouraged from talking. Even the routine of the office would change to allow workers to take breaks whenever they wanted, rather than waiting for an office-wide coffee break (a standard feature of American offices at the time).⁴⁶ Special break areas called *pausenraum* were available in each installation so that workers engaged in more personal conversations would have a comfortable place to go and talk.⁴⁷

Yet even as it allowed for some new freedoms, the office landscape also maintained certain controls. For example, the Quickborner Team believed strongly in limiting individual storage space. They argued that workers have a tendency to “hoard” information that might be relevant and useful to others, and further that storage becomes a means of procrastination. For this reason, office landscaping relied on simple tables rather than conventional desks in order to reduce this tendency of workers to stockpile records and files in their workspace.⁴⁸ While the old office had rules in place about how workers used their desk space, this new kind office design eliminated that desk space altogether. With only a simple tabletop (not even a modesty panel),

46. “Chaos as a System,” 164.

47. Pile, “The Nature of Office Landscape.”

48. “Chaos as a System,” 169.

workers could not store much of anything, work related or otherwise at their desks. This lack of interior space in the desks, mirrors the lack of privacy in the office landscape as a whole. With only a few curved screens and plants to demarcate the space, the office landscape was very open and left most workers with little if any visual privacy. Though the layout was considered less oppressive than the old bullpen in that it was designed to prevent workers from all facing one direction or having eye contact with each other, the complete openness combined with the lack of desk storage actually enabled greater supervision in that all of a worker's activities (even the tapping of their feet under the desk) were rendered entirely visible.

This complete openness, and lack of spatial definition was one of the most common complaints against the office landscape. As a solution to this perceived problem, the architectural press pointed to the development of a new kind of furniture that might give the office space a bit more definition without compromising communication: Herman Miller's Action Office (Figure 2).⁴⁹ The research and development of Action Office began in the late 1950s, when Robert Propst was first hired to head Herman Miller's new research division. Early on he became obsessed with the problems of contemporary office design, and began developing a new kind of furniture that would offer greater flexibility for the individual needs of workers.⁵⁰ Although the first version of this new kind of furniture, Action Office I, actually came to the market in 1965, it was not very successful in the market until the late 1960s when the release of Action Office II

49. "Chaos as a System," 169.

50. Hugh DePree, *Business as Usual: the People and Principles At Herman Miller* (Zeeland, MI: Herman Miller, 1992), 84.



Figure 2: Herman Miller's Action Office shown with workers and work materials "in action."
Reproduced from John Pile, *Interiors 3rd Book of Offices* (New York: Whitney Library of Design, 1976), 194. Permission requested.

corresponded with the sudden interest in office landscaping in the United States.⁵¹ Though Action Office did share some significant commonalities with office landscaping including the elimination of private offices and the emphasis on communication, Action Office was also markedly different particularly in its commitment to supporting the needs of the individual worker.

Unlike the simple tables and panels of office landscaping, or the more substantial desks and semi-movable partitions of the conventional office, Action Office was a fully integrated furniture system that used panel walls not as mere dividers of space, but also as component pieces that supported writing surfaces, shelves and open storage. Fully adjustable, workers could alter their table height, move shelves, rearrange storage, and even move the panel walls to suit their own unique manner of working. Because the furniture required no special tools or equipment to make simple modifications, each workstation could truly be adjusted by the individual worker as needed, without any assistance from a facilities manager. In designing Action Office this way, Propst argued that the workstation should meet the needs of different workers, as well as meet the changing needs of a single worker whose workspace needs might change at any time depending on what projects he or she is working. Thus, Propst suggested that work itself was not an unchanging routine (in which one worker did the same monotonous task over and over), rather it was a complex and evolving series of activities and projects that could take place over hours, days, or weeks.⁵²

51. Stanley Abercrombie, "Office Supplies: Evolving Furniture for the Evolving Workplace," in *On the Job: Design and the American Office*, ed. Donald Albrecht, and Chrysanthé Broikos (New York: Princeton Architectural Press, 2000), 89-90.

52. Robert Propst, *The Office: a Facility Based on Change* (Elmhurst, IL: The Business Press, 1968), 22.

In contrast to the neat and controlled environment of the conventional office and the office landscape, Action Office invited workers to populate their work areas with the information and materials that they needed to do their work. The panel walls of Action Office were a canvas for display and reflection, allowing workers to keep important information constantly accessible and visible. To convey this idea, images of Action Office featured workers at the desks, surrounded by papers in piles and pinned to the walls reflecting the highly messy nature of true work. Unlike the traditional architectural style photos of offices that often featured clean sparse landscapes with empty desks reflecting an ideal image of control and order, Action Office was portrayed as a bustling nerve center with papers and people in constant motion (Figure 2). As a workspace, Action Office helped support the necessary clutter, but also helped to organize it, by allowing workers to effectively manipulate information in different ways. While Propst shared some of the Quickborner Team's concerns about workers storing too much at their desks, he did not believe that the elimination of all storage was the best way to deal with that issue. Instead Propst eliminated deep storage where papers tend to "hide and die" and created a new kind of active storage that kept materials visible and accessible for as long as they were needed.⁵³

In addition to providing vertical space for storage and display, the panel walls also served the important function of enclosure. Like the Quickborner Team, Propst believed that private offices hindered communication and "environmental awareness" and argued that most workers would be better served by being in an open office environment. However, in contrast to Quickborner who used panel walls sparingly to define the space generally and to give privacy and symbolic status to upper level workers, Propst believed that all workers required some

53. Propst, *The Office: a Facility Based on Change*, 59.

degree of enclosure. According to Robert Propst, workers sought an optimum combination of privacy and involvement, allowing them to have a sense of personal territory and control while also enabling them to stay connected to their organizational community. By combining panels of varying heights, not only throughout the office as was fairly conventional at the time, but even in a single workstation, Action Office provided an optimum combination of enclosure and access depending on a worker's individual need.⁵⁴

Yet, even in his emphasis on the needs of the individual, Propst also spoke to organizational needs arguing that Action Office was adaptable to a variety of different kinds of organizations, and was flexible enough to allow frequent and easy change whenever necessary. While in the past organizations may have sought out permanence and stability, the new economic, social and technological climate required the ability of organizations in all industries to accept and adapt to change. Propst further argued that the "grow or die" state of contemporary organizations required companies to continuously expand and develop or else become obsolete.⁵⁵ To this end, Propst asserted that offices too needed to be highly flexible and adaptable to changes; by enabling offices to change as quickly, easily, and as often as necessary the Action Office was ideally suited to this new reality.⁵⁶

Action Office was also highly flexible in that it was adaptable to both the "conventional rectilinear layout or the free-form, communications-oriented office landscape."⁵⁷ In this way,

54. Propst, *The Office: a Facility Based on Change*, 42-43.

55. Propst, *The Office: a Facility Based on Change*, 12-13.

56. Propst, *The Office: a Facility Based on Change*, 29.

57. "Chaos as a System," 169.

Action Office was adaptable to different kinds of arrangements depending on the unique needs of a particular organization and its workers. Further, while the Quickborner Team emphasized following their rules of layout (no private offices, organized around communication, etc), Action Office was not as rigid in its suggestions for use. In using Action Office, organizations could selectively adopt some aspects of office landscaping (for instance organizing the office based on communication) without necessarily adopting the irregular “free-form” layout.

The Development of Systems Furniture

Together Action Office and Office Landscaping became the basis of what generically became known as “the open plan” or the “open office.” Early on there were few furniture options for open planning, and many companies and organizations adopting the open plan did so using conventional furniture and free-standing screens (curved or rectilinear) to wrap desks, and create spatial divisions.⁵⁸ Starting in the early 1970s, a number of furniture manufacturers began producing their own versions of open plan furniture which came to be called “modular furniture,” “unit furniture” and “systems furniture.” The idea of “modular” furniture was that it would create a “uniform structural component” that could be used repeatedly in different configurations throughout the office. The modular work stations were designed to have

⁵⁸for example see: Rodgers, “Modularity Versus Free-form Clustering.”; “Chaos as a System.”

interchangeable components so that offices could easily swap out parts or make repairs. Some lines still featured free-standing screens, and others had elements of the workspace hung from the partitions.⁵⁹

Interior designer and author John Pile wrote extensively about the open plan and the benefits and challenges of different furniture options. In his book, *Open Office Planning*, he describes how many of the systems furniture lines of the 1970s were actually based on “conventional desk systems,” just paired with storage elements and flat panels to create workstations. In fact, these types of systems were not very different from the partition systems used in the 1950s and 1960s, the only difference being that they were marketed and sold by furniture manufacturers rather than partition manufacturers.⁶⁰ According to Pile another significant difference between conventional planning and the new open planning was that while furniture for conventional planning was largely standardized (set sizes that were used across multiple manufacturers), the new systems furniture lines were each completely different and required a total investment by the client. Further the client was effectively “locked in” to whatever system line they selected for some time, “regardless of what happens with price or availability.”⁶¹ When a major company invested in a systems furniture line, there was an expectation that the furniture system would continue to be available for the foreseeable future (at least 20 years).⁶² While the company purchasing the system was locked in, so too was the

59.Dwayne Meisner, “Work Stations Move Into the Mainstream,” *Administrative Management* (1973), 23.

60.Pile, *Open Office Planning : a Handbook for Interior Designers and Architects*, 109.

61.John Pile, “The Open Office: Does it Work?,” *Progressive Architecture* (1977): 68-81.

62.John Pile, “The Second Generation,” *Interiors* (1981): 88-89, 114.

manufacturer, who would need to continue to produce their furniture line (and update it) to meet the on-going demands and needs of their clients. In other words, systems furniture was a significant long-term investment for any company choosing to adopt the open plan. However, one of the biggest benefits of investing in systems furniture was that it had a significant tax advantage over conventional planning in that the “movable walls and partitions” qualified for “accelerated depreciation” for tax purposes.⁶³ Advertisements for systems furniture often also emphasized the reduction in space costs, and the increase in efficiency that would result from implementing an open plan.

Meanwhile, the number of systems furniture lines continued to grow through the 1970s. An article from 1974 identifies 37 different “open plan” furniture systems by 27 different manufacturers, and surveys their various attributes including panel heights, finish options, design elements (leg styles, edge details), and various other characteristics. In his review, Hugh Keiser found that “manufacturers have addressed themselves to developing furniture to meet an idealized, standardized version of work station operations. Each manufacturer has its own theories about the proper placement of files or telephones for maximum efficiency.”⁶⁴ Keiser suggests that although systems furniture was viewed as a means of improving the quality and efficiency of white-collar work, the claims were mostly untested and unsubstantiated. Rather they were based on an idealized vision of work and workers, based on the limited perspective

63. Carter Horsely, “Some Second Thoughts on Open-plan Offices,” *New York Times (1923-Current File)*, March 18 1979, R1, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/120968376?accountid=465>.

64. Hugh Keiser, “Open Plan Furniture: an Investigation and Analysis,” *Progressive Architecture* (1974), 94.

and company motivations of the furniture manufacturers and their designers. By 1984 it was estimated that there were more than two hundred different systems furniture lines on the market.⁶⁵ In this way, the marketplace was a mess of competing ideas about effective workplace design.

One of the main debates among the different open plan options was the role of storage. Office landscaping and Action Office were both developed with a preference for minimal storage in the workspace; the Quickborner Team advocated for almost no storage in the workspace, preferring the use of simple tables that lacked even a pencil drawer, and Action Office encouraged the use of “active storage” elements that only allowed workers to keep materials at hand that they were actively using. Yet, later furniture lines used “storage walls” as central components of the workspace.⁶⁶ According to Hugh Keiser, various manufacturers had different theories on the “superiority of one storage device above another” for example “the difference between wall-hung components and vertically integrated storage units.”⁶⁷ Some of the early systems furniture lines actually used storage as the panels dividing work stations, for example Steelcase’s first “comprehensive” system furniture line called Movable Walls used storage elements as the infrastructure of each workstation.⁶⁸

Ideal panel height was also debated extensively. For example Interior Designer John Pile favored mid-height panels of around 60 inches (5 feet); high enough to create privacy while

65.Abercrombie, “Office Supplies: Evolving Furniture for the Evolving Workplace,” 90.

66.Pile, “The Open Office: Does it Work?”

67.Keiser, “Open Plan Furniture: an Investigation and Analysis.”

68.Abercrombie, “Office Supplies: Evolving Furniture for the Evolving Workplace,” 90.

seated, but tall enough that while standing most people could see over the panel. In his 1978 book, he argues that “lower screens have the advantage of allowing a person to look across longer distances when standing in order to locate people far away, and they increase this sense of openness.”⁶⁹ Others, however favored slightly higher panels. In Keiser’s survey of systems furniture from 1974, he identifies 65 inches (approximately 5.5 feet) as the ideal median height for systems furniture partitions in that it was “sufficiently high to satisfy the acoustical consultants who preferred floor-to-ceiling enclosures” and “sufficiently low to gain approval from the air conditioning and lighting consultants who preferred no barriers.”⁷⁰

Thus, although office landscaping had encouraged minimal usage of somewhat low curving panels and plants to create privacy in the open plan, these new systems furniture lines used these taller partitions sometimes exclusively. The result was open plan installations that often lacked visibility of the overall space. Lila Shoshkes, author of *Space Planning: Designing the Office Environment* from 1976 argued that open plan furniture “often dominates the space, creating honeycombs, roomettes or mazes, cutting off contact and communication.”⁷¹ John Pile similarly argued that the use of too many screens (even low partitions) could create “corridor-like spaces” that were “particularly undesirable.”⁷² While the use of tall panels created individual

69.Pile, *Open Office Planning : a Handbook for Interior Designers and Architects*, 109.

70.Keiser, “Open Plan Furniture: an Investigation and Analysis.”

71.Andrea O Dean, “The Pros and Cons and Future Prospects of Open Landscaping,” *AIA Journal* (1977): 46-47, 82.

72.Pile, *Open Office Planning : a Handbook for Interior Designers and Architects*, 119.

privacy in each work station, the visual effect was considered even in the 1970s to be somewhat “Kafkaesque” and alienating.⁷³

The debate over appropriate panel height was significantly influenced by the growing body of research on worker's preferences. For instance, in their enormous study of office design that surveyed thousands of workers in the late 1970s and early 1980s, BOSTI (Buffalo Organization for Social and Technological Innovation) found that workers in general seemed to prefer greater degrees of enclosure, and found higher panel walls more conducive to achieving privacy than low panel walls.⁷⁴ The influence of the BOSTI study in the mid-1980s on office design was significant in architecture and design circles, as well as the business press including the periodicals *Office Administration and Automation* and *Fortune*.⁷⁵ In the summarized findings in the press, one of the highlights of the study that was repeated again and again in every report was the finding about the positive correlation between high levels of enclosure (partitions on at least three sides of the work station and higher than standing eye height) and ease of communication, sense of privacy, and even the attractiveness of the office. Thus, one of the biggest lessons from the study was that workers preferred more enclosure to less, and further that higher degrees of enclosure actually improved “ease of communication.”⁷⁶ In other words, this research suggested that higher degrees of enclosure could improve communication rather than hinder it. In light of

73. Horsely, “Some Second Thoughts on Open-plan Offices.”

74. Brill, et al., *Using Office Design to Increase Productivity*, 89-102.

75. “The Office of the 80’s: Designing for People and Productivity,” *Fortune* (1980): 29-38; “Using Office Design to Increase Productivity,” *Progressive Architecture* (1985): 154-55; “Using Office Design to Improve Productivity,” *Office Administration and Automation* (1985): 26-29, 77.

76. “Using Office Design to Improve Productivity.”

these findings, it is not surprising perhaps that by 1988, 65-inch tall partitions were the most popular panel height, because it provided privacy at standing height.⁷⁷

The layouts of open plan offices in the 1970s and 1980s were extremely diverse, and were largely dependent on the particular type of furniture selected. While there were some organizations that still adopted the more irregular office landscaping type arrangement, most installations favored a more regular geometric arrangement, often evoking conventional office planning, but using the new systems furniture as the basis for planning. A number of manufacturers (including Herman Miller, Steelcase, and Westinghouse) allowed arrangements in which each workstation was effectively enclosed on three or even four sides. While some of these systems had the option for curving details to “soften” the appearance, the overall visual effect was very orderly and grid-based, often with long rows of individual workstations. For example in the Asiatic Petroleum installation (see Figure 3), Herman Miller’s Action Office is used to create rows of individual workstations; looking at the original plan, there are both main walkways throughout the floor, and then internal passageways within individual divisions so that the openings for each workstation faced “inwards” towards the other workers in the group, rather than outwards towards the more heavily trafficked walkways.⁷⁸

Other installations featured individual workstations using systems furniture or case-based furniture (conventional desk and built storage) set in rows on an angle. In these arrangements, the workstations were generally not directly connected to one another, and “privacy” was

77.Stevens Anderson, “Open Office Systems Gain Flexibility,” *AIA Journal* (1988), 119.

78.Pile, *Open Office Planning : a Handbook for Interior Designers and Architects*, 170.

Figure 3: Detail of the Asiatic Petroleum Office Layout featuring Action Office furniture. The layout shows workspaces oriented away from major traffic paths creating internal divisions within the office. The furniture features rounded corners which soften the rectilinear arrangement. Reproduced from John Pile, *Open Office Planning* (New York: Whitney Library of Design, 1978), p.170. Permission requested.

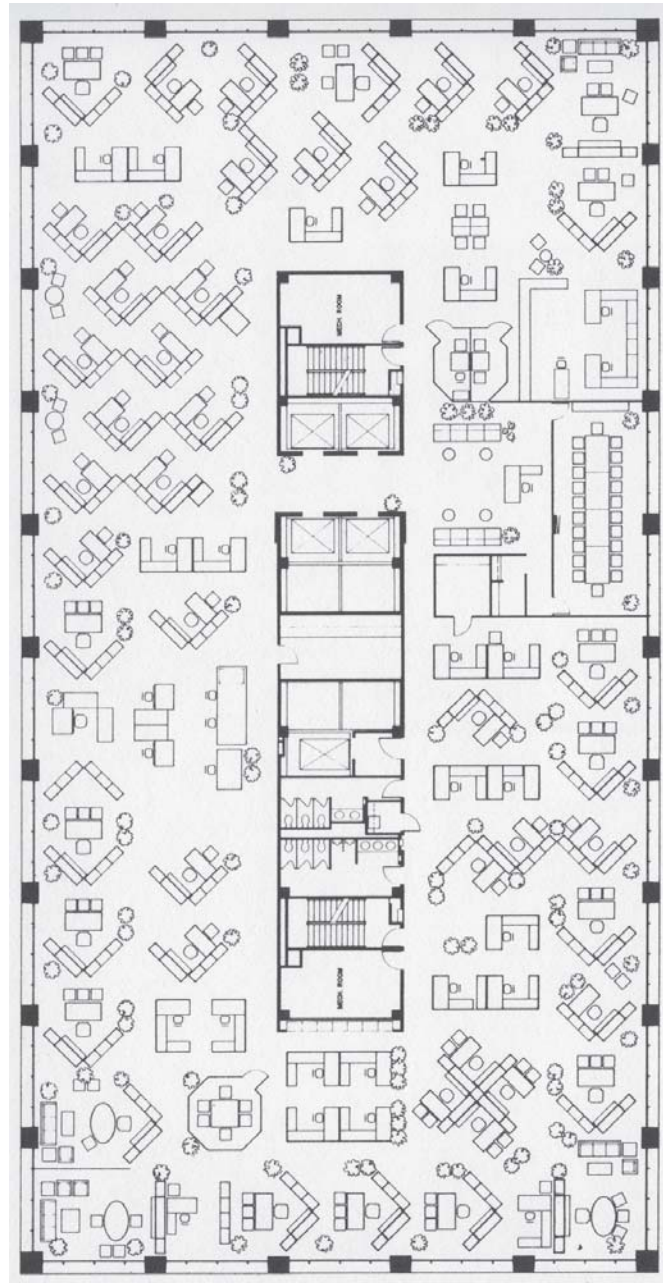


Figure 4: Image of Transcontinental Gas Pipe Line Layout featuring individual workstations with partitions in an "L" formation set on a 45 degree angle to create some visual privacy and an overall sense of order. Reproduced from John Pile, *Open Office Planning* (New York: Whitney Library of Design, 1978), 186. Permission requested.

achieved by the strategic angling of the workstations to create a sense of individual space. For example, the arrangement for Transcontinental Gas Pipe Line (Figure 4) in Houston featured workstations set on a 45 degree angle; looking at the arrangement you can see an overall regular pattern, and workstations are strategically oriented to create visual privacy.⁷⁹ Some installations a blend of these styles with some work areas set in the open with no panels (particularly for clerical or secretarial positions), and other work stations with varying degrees of enclosure. For example, an image of a Steelcase Series 9000 arrangement in an advertisement for the furniture shows a woman working in an open desk in the foreground, and a man in a larger and more enclosed workstation in the background.⁸⁰

In the late 1970s as the open plan became more common, there were growing reservations among workers about the experience of working in an open plan environment. In 1977 profiles on two “open office landscape” installations ran side-by-side in *AIA Journal* that explored the experience of workers in offices that had implemented the open plan, and both revealed some negative reactions to the open plan. The first article featured the AIA (American Institute of Architects) headquarters in Washington D.C. which used a system furniture line called “Reff” with 56-inch high partitions and filing cabinets which served to separate work stations and define corridors. The workers in the AIA installation reportedly had largely negative or mixed reactions to the new arrangement, and according to the article two-thirds of the respondents said they would prefer to work in a “conventional, closed office plan.” One worker commented “only the courtesy and general amiability of the staff members prevent this from being an absolute snake

79.Pile, *Open Office Planning : a Handbook for Interior Designers and Architects*, 186.

80.Steelcase advertisement, *Architectural Record*, February 1975, 80.

pit. Seated at a desk in a cubicle, one is visually isolated but continually assaulted by distracting noise.” Other complaints from workers in the AIA installation included over-crowding, insufficient storage space, and inadequate lighting.⁸¹ It is significant that even with 56 inch high partitions and storage units, workers still felt isolated in their workspaces in the AIA installation.

The second profile examined the Weyerhaeuser Company offices in Tacoma, Washington which featured a specially designed wood-based system by Knoll that would later be marketed as the Stephens System. Like the AIA offices, workers complained about overcrowding in the open plan particularly with the increasing density of workstations; as one worker said “it was lovely when we first moved in” but “then it got jammed and we had to cubicleize.” Despite some mixed reactions, the company was very positive about “open landscaping” saying that it is lower cost and facilitates change.⁸² This reveals the growing gap between the interest of organizations in using the open plan concept to reduce space usage and cost, and the workers who despite the attractive surroundings were feeling cramped into increasingly smaller workspaces.

It is significant that both articles include workers referring to the open plan workspace as a “cubicle,” in both cases describing negative aspects of the open plan. In the first article, the term is associated with the contrast of visual isolation and noise, and in the second it is used to describe a large number of workspaces crammed together. Although the word “cubicle” as a term for open plan workspaces would not become common for another decade (more on this in Chapter 4), it already was associated with the perceived negative aspects of the open plan.

81.Andrea O Dean, “Evaluation of an Open Office Landscape: AIA Headquarters,” *AIA Journal* (1977): 32-38.

82.Donald Canty, “Evaluation of an Open Office Landscape: Weyerhaeuser Co.,” *AIA Journal* (1977): 40-45.

Nonetheless, the open plan was becoming the new standard of office design. In 1980, a study by Louis Harris Associates and the furniture manufacturer Steelcase found that half of the white collar workers in the United States were working in some kind of “open office” arrangement.⁸³

Over the course of the 1980s, reports on the open plan in the news media also became more negative: a 1982 article titled "Offices Don't Sit Well With Workers," emphasized workers' frustration with the lack of privacy in the open office; and a 1985 article from the *Chicago Tribune* described how despite the widespread popularity of open plan offices, "many companies are having second thoughts about life without walls" and some companies were even returning managers to private offices while leaving "lower-level employees at work stations."⁸⁴ Similarly, a news article from 1986 documented the increasing cost of office space and the need for businesses to "shoehorn" workers in the smallest space possible.⁸⁵ By the early 1990s some news stories even began proclaiming the "death" of the cubicle as companies like Apple Computer chose private offices in the construction of their newest building.⁸⁶

83.Andrea O Dean, “Workplaces: the Open Office Revisited,” *ALA Journal* (1980): 50-56.

84.Michael Millenson, “Office Don’t Sit Well With Employees,” *Chicago Tribune (1963-Current File)*, June 7 1982, D9, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/172718177/13E4D9EEEEC4D197FE2/2?accountid=465>; Timothy Smith, “Design: Day of the Open Office is Up Against the Wall,” *Chicago Tribune (1963-Current File)*, October 13 1985, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/176312781?accountid=465>.

85.Laura Green, “Wanted: Offices That Work,” *Chicago Tribune (1963-Current File)*, April 6 1986, H32, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/176302512/13E4DA6E3DF22845232/1?accountid=465>.

86.John Markoff, “Where the Cubicle is Dead,” *New York Times (1923-Current File)*, April 25 1993, F7, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/109190199?accountid=465>.

The transformation of the open plan into the "cubicle" is a story about the transformation of the structure, culture, and meaning of white-collar work in this period. The open plan is more than just an office design, it is a set of ideas about the meaning and experience of office work. In the next chapter I will examine the progressive idealism of the architects and designers of the late 1960s who first promoted the open plan. I argue that their vision of the future of work which emphasized the importance of communication, the inevitability of radical and constant change, and the end of organizational hierarchy was tied to a larger intellectual movement called post-capitalism.

Chapter 2

Building Utopia: The Open Plan as a Post-Capitalist Artifact

When the open plan first appeared in the United States, the architectural press embraced it with starry-eyed optimism. Articles describing both office landscaping and Action Office in the late 1960s articulated a new kind of workspace and an entirely new kind of office culture.¹ These architects and designers forecast a radical change in the culture and structure of office work, and argued that the open plan was the ideal spatial counterpart to this utopian vision of the future of office work. It is tempting to dismiss the idealistic future these architects and designers were promoting as simply naive fantasies, but in the 1960s, these architects and designers were not the only ones articulating this kind of radical vision for the future.

In this chapter, I situate the idealism of the architects and designers who promoted the open plan and the organizations who adopted it within the broader context of an intellectual movement called “post-capitalism.” According to historian Howard Brick, post-capitalism was based primarily in the social sciences, and its proponents envisioned the radical transformation of America from a society driven by economics to one driven by social relations. Central to this new vision was an embrace of dynamic change as a constant force and an engine of growth. While technological change would remake the means of production and the nature of work,

1. For example: Pile, “The Nature of Office Landscape.”; “The TV Generation Turns on the Office,” *Progressive Architecture* (1973): 78-85; “Chaos as a System.”; “From Grid to Growth.”

society would harness the resulting economic growth to create a more equitable society. Yet change was not just something in the future, it was a reality of the present. Set amidst the social and cultural upheavals of the 1960s that produced sweeping social legislation, post-capitalists of that period saw their own era infused with the drumbeat of change. In this way, American society was already being transformed, and post-capitalists such as David Riesman, Daniel Bell, and John Kenneth Galbraith saw no end to the possibilities of progressive social change.² Much of the research on post-capitalism focuses on the intellectual ideas about post-capitalism, but rarely considers how those ideas might have filtered into mainstream culture or everyday experience.³ Although associated with academia, I argue that the post-capitalist vision was widely circulated and deeply influential in articulating the changes of the present and the expectations of the future, particularly with respect to the future of work as forecast in the late 1960s.

Brick argues that the post-capitalist vision was characterized by several common themes including: a shift away from economics and towards an emphasis on social relations, a belief in the potential for dynamic social and technological change, and the goal of creating a more inclusive and egalitarian society.⁴ Drawing on this framework, I argue that the open plan was a post-capitalist artifact that was designed to enhance communication and social interaction in the

2.Howard Brick, *Transcending Capitalism: Visions of a New Society in Modern American Thought* (Ithaca: Cornell University Press, 2006), 186-218.

3.Nelson Lichtenstein, ed. *American Capitalism: Social Thought and Political Economy in the Twentieth Century* (Philadelphia: University of Pennsylvania Press, 2006); Brick, *Transcending Capitalism: Visions of a New Society in Modern American Thought*.

4.Brick, *Transcending Capitalism: Visions of a New Society in Modern American Thought*, 268-69.

workplace, give greater flexibility for change (particularly in response to social and technological changes in office work), and eliminate spatial hierarchy. By situating the open plan within the network of research and ideas circulating in the 1960s, I illustrate how the ideas about the future of office work promoted by the architects and designers and embraced by the organizations that adopted the open plan were not mere fantasies, instead they represented a shared belief in the possibility of radical social and cultural change.

The New Social Research: Communication, Interaction, and Motivation

The architects and designers who promoted the open plan were strongly attracted to the new research in social sciences about the importance of communication, social interaction, and worker motivation. Drawing on this new research, the open plan was often portrayed as the spatial embodiment of these new ideas circulating in the post-war period. This new social research was also hugely influential in business circles particularly the emphasis on understanding human motivation, communication, and group interaction in creating an effective work organization.

In their emphasis on the social sphere, this new research in the social sciences challenged the primacy of economics as the driving force of American society. Building on the ideas of economist John Maynard Keynes, who posited that the economy was deeply embedded in the social sphere, these social scientists viewed the economic growth of the post-war period as an opportunity to promote public investment in social services. Rather than conceptualizing the

economy as a self-correcting system, these theorists argued that shared social values must direct the economy towards meeting larger social needs such as welfare, equality, and growth.⁵

In the early post-war period, one of the new areas of research was on human motivation, particularly in the context of work. Building on Maslow's concept of the "hierarchy of needs" the new psychology of work suggested that people were not motivated by money but rather by the psychological need for self-actualization. In 1959, psychologist Frederick Herzberg published his seminal book *The Motivation to Work* in which he argued that the factors that produced job satisfaction were different from those that produced dissatisfaction, and this was significant in job design circles because it suggested that the variables that would keep workers from being unhappy (pay, environment, supervision) were not the same as those that would allow them to achieve satisfaction (achievement, advancement, recognition and responsibility).⁶ In other words, while good pay could keep a worker from being unhappy, it would not actually produce satisfaction, instead organizations needed to support these internal motivations in order to help workers achieve satisfaction. Thus, Herzberg's research revealed that workers were motivated by an internal desire for personal and social recognition and success rather than by external factors such as a paycheck.⁷

5. Brick, *Transcending Capitalism: Visions of a New Society in Modern American Thought*, 152-85.

6. Sharon Parker, and Toby Wall, *Job and Work Design: Organizing Work to Promote Well-being and Effectiveness* (Thousand Oaks: Sage Publications, 1998), 10.

7. Frederick Herzberg, et al., *The Motivation to Work* (New York: John Wiley & Sons, 1959).

Parallel with this research on human motivation was new research on the importance of communication and team work in both industrial and office settings. In the late 1940s Eric Trist examined the social relationships embedded in the traditional coal mining process in Britain, and found that the interaction among different workers throughout the manual coal-getting process resulted in greater job satisfaction, higher productivity, and increased quality over the mechanized coal-getting process which stripped away social interaction and produced dissatisfaction as well as various production problems. As a result of this research, Trist advocated for the use of autonomous working groups in industrial settings in order to give workers greater control over the work process, and foster a more cooperative social environment.⁸ Meanwhile, in the United States, social psychologist Kurt Lewin began studying group interactions, and experimented with “T-Groups” as a way to create cultural change and stimulate creative problem solving. These “T-Group” sessions brought various people together to engage in role playing and training exercises that were intended to encourage open communication and improved group interactions. Both of these ideas were hugely influential, and by the early 1960s, American businesses such as Proctor and Gamble began experimenting with Trist’s concept of creating autonomous work groups in their factories, and at the same time bringing teams of managers and executives to Harriman, NY to participate in “T-Groups” with the National Training Laboratories for Group Dynamics (NTL).⁹ Together, these two trends marked an increasing emphasis on communication and social interaction in the work process.

8.E.L. Trist, and K.W. Bamforth, “Some Social and Psychological Consequences of the Longwall Method of Coal-getting,” *Human Relations* 4 (1951): 3-38.

9.Kleiner, *The Age of Heretics: Heroes, Outlaws, and the Forerunners of Corporate Change*, 44, 67,69.

Connected to this sea change in thinking about work organization was a new recognition of the negative implications of poor social relationships in the office. In the late 1950s, the new Institute for Social Research at the University of Michigan conducted a study on worker's health and found a strong correlation between the incidence of job-related stress and an increased risk of coronary heart disease. In particular, the social scientists at the University of Michigan found that social stresses involving worker's relationship with others were often the greatest sources of both stress and job dissatisfaction.¹⁰

The adoption of these new ideas about the sources of job satisfaction and the importance of social relations in the office, reflected a gradual shift in thinking in corporate America about the bottom line. During the post-war period, American corporations were largely driven by “the numbers;” according to management historian Art Kleiner, management used complex formulas to translate raw human effort into numbers that could be manipulated and analyzed in terms of efficiency and profit.¹¹ While cost-benefit analyses and return on investment (ROI) continued to be a driving force of American business throughout the post-war period, the acceptance of these new ideas about the concept of job satisfaction and the value of social relations in the workplace generated a more complex concept of relative cost and profit. Although American corporations largely adopted these new ideas as a way of improving productivity and profit, their acceptance required businesses to recognize that sometimes the solution that carried the greatest short-term cost benefit, may have long-term consequences that could be offset by spending more time and

10. Bruce Margolis, and William Kroes, “Work and the Health of Man,” in *Work and the Quality of Life*, ed. James O'Toole (Cambridge: MIT Press, 1974), 137.

11. Kleiner, *The Age of Heretics: Heroes, Outlaws, and the Forerunners of Corporate Change*.

money up front. This required a more nuanced conception of cost that included issues such as healthcare costs (as a result of worker injury, physical and mental stress, and illness, etc.), quality issues, and job dissatisfaction (resulting in absenteeism and job turnover) as equally important as the more standard costs of production.¹² In other words, the social aspects of work could affect the economic success of an organization, and only by factoring in those human costs could businesses recognize the full negative consequences of the old corporate culture.

This shift away from a purely economic approach and towards a more social understanding of corporate culture was also reflected in the design and promotion of the open plan. First of all, both office landscaping and Action Office rejected the primacy of the organization chart as the basis for office planning. In their rejection of the organization chart, proponents of the open plan were recognizing a fundamental disconnect between the image of the organization as symbolized by the organization chart, and the genuine relationships among individual workers and even divisions of the company as revealed by patterns of communication. This represents a significant departure in thinking about the meaning and purpose of the organization chart itself. According to historian Alfred Chandler, in the post-war period, the modern corporation became increasingly decentralized as companies expanded through the acquisition of other companies. In this new model each company would retain its own operational divisions (purchasing, traffic, research and development, sales, advertising, and production) allowing each unit to maintain significant independence from the larger corporate behemoth. Instead of managing operational functions of each division of the corporation, the central offices were devoted almost exclusively to financial

12. James O'Toole, *Work in America* (Cambridge: MIT Press, 1981), 27-28.

planning for future investments and expansion.¹³ As a snapshot of the organization, the organization chart reflected the formal bureaucracy and economic scope of the company, but did not convey the day-to-day operational activities and interactions of each individual unit. By abandoning the organization chart as the best method of organizing the physical layout of the office, proponents of the open plan were acknowledging that the day-to-day social interactions among workers were a more accurate reflection of the work process than the economic and bureaucratic structure revealed by the organization chart.

Yet more than just a symbolic rejection of the economic and bureaucratic approach to business, the open plan was portrayed as a genuine solution to the communication problems embedded in the modern corporation. By tearing down the walls of private offices and placing everyone out in the open area of the office, advocates of the open plan argued that workers would be more connected to one another and be free of the social constraints perpetuated by conventional office design. For example, executives freed of the burden of a private office would no longer feel the need to communicate through their secretaries, and instead would be empowered to communicate directly to one another making them more efficient as well as more effective.¹⁴ No longer hindered by walls and the formalities of the corporate hierarchy that those walls symbolized, workers would be free to have informal meetings and conferences as needed. Arranging workers based on their patterns of communication would only further improve

13.Chandler, *The Visible Hand: the Managerial Revolution in American Business*, 480-82.

14.Robbins, “‘Radicals’ Offer New Office Plan.”

communication and interaction in the office. As one writer described, such an arrangement was a “genuine embodiment of the form-follows-function dictum,” and must therefore be an ideal office design solution.¹⁵

While the new design would better support communication and interaction, it would also help alleviate some of the social anxieties of the office by making people more “readily accessible to each other, across departmental lines, without the psychological awkwardness that arises from have to ascertain, indirectly, whether the person one wants to see is busy. One can simply look over to his desk and see if he is busy.”¹⁶ In this way, the open plan would benefit the entire organization by improving communication as well as the individual worker who would no longer suffer the “psychological awkwardness” of not knowing if a colleague is too busy to be interrupted. Another article describes how the supposed privacy in the conventional office was actually an illusion in that one has the “constant unknown of whether somebody—friend or foe—is on the other side of the wall” listening in on private conversations. In the open plan workers have greater environmental awareness, and are always conscious of who is nearby protecting them from being accidentally overheard by the wrong person.¹⁷ The office worker was thus described as a slightly paranoid and psychologically sensitive being, needing the environmental support of the open plan to feel comfortable interacting in the office.

15. “From Grid to Growth,” 103-04.

16. “Chaos as a System,” 164.

17. Robert Propst, “The Landscape is Based on Thorough and Technical Data,” in *Office Landscaping: an Open Plan Concept of Office Design* (Elmhurst: The Business Press, 1969), 30.

In supporting their claims of the benefits of the open plan, architects and designers relied heavily on research in the new social sciences. In the 1960s, architects were already drawn to the ideas coming out of behavioral psychology and sociology about the interaction of people and their environment. Indeed, some architecture firms even experimented in collaborating directly with behavioral psychologists and sociologists in their architectural designs in order to create more effective spaces.¹⁸ The new social sciences were also used by architects and designers to support design decisions, particularly the implementation of the open plan. In describing the advantages of the open plan, its proponents drew on the research and ideas emerging out of the behavioral sciences, often suggesting that a simple change in the physical arrangement of the office could fundamentally alter the culture and operational habits of the organization.

Yet, not everyone agreed with the belief that the open plan could radically transform the experience of work. One particularly vocal dissenter was a human factors engineer, Malcolm Brookes, who wrote several articles debunking many of the assumptions underpinning the open plan. Brookes argued that without "hard data" revealing a correlation between worker's behavioral patterns and the design of the office, the claims of architects and designers regarding the benefits of the open plan were invalid.¹⁹ In another article, Brookes describes these supposed benefits of the open plan as "sociological claptrap" and "hokum."²⁰ Despite this lack of "hard data," proponents of the open plan relied on this new research regarding the benefits of social interaction and communication in the workplace to highlight the perceived organizational merits

18.C.M. Deasy, "When a Sociologist Gets in the Act," *AIA Journal* (1968): 72-76.

19.Malcolm Brookes, "A Maze of Contradictions," *Progressive Architecture* (1969), 131.

20.Malcolm Brookes, "They Call it Landscape, I Call it Hokum," *Administrative Management* (1971), 94.

of the open plan. Indeed, the most vocal advocates of the open plan treated the social benefits of the design as common sense arguing that simply removing walls and placing workers close to those they work with most frequently must logically improve communication and productivity.²¹

Though the open plan had some cost savings associated with it, those economic benefits were almost always described as “incidental” to the primary goal of improving communication and interaction among workers.²² Further, recognizing that workers were no longer driven by “money” but rather by a desire for “meaningful achievement and purpose,” the open plan was also described as a way of making workers feel connected and important to the organization.²³ With proper planning for environmental factors, even clerical workers will be “encouraged to take interest and pride in their duties,” and by “humanizing” the back office, companies in turn would see “heightened job satisfaction, more and better productivity, and a significant decrease in turnover.”²⁴ In this way, the social and psychological benefits of the open plan would only further enhance the economic benefits of the open plan.

Underneath their embrace of these new sociological ideas about the value of communication and interaction in the workplace, architects and designers were echoing themes that were broadly circulating among the major American corporations. As American businesses sought ways to change the social climate of their organization in order to continue to grow, they were particularly open to the suggestion that simply changing their office design would change

21. “From Grid to Growth,” 105; Rodgers, “Modularity Versus Free-form Clustering,” 13.

22. Pile, “The Nature of Office Landscape,” 43-44.

23. Propst, “The Landscape is Based on Thorough and Technical Data,” 30.

24. Maurice Mogulescu, *Profit Through Design; Rx for Effective Office Space Planning* (New York: American Management Association, 1970), 102-07.

their office communication. However, in their attempts to link the open plan with these new ideas, the architects and designers who promoted the open plan were not trying to falsely claim its social benefits, rather they genuinely believed that the open plan was an ideal solution to some of the problems of the modern corporation. Deeply influenced by the new research in the social sciences, these architects and designers saw the open plan as the spatial manifestation of the same ideas that were already making significant waves among American companies.

Changing Corporate Culture; Changing Management Culture

In their adoption of these new ideas about social relations in the office, businesses were seeking a fundamental change in their organizational culture. Within the context of the radical social and cultural changes of the 1960s, American corporations began to recognize change as an imperative organizational value, and sought to reinvent their corporate culture in order to fit with the countercultural spirit of the times. Indeed, according to writer and journalist Thomas Frank, American business viewed the new counterculture as a "symbolic ally" in their battle against corporate red tape and bureaucracy.²⁵ In their efforts to reinvent their corporate culture and eliminate bureaucracy, many organizations turned to new ideas of management that were just becoming popular as a means of redefining their organization, and the open plan was promoted by architects and designers as a means of achieving this new corporate

25. Thomas Frank, *The Conquest of Cool: Business Culture, Counterculture and the Rise of Hip Consumerism* (Chicago: University of Chicago Press, 1998), 9.

and managerial culture. The architects and designers who promoted the open plan believed that these new ideas in management theory were already transforming organizational culture, and argued that the open plan would perfectly support this new office culture.

On the heels of Herzberg's research on the sources of job satisfaction, psychologist Douglas McGregor published *The Human Side of Enterprise* in 1960, a book that would become a primer of new management theory. Drawing on both Herzberg and Maslow, McGregor argued that there were two main management philosophies: Theory X and Theory Y. According to McGregor, Theory X assumes that workers have an innate dislike for work and responsibility and they must be coerced, directed, and ultimately controlled in order to achieve organizational goals. Managers using this method create a tightly controlled and autocratic work environment in which workers are given little if any responsibility because of their perceived ineptitude and lack of self-motivation. Theory Y, on the other hand, argues that people do not inherently dislike work, and further that people actively seek responsibility and the opportunity to exercise their creativity and ingenuity in order to achieve both self-actualization and organizational goals. A Theory Y approach to management harnesses these innate tendencies towards the positive fulfillment of organizational goals by giving workers the opportunity for responsibility and the freedom to exercise their natural creativity and ingenuity.²⁶ McGregor's concept of Theory Y was hugely influential in business circles throughout the 1960s, and became a common term in modern business lingo.²⁷ For example, in his 1970 book *Up the Organization* former Avis CEO

26. Douglas McGregor, *The Human Side of Enterprise* (New York: McGraw-Hill, 1960).

27. Frank, *The Conquest of Cool: Business Culture, Counterculture and the Rise of Hip Consumerism*.

Robert Townsend argues that McGregor's Theory Y was the secret of his own success with Avis, and the only way to truly motivate workers.²⁸ A *New York Times* bestseller, *Up the Organization* was just one of a number of popular management books and articles that drew on McGregor's concept of Theory Y.

The adoption of Theory Y as a management ideology did not happen overnight; however, its legacy in the post-war period is significant in that it reflected a larger change in the structure and content of work. By the early 1970s, these ideas had become so widespread that younger workers in particular began expecting greater autonomy and responsibility in their jobs even fresh out of college. Research on high school student attitudes from 1960 to 1970 showed that students were shifting from wanting stable and long-lasting jobs, to "work that seems important to me" and offers the "freedom to make my own decisions." No longer content with just a job, young workers now demanded a work environment that fostered independence and creativity. These rising expectations of the value and content of work among youth were often attributed to the general affluence of the 1960s—in a booming economy workers could afford to consider factors other than pay and stability regarding where they worked. Whatever the cause, companies hoping to attract this new youth culture were told that they would not be able to motivate workers, but instead "must create conditions in which the worker can motivate himself."²⁹

28. Robert Townsend, *Up the Organization: How to Stop the Corporation From Stifling People and Strangling Profits* (London: Michael Joseph, 1970), 128-33.

29. O'Toole, *Work in America*, 43-51.

Parallel with this shift in management culture and the influx of this new generation of workers, was the rise of a new kind of worker: “the knowledge worker.” First coined by management guru Peter Drucker, the “knowledge worker” was a new class of worker who relied on the development and exchange of knowledge and ideas rather than manual labor.³⁰ Highly educated with essential organizational functions particularly in the areas of research and development, technology and engineering, finances, and creative work, the knowledge worker was portrayed as a central part of the modern corporation whose needs and interests were extremely different from his predecessor, the manual worker. According to Drucker, the knowledge worker was motivated by a sense of achievement and personal satisfaction, and could not be assessed on any of the conventional standards of productivity, nor be motivated by fear.³¹

Together, Theory Y management and “knowledge work” represented a significant shift in the culture of white-collar work in the 1960s. Proponents of the open plan were particularly drawn to these two ideas, and used them in both the design philosophy of the open plan as well as its marketing. In an article from 1968, Robert Propst describes a conference that he attended about office landscaping in which Wolfgang Schnelle, the founder and president of the Quickborner Team, directly related office landscaping as a concept to the developmental management philosophy embodied by McGregor’s Theory Y.³² In his own work for Herman

30.Nils Gilman, “The Prophet of Post-fordism: Peter Drucker,” in *American Capitalism: Social Thought and Political Economy in the Twentieth Century*, ed. Nelson Lichtenstein (Philadelphia: University of Pennsylvania Press, 2006), 110; Peter Drucker, *The Essential Drucker*, Kindle ed., (New York: Harper Collins, 2001).

31.Peter Drucker, *Management: Tasks, Responsibilities, Practices* (New York: Harper Colophon Books, 1985), 176-77.

32.Propst, “The Landscape is Based on Thorough and Technical Data,” 27.

Miller, Propst also drew heavily on McGregor's Theory Y to articulate the new culture of management dominating business in the late 1960s.³³ Advocates of the open plan saw business moving towards a Theory Y culture, and capitalized on that shift by describing the open plan as the ideal space solution for the "developmental" approach to management. In a report on office design produced jointly by Hauserman (a furniture manufacturer) and the research group Buffalo Organization for Social and Technological Innovation (BOSTI) called *The Management of Change and Productive Interiors*, the authors argue that "most of the 'new' managers" accept McGregor's Theory Y concept of management and have been putting its central tenets to use "in very practical ways."³⁴ Theory Y was thus described as more than a pie in the sky theory, but a real and practical change in the culture and structure of management.

While Theory Y was reinventing management culture, the knowledge worker was reinventing the structure and meaning of white-collar work. Proponents of the open plan were strongly attracted to these technical and creative types that were supposedly overtaking the office in high numbers. Robert Propst was particularly interested in the knowledge worker, and used knowledge work as the basis for his design of Action Office. According to Propst, the modern office was becoming a "thinking place" and the modern worker required a work space that could help them manage and manipulate information in order to generate ideas. According to Propst, this visual clutter was a central aspect of the work process. In an article from 1974, Propst described how the work artifacts that make up clutter "are the feedback structure that keep telling

33.Propst, *The Office: a Facility Based on Change*, 18.

34.Michael Brill, et al., *The Management of Change & Productive Interiors* (Cleveland: Hauserman, Inc, 1972), 7-8.

us what we are doing—key to what we know—the rearrangeable signals in the building block structure of a thinking process.” To this end, what appears disorderly and chaotic to the “casual observer” are in fact the “subtle feedback assemblies that sustain the progress of many kinds of knowledge work.”³⁵ Propst quotes Drucker directly stating, “the productivity of knowledge has already become the key to productivity, competitive strength, and economic achievement”... “What matters is that knowledge work has become the central factor of production in an advanced economy.”³⁶ Thus arguing that knowledge work is essential to modern business, Propst argued for what he called “the process aesthetic” that embraced the messy reality of knowledge work and allowed individual workers to adjust their spatial environment to meet their unique needs.³⁷

These changes in the culture of management and the structure of white-collar work were only the tip of the iceberg. While companies were already struggling to adapt to these most recent changes, proponents of the open plan argued that more change was on the horizon. In their promotion of the open plan, architects and designers insisted that modern business culture required an office that could enable rapid adaptation to cultural, technological, economic, and social change. Robert Propst argued that change was “a new constant force” that was radically altering the organizational culture of American business.³⁸ Similarly, in their study with the furniture manufacturer Hauserman, BOSTI argued that “change has been occurring at an

35. Robert Propst, “Process Aesthetic: Some Thoughts on the Thinking Process,” *Progressive Architecture* (1974), 79.

36. Propst, “Process Aesthetic: Some Thoughts on the Thinking Process,” 78.

37. Propst, “Process Aesthetic: Some Thoughts on the Thinking Process,” 80-81.

38. Propst, *The Office: a Facility Based on Change*, 29.

accelerating rate” and “responding to changing conditions has become a more regular event in the lives of both organizations and individuals.” According to BOSTI, “often the very *survival* of an organization is based on its ability to make timely and accurate responses to increasingly complex forces of change.”³⁹ In these references, change is an abstract “force” that is unpredictable yet constant. Rather than suppressing or preventing change, the modern corporation was expected to embrace change as a positive value in organizational culture.

According to Propst:

the acceptance of change is in itself a structure in our society dominated by a critical minority, the change agents. Change agents are those individuals who have this ‘inordinate self-confidence.’ They in turn are watched closely by the majority as new idea testers.”⁴⁰

The change agent was thus both a hero of the modern business man, and a rebel of the conventional corporate bureaucracy who could promote change from within the organization.⁴¹

The open plan was treated as an expression of this heroic change culture, requiring the leadership and tenacity of a change agent to be properly implemented.⁴²

In addition to representing the cultural change in the office that was already well underway, the open plan also embodied this ability to adapt to change. While the conventional office supposedly hindered change, the open plan was portrayed as highly flexible and adaptable, able to be rearranged as needed with minimal cost. In an article on the first office landscape

39.Brill, et al., *The Management of Change & Productive Interiors*, I.

40.Propst, *The Office: a Facility Based on Change*, 30.

41.Kleiner, *The Age of Heretics: Heroes, Outlaws, and the Forerunners of Corporate Change*, 38.

42.Propst, *The Office: a Facility Based on Change*, 30.

installation at DuPont, the arrangement was praised for its incredible flexibility allowing the company to completely rearrange the floor just two months after the original installation in just 12 hours costing only 33¢ per square foot in contrast to the \$5 to \$10 cost per square foot on DuPont's conventional floors.⁴³ In a similar vein, Action Office was also described as highly adaptable to change at the organizational level, as well as for the individual worker who could make adjustments to their workspace to suit their own changing needs. This concept of change as an important value in office design was echoed in BOSTI's recommendations to Hauserman regarding the ideal specifications for a new furniture system; of their eight recommendations, five specifically dealt with the importance of supporting and enabling change in the modern office environment.⁴⁴

Unlike the fixed partitions of the conventional office which gave an air of permanence, the open plan was a symbol of change and even "disposability." According to an article in the journal *Progressive Architecture*:

Since office processes are changing and changing at an ever-faster pace, it is essential to avoid large investments in permanent fixtures whose obsolescence was inevitable. The use of plastic or even paper equipment—cheap materials, cheaply fabricated—would greatly facilitate these necessary changes.⁴⁵

In this way, just as disposability had become a common feature in many household objects in the 1960s, so too was it a virtue in the workplace where the rapid pace of change made disposability a reflection of a more modern approach to business that embraced change as a way of life.

43. "Burolandschaft USA," 176.

44. Brill, et al., *The Management of Change & Productive Interiors*, 41-44.

45. "From Grid to Growth," 106, 108.

Writing in 1970, sociologist Alvin Toffler described this culture of disposability arguing that “as change accelerates and complexities multiply, we can expect to see further extensions of the principle of disposability.”⁴⁶ Emphasizing its inherent impermanence and flexibility due to its lack of fixed partitions, proponents of the open plan argued that this new kind of office design was ideally suited to this culture of change.

Forecasting the End of Corporate Hierarchy

Among the various changes happening within the culture and structure of white-collar work in the postwar, perhaps the most significant was the fundamental rejection of corporate hierarchy. In the 1950s, the organizational hierarchy was treated as the immutable backbone of the corporation that codified the chain of command from the president of the company all the way to the lowest clerical worker. Embodied by the organization chart and reinforced by conventional office design, which rewarded organizational status through increasingly larger and more opulent office space, the early post-war corporation was characterized by a culture of bureaucratic conformity. However, in the 1960s the anti-establishment mood of the new counterculture began to influence the structure of American business.⁴⁷ While companies began experimenting with the new ideas about communication in the workplace, and embracing a Theory Y approach to management, many organizations also began to reconsider the meaning and use of the corporate hierarchy. What once represented the pinnacle of successful management and control, was now

46. Alvin Toffler, *Future Shock* (New York: Bantam Books, 1971), 52-58.

47. Kleiner, *The Age of Heretics: Heroes, Outlaws, and the Forerunners of Corporate Change*, 20.

an albatross that limited creativity and inhibited effective communication. By eliminating private offices, and removing the organization chart as the basis of planning, architects and designers argued that the new open plan would help companies redefine their corporate hierarchy.

By the late 1960s, the ideal modern corporation was no longer an army, but rather a complex organic system in which all of the parts were equally important to the proper functioning of the whole organization. In contrast to the more traditional bureaucracy that emphasized stability and control, the “organic” organization was ideal for an era of dynamic change. In their book *The Management of Innovation* from 1961, Tom Burns and G.M. Stalker described how the “organic” style of organization which eschews a conventional hierarchical structure in favor of a more flexible and participative organizational structure enabled greater “technological creativity” and innovation by giving workers increased individual responsibility and autonomy.⁴⁸ The comparison of the corporation to an organic system was a fairly widespread analogy that originated at least in part with the work of biologist Ludwig von Bertalanffy, who in 1950 first developed the “open systems theory” in which he compared living systems to single cells that survive through continuous interactions with its physical environment. Applied to organizational theory, open systems suggested that corporations like cells and ecosystems must be able to respond efficiently to changes in their “environment” in order to survive.

In their 1965 article “Causal Textures of the Environment,” Fred Emery and Eric Trist draw on Bertalanffy’s concept of open systems, and argue that a “turbulent” organizational environment required an organizational structure that that was “essentially different from the

48. Jay Shafritz, and J. Steven Ott, *Classics of Organization Theory* (Pacific Grove, CA: Brooks/Cole Publishing Company, 1992), 202.

hierarchically structured forms” that characterized the typical post-war organization. According to Emery and Trist, this new type of structure was strongly correlated with the managerial approach typified by McGregor’s Theory Y, particularly in setting larger group values that can help the organization effectively respond to environmental turbulence.⁴⁹ Rather than favoring a hierarchical organization that emphasized control and order, companies in a “rapidly changing environment” were encouraged to adopt a flatter and more flexible organization that emphasized lateral communication and creative problem-solving.⁵⁰

Architects and designers promoting the open plan were also drawn to these ideas of the “organic” organization, and portrayed the open plan (particularly office landscaping) as the perfect spatial counterpart to this new organizational structure. By using patterns of communication instead of the formal organization chart, the open plan represented a radical re-imagining of the processes and structure of the organization. Central to the architects' and designers' argument was the idea that these organizations were already “organic” in their structure, they just required an office space to reflect and support it. The American architectural press focused heavily on the perceived tension between the genuine “organic system” of the office and the false “order” imposed by the conventional office design with its basis in the organizational chart.⁵¹ Even the visual chaos associated with the office landscape in particular was promoted by the architects and designers as a new kind of order that more accurately

49.F.E. Emery, and E.L. Trist, “The Causal Texture of Organizational Environments,” *Human Relations* 18 (1965): 21-32.

50.Paul Lawrence, and Jay Lorsch, “”Organization-environment Interface”,” in *Classics of Organization Theory*, ed. Jay Shafritz, and J. Steven Ott (Pacific Grove, CA: Brooks/Cole, 1992), 231.

51. “From Grid to Growth,” 101.

reflected the organization. The seeming chaos of the open plan thus did not merely disguise a natural order, rather the visual chaos itself embodied order. One article describing this seeming paradox summed it up thus: “disorder is made up of all the kinds of order we are not conditioned to recognize.”⁵² The designer’s role in creating the open plan was to serve as “nature’s guide” in shaping the office to ultimately reflect the organic order implicit to the organization.⁵³ In this way, the formal organization chart was treated as an archaic form that was no longer relevant to the true needs and functions of the organization. By rejecting hierarchy as the foundation of office design, the open plan would effectively support the true organizational structure.

The new conventional wisdom of the 1960s also indicated that bureaucracy and hierarchy were the enemy of progress. According to sociologist Alvin Toffler, the “collapse of hierarchy” was coming, and he argued that to survive, “organizations must cast off those bureaucratic practices that immobilize them, making them less rapidly responsive to change.” The conventional hierarchy was thus portrayed as a deterrent to change, and organizations were encouraged to eschew the markers and habits of hierarchy in order to keep up with the modern pace of organizational change. According to Toffler “the chain of command” was slowly deteriorating as companies were moving towards flatter hierarchies and lateral communication.⁵⁴

52. “Chaos as a System,” 161.

53. “From Grid to Growth,” 103.

54. Toffler, *Future Shock*, 137-42.

Through the new strategies of job design and new theories of management, companies were urged to give workers at all levels greater autonomy and responsibility in their jobs in order to foster greater job satisfaction as well as improve the quality of work.⁵⁵

By some accounts, the development of new kinds of office automation would only help flatten the organization even more. These “exuberant technologists” believed that new technologies would liberate workers from “menial jobs” and new jobs would be created requiring a “better educated and more autonomous work force.”⁵⁶ For instance, Alvin Toffler predicted that new technology would eliminate routine tasks and require workers to be responsible for solving “non-routine” problems as they arise.⁵⁷ With the resulting elimination of “drudgery,” workers would be free to engage in more “creative tasks such as problem solving” and “dealing with clients.”⁵⁸ According to Toffler, this new kind of worker “rather than occupying a permanent, cleanly defined slot and performing mindless routine tasks in response to orders from above” will find “increasingly that he must assume decision-making responsibility.”⁵⁹ New technology would thus enable a transformation of the corporate hierarchy, eliminating routine and repetitive jobs, and creating a new kind of organizational structure in which all workers had the ability and autonomy to make decisions.

55.O’Toole, *Work in America*, 96-105.

56.Roslyn L. Feldberg, and Evelyn Nakano Glenn, “Technology and the Transformation of Clerical Work,” in *Technology and the Transformation of White-collar Work* (Hillsdale: Lawrence Erlbaum Associates, 1987), 77-78.

57.Toffler, *Future Shock*, 141.

58.Rosemary Crompton, and Gareth Jones, *White-collar Proletariat: Deskilling and Gender in Clerical Work* (London: Macmillian Press, 1984), 45.

59.Toffler, *Future Shock*, 142.

Proponents of the open plan were similarly optimistic about the impact of new technology on white-collar drudgery. While the open plan was not “so revolutionary that it dispenses with mindless repetitive tasks,” nonetheless its most vocal advocates argued that the end of such tedious “mental drudgery” was near as those tasks were increasingly being fulfilled by computers.⁶⁰ In a 1969 article, interior designer John Pile argues that “routine work is becoming truly mechanized, and the staff member, no longer a mere cog in a machine is becoming a thinker, a creator and a decision maker.”⁶¹ Similarly, Robert Propst argued that the new concepts in job design and management theory that emphasized increasing the responsibility and autonomy of workers mean that “tasks of rote” were being replaced by “tasks of judgement.”⁶² In this way, proponents of the open plan saw the rise of new technology as a liberating force that would transform the nature and culture of white-collar work.

In the future, all workers would be knowledge workers whose work would be dominated by thinking and problem solving rather than routine busy work. The knowledge worker represented the disintegration of the conventional corporate hierarchy. According to Peter Drucker, “in knowledge work there is essentially no ‘superior’ and no ‘subordinate’. There is only a ‘senior’ and a ‘junior.’”⁶³ In his other writings, Drucker argued that although knowledge workers did not generally carry managerial authority, they nonetheless retained significant autonomy and decision-making responsibility giving them the equivalent status of an executive.⁶⁴

60. “From Grid to Growth,” 105; “Chaos as a System,” 164.

61. Pile, “The Nature of Office Landscape,” 46.

62. Propst, *The Office: a Facility Based on Change*, 19.

63. Peter Drucker, “Worker and Work in the Metropolis,” *Daedalus* 97, no. 4 (1968), 1251.

64. Drucker, *The Essential Drucker*.

Echoing Drucker's ideas, proponents of the open plan similarly argued that the rise of the knowledge worker indicated a fundamental shift in thinking about hierarchy particularly as it related to the physical arrangement of office space. John Pile argued that in the modern office "everyone is an executive" and the open plan in which "everyone shares space of considerable luxury with a minimum of rank distinctions" is the optimum solution to express this "emerging reality."⁶⁵ Robert Propst argued that as the "developmental" approach to management embodied by McGregor's Theory Y "gains momentum" the layout of the office would need to change to support this shift in thinking from an emphasis on status and hierarchy towards goal setting and autonomy.⁶⁶ In this way, the old conventional office that privileged status and hierarchy was no longer an appropriate design for the modern office. By forcing the executive out of his or her private office and making everyone more accessible, the open office created a more egalitarian office environment that could better support the goals and values of the developmental approach to management.⁶⁷ In this modern workplace, the conventional symbols of status associated with post-war office design were out of date and out of step with the new culture of white collar work.

According to architectural historian Olga Pantelidou, architect Kevin Roche was a particularly strong proponent of office design that eliminated a sense of hierarchy. Pantelidou argues that Kevin Roche's architectural designs of the late 1960s and 1970s were themselves contributions to the broader discourse about management theory and organizational culture, particularly in its emphasis on removing the traditional symbols of hierarchy. She describes how

65.Pile, "The Nature of Office Landscape," 46.

66.Propst, *The Office: a Facility Based on Change*, 18.

67. "Chaos as a System," 164.

his design for the Ford Foundation Building treated all workers as “executives” by giving everyone a private office with a view of the outside or of the interior atrium. Later Roche’s design for the Cummins Headquarters from 1972 placed all workers (even top executives) in an open plan arrangement. Pantelidou argues that “by placing the highest-ranking employees in the same workstation as the lowest” Roche’s design “communicates that everyone is equally indispensable.” Pantelidou describes these two polar opposite approaches to achieving spatial equality as an indicator of Roche’s genuine commitment to achieving the elimination of hierarchy in corporate architecture. In her analysis, Pantelidou treats Roche’s designs as artifacts of his progressive intentions, irrespective of how the design was implemented or used later.⁶⁸ In his commitment to eliminating spatial hierarchy, Roche was echoing the spirit and idealism of post-capitalist thought, and creating a vision of the workplace that was intended to foster greater equity among all white-collar workers.

The proponents of the open plan were similarly committed to the radical transformation of the workplace. From their perspective, the open plan was both the reflection of this new culture of work and the embodiment of it. Embedded with these ideas about “developmental management” and “knowledge work,” the open plan promised to dissolve the old corporate hierarchy and create a new kind of workplace that fostered communication and innovative thinking. As companies sought ways to adopt the new “organic” organizational culture that emphasized lateral communication and worker autonomy, the open plan became a familiar symbol of this new kind of work culture.

68.Pantelidou, “Designing for the Workflow.”

In the popular press, companies that adopted the open plan were celebrated as paragons of this new youthful counter-cultural and anti-hierarchical spirit. According to Thomas Frank the counter-culture of the 1960s characterized by "hip" clothing and rock music appealed to American business as they sought to embrace change and attract youth both as consumers and as employees, and many of them adopted the symbols of the counter-culture to signify their connection to that spirit.⁶⁹ For example, in 1972, an article in the *New York Times* described McDonald's new corporate headquarters with its giant waterbed room for reflection and relaxation (available to all white-collar employees), rock music playing over the intercom, and an open plan arrangement where even the corporate chairman, Ray Kroc, did not have a private office. In the article, the new office arrangement is described as making workers "more open" and "less bitchy" and even changing the way workers dress: "both male and female employees have taken to brighter, more mod apparell [sic] since the move here. Many more men, including Mr. Kroc now work coatless."⁷⁰ In this way, the open plan was described as a means of symbolically removing the symbols of hierarchy and engendering a flatter and more casual corporate culture where workers are happier and even the corporate chairman feels comfortable working "coatless."

By eliminating private offices, the open plan was portrayed as the spatial manifestation of this new office culture. Though its proponents often emphasized the economic benefits of the open plan, their own support of the open plan was often tied to a liberal social agenda that

69. Frank, *The Conquest of Cool: Business Culture, Counterculture and the Rise of Hip Consumerism*, 26.

70. Andrew H. Malcolm, "McDonald's Sold on Its 'think Tank' Office," *New York Times* (1923-Current File), January 31 1972, 29, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/119387332?accountid=465>.

avored equality and revolution over hierarchy and the status quo; according to Frank Duffy, in the office landscape “egalitarianism was worn conspicuously as a badge of progress.”⁷¹ Along the same lines, another architectural writer described how:

open office planning proclaimed the liberation of the file clerk, the socialized, revolutionized, non-hierarchy, windows-for-everyone environment. No longer did the corner office necessarily belong to the Chairman of the Board; no longer were the clerks lined up at neat rows of desks recalling the fantasies of Fascist Germany.⁷²

Despite the grotesque imagery and sardonic tone, the author conveys the unabashed enthusiasm that much of the architectural community conveyed in their promotion of the open plan. Advocates of the open plan thus envisioned a utopian future in which hierarchy was truly eliminated, and the clerical worker was liberated from the chains of menial labor.

Yet even as they projected a vision in which "all" workers would be equal, the architects and designers only mean "white-collar" workers. In other words, although their vision of the office emphasized a flattened hierarchy, the service workers, cleaning service, mail room employees, and cleaning staff for example were not a part of this progressive vision of the future. This limitation of their view suggests a failure to fully understand the enormous social divisions that existed not only in American society as a whole, but even in American offices. Although their vision emphasized equity, not all workers were truly treated equally even in this progressive utopian future.

71. Duffy, et al., *Planning Office Space*, 72.

72. “The TV Generation Turns on the Office,” 78.

The Open Plan and the Future of Office Work

The architects and designers who promoted the open plan in the late 1960s were not articulating a unique vision of the future, rather they were echoing the research and ideas that were widely circulated in this period. By setting the open plan within the context of post-capitalist thought, I argue that the open plan idea was more than a naive fantasy perpetuated by architects and designers alone. Just as society would be liberated from the threat of scarcity and inequality through the harnessing of dynamic social and technological change, so too would the worker be freed from the alienating and dehumanizing aspects of work. The new office would foster communication and interaction, readily accept and adapt to social and technological change, and emphasize equality. The architects and designers promoting the open plan genuinely believed in the possibility of such radical social change, and they saw Theory Y management, knowledge work, and the organic organization as more than pipe dreams of the future—they were the reality of the present.

As an artifact of the 1960s, the open plan was deeply embedded with these ideas. By tearing down the walls of private offices and arranging workers based on communication patterns, the open plan promised to improve communication, create a space that was optimally adaptable to imminent social, technological and cultural change, and reduce hierarchy. An easily recognizable symbol of these new ideas about work, the open plan was closely associated with the central tenets of Theory Y style management, knowledge work, and the organic organization. As companies were struggling to adapt to the new culture and climate of white-collar work,

particularly the new ideas about management and job design and the new organizational structure, they sought ways to display their own cultural change. Simply by adopting the open plan, American corporations could show their commitment to this new kind of work culture.

Most importantly, the open plan was a part of the cultural discourse about the importance of communication and social relations, the possibility of social and cultural change, and the elimination of hierarchy in American society and by extension, American offices. As an artifact of its era, the open plan was a part of the larger debate of the late 1960s about the future of American society and the future of work. What might appear as naive idealism was in fact a genuine belief in the possibility of change, and a commitment to achieving that change by transforming the physical design of offices. Most importantly, the open plan became a means of expressing and encouraging change, and was an intervention into the entrenched bureaucracy and hierarchy of post-war corporate culture. The companies and organizations that chose to adopt the open plan in its earliest iterations did so in part to embrace this progressive vision of the future and of their organization as well. The open plan reorganized and restructured the office in order to achieve the central ideals of the post-capitalist vision by fostering social relationships, enabling change, and eliminating hierarchy.

At its most mundane, the open plan was about the needs of workers, and their everyday experience of work. As a post-capitalist artifact, the open plan brought the esoteric and highly intellectual ideas associated with the post-capitalist vision into the realm of everyday experience. The open plan was designed to organize workers and their relationships with each other and with the organization in a very intentional and strategic way. By creating a landscape of the office that was meant to reflect as well as determine social relations in the office, the open plan

simultaneously described the hidden relationships of the office and created entirely new relationships. As workers used these early open plan installations, they were inhabiting a utopian-like fantasy of the future of office work and a future of American society. Yet despite the good “intentions” of the architects, designers and organizations who promoted and adopted the open plan, the truth is that the open plan concept could never have single handedly transformed the organization or the experience of work for its inhabitants. Howard Brick argues that the “Achilles heel of the postcapitalist vision was its reluctance or inability to grasp the limits of change under given conditions while it exposed the potential for change in contemporary social structures.”⁷³ Similarly, the architects and designers who promoted the open plan failed to anticipate the limits of organizational, cultural, and social change in the workplace. In the next chapter, I situate the implementation of the open plan within the context of a real organization, Wisconsin State Government, in order to examine the complex influence of social, cultural, economic, organizational, and technological change on the structure of work and the design and use of the open plan over the course of the 1970s and 1980s.

73. Brick, *Transcending Capitalism: Visions of a New Society in Modern American Thought*, 218.

Chapter 3:

Changes in Work; Changes in Offices: Wisconsin State Government

In their willingness to accept and promote the power of space to create and transform social relations, the architects and designers who promoted the open plan in the late 1960s did not anticipate the power of organizations and individuals to resist or otherwise challenge their vision for the future. In his book *The Practice of Everyday Life*, Michel de Certeau describes the dynamic interplay of people and space. According to de Certeau, space and place serve to organize and create structure in the everyday, but people live dynamically within space creating their own “stories” and their own everyday experiences within those spaces and places.¹ While the open plan concept presented an opportunity for transformation and change in the experience of work, in truth the workers and organizations that inhabited the open plan were bound by the ‘real’ experience of work in those spaces as defined by the social, technological, economic, organizational, and cultural context which directly informed the way the open plan was implemented, used, and understood. Using a case study of the implementation of the open plan in Wisconsin State Government Office Buildings, this chapter will examine how the myriad changes in the culture and structure of work (organizational, social, economic, technological, etc) informed the implementation, meaning and use of the open plan.

1. Michel de Certeau, *The Practice of Everyday Life*, trans. Steven Rendall (Berkeley: University of California Press, 1988).

A central aspect of this chapter is the analysis of change in the context of office work. Adapting a model from Industrial Systems Engineering called the Work System Model, this chapter will examine various points of change and non-change in the culture and structure of work. Developed by Michael Smith and Pascale Carayon, the Work System Model is a concept that emerged out of the human factors discipline and the socio-technical systems approach which emphasizes the interaction of people and the technical elements of the workplace. The Work System Model expands this idea of interactivity to include the organization, the environment, and the tasks along with people and technology. According to the Work System Model concept, these five basic elements of work are fundamentally connected such that a change in any one area will result in a change in the others. At its most basic level, the Work System Model is a tool to describe work, and was designed to conceptualize the influence of intended and unintended changes in the workplace.² Although it was not designed to be used to analyze cumulative patterns in work situations over extended periods of time in the past, I am adapting it to analyze the changes in the context of office work in the 1970s and 1980s. The benefit of this model is that it provides a means of seeing various points of change, and understanding how those changes interact with each other.

Applying this model to archival material at the Wisconsin Historical Society, this chapter examines the changes in white collar work and office design in the 1970s and 1980s in the context of Wisconsin State Government offices. Due to open records laws in Wisconsin, there are significant public records of the internal debates and decisions related to changes in office work

2. Smith, and Sainfort, "A Balance Theory of Job Design for Stress Reduction."

and office design in this period. The Department of Administration Secretary's Subject Files held in the Wisconsin Historical Society Archives are the primary source of material for this chapter along with other supporting documentation from the Wisconsin Historical Society such as newsletters, reports, and other public record materials. The Department of Administration (DOA) is the administrative backbone of Wisconsin State Government. Their department, which was founded in 1959, is concerned with managing a wide range of work related purchases and activities including: personnel, management, technology, office space, and budgeting. While they are a large department with various internal divisions, they also interact closely with all of the other departments and agencies in Wisconsin State Government as well. The Secretary of the Department of Administration, whose files were the basis of this research, has the challenging role of overseeing all of these different areas of work for Wisconsin State Government. Because of the inherently public nature of government work, the various agencies of Wisconsin State Government were often forced to articulate ideas or thoughts on paper that in a private industry work setting might never have been documented. Nonetheless, Wisconsin State Government experienced many of the same social, cultural, political, economic, and technological challenges that many other government and private enterprises struggled with in this period in the United States. Some of these national issues included: the energy crisis, a drop in productivity, the influence of political conservatism, the increasing diversity of the white-collar workforce, the increasing reliance on temporary workers, and the spread of computers and other forms of

automation in offices.³ In this way the changes that occurred in Wisconsin State Government reflect larger changes that were happening in offices across the United States in this period.

A History of the Open Plan in Wisconsin State Government

In early 1973, the Board of Vocational Technical and Adult Education had a “delicate internal management problem” that only office design could fix. Their office on the 7th floor of the Hill Farms Building was reportedly a very “pleasing environment” with a “judicious use of color and plant material,” but there were “dimensional worms in Eden.” According to a series of memos between the Board and the DOA, a group of directors were unhappy because one director among them had an office that was six linear feet larger than the others due to the unfortunate placement of a radiator. In order for the offices to be made the same size, the radiator would have to be relocated, a significant expense. Most tellingly, the memos indicate a recognition that all too often “space layout matters tend to take on a seriousness of concern disproportionate to their relative importance in the overall responsibility of governmental management.” While the

3. Judith Stein, *Pivotal Decade: How the United States Traded Factories for Finance in the Seventies*, Kindle ed., (New Haven: Yale University Press, 2010); National Commission on Productivity, *A National Policy for Productivity Improvement* (Washington D.C.: U.S. Government Printing Office, 1975); Rosabeth Moss Kanter, “Work in a New America,” *Daedalus* 107, no. 1 (1978): 47-78; Andree Brooks, “‘Temp’ Work Becomes Full-time Way of Life,” *New York Times* (1923-Current File), October 17 1982, 6, <http://search.proquest.com.ezproxy.library.wisc.edu/hnpnewyorktimes/docview/121926626/13D9F34E3E1BEB36CE/1?accountid=465>; National Association of Office Workers Working Women, *Race Against Time: Automation of the Office: an Analysis of the Trends in Office Automation and the Impact on the Office Workforce* (Cleveland: National Association of Office Workers, 1980).

change was eventually approved, the memos from the DOA indicate significant “reluctance” to make the change and “sorrow” for the unfortunate example of “negativism” and “bureaucracy.”⁴

At around the same time just a few floors away, another office in the same building had requested to have their private offices eliminated altogether in favor of the new “office landscaping approach.” In their justification, the Bureau of Planning and Budget described how the office landscaping idea would work well with their office’s management philosophy and the “developmental nature” of their work, and would ultimately improve their team interaction and communication. They further argue that the implementation of office landscaping will serve as a “modest test of the applicability of the office landscaping concept to this building.” Most tellingly, the request indicates that “expectations” have been “built up” such that the “staff morale” might be harmed if this plan for a remodel were not approved. According to their proposal, this early implementation of office landscaping would only require some “modest cost” (estimated about \$9,900), specifically the purchase of 24 new “movable partitions” that would be used in conjunction with existing furniture to minimize screening costs and promote privacy.⁵ Together these two stories reveal the contrasts of the period. On the one hand there was a strong culture of bureaucracy and hierarchy in Wisconsin State Government that was supported and reinforced by office design, and on the other an interest (at least in some offices) in throwing out such old symbols of status in favor of the new open plan.

4. Correspondence between Wayne McGown and Fred Hiestand, February 2, 1973 and February 13, 1973 and Correspondence from John Short to Wayne McGown February 6, 1973, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 14 Folder 12, Wisconsin Historical Society.

5. Correspondence from Jack Schoop to Harry Schmidt, August 15, 1972 Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 11 Folder 18, Wisconsin Historical Society.

Through the mid-1970s the office landscaping concept remained a fairly experimental one in Wisconsin State Government that was used in only a few offices that elected to make the change on their own, but by 1978, the open plan was transformed from an unconventional idea into the new standard for all Wisconsin State office buildings. At the time, the DOA was in the process of designing and constructing two new office buildings in Madison: General Executive Facility (GEF) 2 and 3. The new space guidelines called for the elimination of internal fixed walls and ceiling height partitions in favor of an “open space landscaping” design using “existing furniture and freestanding screens or movable partitions.” The guidelines further state that “enclosed space will be restricted”...“to those areas where enclosure is essential to the functions being performed in the space.” Thus, while in 1973 private offices had been a perceived status symbol the subversion of which could incite thorny management problems, by 1978 they were refashioned (at least in the official guidelines) as primarily functional spaces.⁶

The new space standards from 1978 also make the first reference to the new “Systems Furniture.” According to the guidelines, units would be allowed to purchase systems furniture when its use would “result in more effectively used space.”⁷ In other words, if agencies could demonstrate a reduction in their space needs as a result of systems furniture usage they could purchase new furniture instead of using conventional furniture with freestanding screens and partitions. At the time, the DOA was not sure they would have the money to afford systems

6. “Guidelines and Standards for Effective Space Management,” December 1978, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 33, Folder 22, Wisconsin Historical Society.

7. “Guidelines and Standards for Effective Space Management,” December 1978, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 33, Folder 22, Wisconsin Historical Society.

furniture, and thought that they would likely end up using conventional furniture with screens rather than purchasing systems furniture. In planning mock-ups for workers and agencies to see, they DOA discussed hiding the all systems furniture example from the other mock-ups, presumably to keep from getting agencies' and their workers' hopes up for acquisition of the new furniture.⁸ Indeed, there is other evidence in the archival material that the new "systems furniture" was perceived in a very favorable light by both agency heads and their staff. For example, when the Department of Revenue was negotiating the details of their new space their Secretary wrote to the Secretary of the DOA stating:

if the open office concept is going to work, we have continually felt that a serious commitment to unit furniture should be made. Otherwise the resulting work environment is little more than a hodge-podge of dissimilar desks, chairs and other office equipment. When many of our professional personnel are being asked to give up private offices, the least we can offer in return is pleasant surroundings.⁹

Although he was really fighting for an increase in private offices, his argument that workers would prefer "unit furniture" because it would create a more "pleasant" office environment suggests that its use might off-set the loss of private offices for his staff. The DOA was very firm that only agencies that accepted a reduction in space would receive the new furniture. Most surprisingly, the furniture was desirable enough to both agencies and their staff, that some, including the DOA itself, elected to accept the necessary space reduction in order to acquire the new furniture for their space.

8. Memo from Neal Steinhoff to Paul Brown, October 2, 1978, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 32 Folder 30, Wisconsin Historical Society

9. Correspondence from Mark Musolf to Kenneth Lindner, July 9, 1979, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 32 Folder 30, Wisconsin Historical Society.

In fact, in some agencies, the staff were more supportive of the implementation of the open plan in their buildings than the agency head. When the Wilson Street Office Building was being remodeled in the early 1980s, the Secretary of the the Department of Health and Social Services wrote the DOA Secretary stating that the recommendations for space needs for the agency were too low (particularly in terms of private offices) and that those figures represented "a community of interest between our two staffs and their strong preference for 'open landscape' approaches. A preference I share only in part [sic]".¹⁰ His letter suggests that while his staff were enthusiastic about the open landscape concept, he was actually more reluctant to adopt it, particularly because of the limits on private offices. This is significant because it suggests that at least some workers in Wisconsin State Government were in favor of the adoption of the open plan, even while some upper level managers and executives were against it.

Over the course of the 1980s as offices in older Wisconsin State office buildings were remodeled or redesigned, the new space standards and the "open space landscaping" concept were used as the basis for planning. While some older offices continued to use movable screens to wrap older conventional furniture, more and more offices began acquiring and using the new systems furniture. By 1985, the Department of Corrections, which had been making traditional office furniture for a number of decades, began manufacturing a new systems furniture line in coordination with a private company called Indiana Chair Frame which specialized in prison

10. Letter from Donald Percy to Kenneth Lindner, October 19, 1981, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 39 Folder 27, Wisconsin Historical Society.

industry furniture manufacturing.¹¹ By that time agencies were required to give preference to Badger State Industries (the Corrections manufacturing arm) for all furniture purchases creating an automatic market for the new furniture.¹² The first systems line produced by Badger State Industries (BSI) was fairly simple with only a few laminate and fabric options to choose from, but as BSI expanded its systems furniture lines to meet the changing demands of state agencies (wire management, ergonomics, etc) it also expanded its offerings in terms of finishes, fabrics, and accessories.¹³

By the late 1980s, the open plan office had become a well established design in Wisconsin State offices. In 1989, as the Department of Administration began re-evaluating their space standards in preparation for the construction of a new office building in Madison, they put together a committee of workers to research and produce a new report titled “Goals for the Ideal Office Building.” The report examined a number of factors related to office building design and use, and included several sections specifically dealing with internal design and arrangement of space. According to the study, a “cubicle” should offer the optimum balance of privacy, openness and air circulation in order to maximize “an employee’s [sic] productivity”, but “all three objectives are frequently lost in using modern systems furniture.” First of all, it is significant that the report uses the term “cubicle” to refer to an open plan workspace. It is used in such a way as to suggest that it is by this time a common and familiar term that requires no

11. "ICF Successful Programs for Correctional Industries," undated Pamphlet Courtesy of Badger State Industries.

12. Draft of Ergonomic Policy Statement, 1985, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 52 Folder 17, Wisconsin Historical Society.

13. Badger State Industries Marketing Material, 1985-1989, courtesy of Badger State Industries.

explanation. The report also acknowledges that while the use of systems furniture has certain organizational advantages particularly related to flexibility, efficiency, and ease of change, it also can be used in such a way as to create a negative experience for workers particularly in terms of a lack of privacy, openness, and air circulation. This suggests a fundamental tension in the needs of workers between a desire for privacy and a need for “openness.” To address this tension, the report makes specific recommendations regarding panel heights (60-66 inches high), the use of glass (so that “employees [sic] can see out of their cubicles”), and the placement of workers in relation to one another (so that an employee can not see their adjacent neighbor).¹⁴ All of this is significant in that it shows that by 1989 systems furniture and the open plan concept were no longer perceived in a positive light by Wisconsin State Government workers; the “cubicle” had thus become associated with some of the very attributes (lack of privacy, claustrophobic, etc.) that would become recurring jokes in *Dilbert* cartoons in the 1990s.

Thus, the open plan first appeared in Wisconsin State Government offices as a progressive idea that would offer foster and support a "developmental" approach to management, allow greater flexibility, and reduce hierarchy in the workplace but over time became a source of frustration for workers. In tandem with the use of the open plan was the development and use of systems furniture which was initially treated as a special amenity in the late 1970s, but became a standard office furniture in Wisconsin State Office Buildings by 1992. Over this same period numerous economic, social, technological and organizational changes influenced the way in which the open plan and systems furniture were designed and used in Wisconsin State

14. Goals for the Ideal Office Building: A Quality Improvement Project, March 1989, Wisconsin Department of Administration Secretary's Subject File, Part 1, Box 73 Folder 37, Wisconsin Historical Society.

Government. The remainder of this chapter will use the Work System Model to identify key changes in each area of the work system that had direct or indirect consequences for the design, implementation, and use of the open plan in Wisconsin State Government through the 1970s and 1980s.

Organization: Cutting Costs; Cutting Space

Organization within the Work System Model refers to the “organizational context in which work tasks are carried out.”¹⁵ In the period of this research, 1970-1992, Wisconsin State Government was characterized by a highly bureaucratic organizational structure defined by organization charts and clearly defined hierarchies stretching from the top-most executive position (the governor) down to the lowest level workers. Within this large bureaucracy, each agency also had its own internal working culture that was sometimes at odds with the larger bureaucracy of state government. There were numerous examples throughout the archive of agency heads complaining about the irritating bureaucracy of State Government and frustrations with following tedious bureaucratic procedures. There were also some differences among the agencies with respect to their organizational culture and structure, with some exploring a more developmental approach to management emphasizing worker autonomy, flexibility, and motivation and others maintaining a more strict bureaucracy that emphasized rules, order, and control. Yet, the over-riding organizational issue of this period was the intense pressure to cut

15.M.J. Smith, and P. Carayon, “Balance Theory of Job Design,” in *The International Encyclopedia of Ergonomics and Human Factors*, ed. W. Karwowski (London: Taylor & Francis, 2001), 1182.

costs in Wisconsin State Government by any means necessary. This pressure affected the organization itself, but also all of the people in it, shaping and driving decision making and serving as the justification for most changes.

To understand this organizational pressure, it is important to understand the broader economic context of the period. While the immediate post-war period of the 1950s and 1960s were characterized by enormous economic growth, the 1970s began a period of nationwide belt-tightening with a drop in productivity (both private and public sector), an on-going energy crisis, and growing inflation.¹⁶ In Wisconsin State Government there was significant pressure to increase productivity and reduce the cost of services by any means necessary. Starting in 1972, Governor Patrick Lucey initiated a new “productivity” mandate which required agencies to make a 5% budget cut over the 1973-1975 biennium.¹⁷ This early productivity mandate was only the first of a number of major cuts that agencies would see throughout the 1970s and 1980s. Through this period agencies were under continuous pressure to meet a number of new cuts and in tandem with those cuts they were also required to justify nearly any change made within their organization (new purchases, new office procedures, etc.) in terms of cost savings or increased productivity or both.

From this very first major cut in the early 1970s, agencies were encouraged to look at “waste space” in their offices to help them meet the mandate, and early on the office landscaping

16.Stein, *Pivotal Decade: How the United States Traded Factories for Finance in the Seventies*.

17.Patrick Lucey, “Wisconsin’s Productivity Policy,” *Public Administration Review* 32, no. 6 (1972), 796.

concept was often justified as a means of reducing wasted space.¹⁸ When the Bureau of Planning and Budget put forward their request for a new office landscaping arrangement they specifically mentioned reducing wasted space as one of the proposed benefits of the new plan.¹⁹ By 1978, the economic benefits of the open plan concept was widely accepted in Wisconsin State Government such that when the Public Service Commission requested an office landscaping installation in their offices their request for \$96,063 was authorized by the State Building Commission and increased to \$106,800 in order to more accurately reflect the cost of the redesign.²⁰ In other words, although office landscaping required a significant up front investment to effectively implement, the cost was considered justifiable as an investment that would save money in the long run by reducing space usage.

In this context, it is not surprising then that as the DOA was in the process of designing and constructing two new office buildings in 1978, that they would choose the open plan or “open space landscaping,” as they called it, as the new space standard for all state office buildings. In their justification, they emphasized the cost-saving and space saving benefits of the “open office” concept and made no reference to any of the qualitative benefits associated with the open plan in

18.”Governor’s Productivity Improvement Program,” 1972, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 12, Folder 13, Wisconsin Historical Society.

19. Correspondence from Jack Schoop to Harry Schmidt, August 15, 1972, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 11, Folder 18, Wisconsin Historical Society.

20. Agency Request for State Building Commission Action, September 1978, and cover memo from Dick Peterson to Gary Johnson, WI Department of Administration Secretary’s Subject File Part 1, Box 32 Folder 28, Wisconsin Historical Society.

the early 1970s. Even systems furniture was treated as an inducement to agencies and departments to accept further space reductions in order to acquire the new furniture.²¹

In fact, the DOA itself was facing significant external pressure (from the public and from the legislature) to demonstrate frugality in office space. In November of 1979, the Democratic Party of Milwaukee County even put forward a resolution condemning both the DOA and the governor for their decision to purchase expensive systems furniture for the new office buildings. In his drafted but unsent response to the Milwaukee County Democratic Party, the DOA Secretary responded that the new systems furniture would reduce individual work spaces by 30 square feet, allow more workers to fit into the space, and allow for greater flexibility for the future.²² In this way, the transparency of state government and the need to defend choices both internally (to other agency heads) and externally to the public required the DOA to emphasize the economic benefit of the new furniture.

The early implementation of the open plan represented a “shrinking” of office space for workers as many saw themselves moved from private offices to open plan work stations. The 1978 space guidelines sharply limited private office space to only those positions that needed privacy for functional purposes, and the DOA Secretary at the time held firmly to this policy even responding personally to agency heads seeking exceptions to the rules for various workers. When workers of the Department of Natural Resources (DNR) moved into their brand new office

21. “Guidelines and Standards for Effective Space Management,” December 1978, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 33 Folder 22, Wisconsin Historical Society.

22. Draft of correspondence from Kenneth Lindner to Waldemar Schultz, February 5, 1980, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 35 Folder 18, Wisconsin Historical Society.

space, they reportedly felt their individual workspaces were too small.²³ Similarly, the DOA initial move into GEF 2 was described as a ‘shoehorn’ operation in which the agency was actually forced to move into an even smaller space than initially planned. In fact a number of the individual space allocations for workers were reduced significantly from the official space standards of the time.²⁴

Just a few years after the new state office buildings were completed under the new space allocations, there was continued pressure on all state agencies to continue to reduce office space usage. In 1981, the DOA released an official “Announcement” for the 1981-1983 biennium that sought to further reduce costs in light of upcoming federal budget cuts by reducing space usage particularly in privately leased space, by moving as many agencies and departments into state owned office buildings as possible. Agencies were encouraged to notify the DOA if they were able to further consolidate their office space in order to save money.²⁵ In 1982, the DOA reduced their own office space in GEF 2 by 12,466 feet in order to reduce costs. According to an article in the DOA newsletter, *DOA Today*, the space consolidation “was strictly a matter of economics” with “recent budget reductions, layoffs and job vacancies” the consolidation of space was “not only possible but extremely desirable.”²⁶ It is significant that the DOA made specific reference to

23. “DNR’s Move to GEF #2 Provides Lessons for DOA,” *DOA Today*, October 1979, 3, Wisconsin Historical Society.

24. Correspondence from Victoria Luster to Eileen Polizzotto, February 1, 1982, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 40 Folder 29, Wisconsin Historical Society.

25. DOA Announcement, October 26, 1981, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 39 Folder 1, Wisconsin Historical Society.

26. “Space Consolidation Effort Underway in GEF #2,” *DOA Today*, May 1982, 1, Wisconsin Historical Society.

layoffs and unfilled vacancies in relation to office space. The physical space of the office became a visible symbol of the climate of budget cuts and layoffs.

When the DOA began its space consolidation effort in 1982 they asked all of their workers to participate in the process by purging “everything that is unnecessary” such as “accumulated books, papers, and files.” According to the article in *DOA Today*, “space considerations per employe [sic] basically allow for one desk and one open shelving unit or one filing cabinet.”

Workers were told that if they:

clean out all the junk that’s been eating up your space, you’ll undoubtedly be surprised at how much easier it is to find what you really need to do your job. Burying your work space under tons of paper doesn’t make you a more dedicated employe. It simply gives the impression that you’re a pack rat.²⁷

Although workers had reduced workspace to begin with in the new open plan office, they were asked to continue to reduce their own workspace usage to save money. The open plan thus became a direct expression of this culture of constant budget cuts.

In tandem with this negative usage of the open plan, workers were also subject to constant threats of layoffs also driven by budget cuts. Waves of layoffs through the 1980s resulted in a climate of continuous uncertainty. In 1981 alone, 484 state workers received a layoff notice.²⁸ Even as late as 1988, the DOA put forward a management philosophy that emphasized the importance of increasing productivity and reducing costs. According to the report, the DOA was committed to a “principle of no layoffs of our employes [sic],” but acknowledged that “some

27. “Space Consolidation Effort Underway in GEF #2,” *DOA Today*, May 1982, 1, Wisconsin Historical Society.

28. “Layoff Data Prepared by DER,” July 1982, *DOA Today*, 4, Wisconsin Historical Society.

positions may be eliminated.”²⁹ In this way, the open plan came into common usage in an organizational climate of increasing economic uncertainty, and it became a direct reflection of this climate. Indeed the spatial flexibility offered by the open plan which allowed for quick changes and readjustments to individual workspaces as well as to the whole office layout enabled Wisconsin State Government to efficiently accommodate significant changes in staffing. The interaction of office space with all of these budget cuts is perhaps best represented by a press release from the Wisconsin State Employees Union from 1982. According to the Union, workers in the Department of Industry, Labor and Human Relations had experienced “continual organizational flux” with “catastrophic funding cuts” which resulted in layoffs, reorganizations and various moves and remodeling efforts such that the workers began referring to the process as “rearranging the deck chairs on the Titanic.”³⁰

Meanwhile, over the course of the 1980s, Wisconsin State Workers saw their workspaces actually shrinking around them. While the 1978 space standards allowed 80-90 square feet for para-professional, 90-100 for professional, and 140-165 for management, the 1989 standards give 64-72 for para-professional and professional alike, and 80-120 for management. The 1989 standards also changed the language on private offices significantly, stating that workers of bureau director status or higher would be entitled to a private office (removing the language that

29.”Management Philosophy,” Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 67, Folder 17, Wisconsin Historical Society.

30. Madison Area Fightback Committee, Press Statement, September 23, 1982 Wisconsin Department of Administration Secretary Subject Files Part 1, Box 40 Folder 40, Wisconsin Historical Society.

emphasized function over status in the 1978 guidelines).³¹ Workers in Wisconsin State Government saw this workspace shrinkage occurring around them in real-time, and it was often specifically articulated as a means of cutting costs by any means necessary. The shrinking workspace also reflected the increasing uncertainty and instability of state office work, in which sudden catastrophic cuts could result in dwindling workspaces and most importantly loss of jobs.

People: Landscapes of Power and Status

In the Work System Model, this section usually focuses on the “individual” worker, but in the context of this study I want to look more broadly at the social and cultural aspects of workers in aggregate in Wisconsin State Government in this period, and how those elements interacted with worker’s experience of work and workspaces.

First of all, it would be foolish to attempt to generalize across all state office workers. The total number of state employees in this period (1970-1992) reached its peak in 1977 with 77,706 state workers.³² In the early 1980s employment numbers fell to 61,992, but gradually rose again

31. "Guidelines and Standards for Effective Office Space Management," December 1978, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 33 Folder 22, Wisconsin Historical Society; Wisconsin State Administration Building, 101 E Wilson Street Space Guidelines, January 30, 1990, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 87 Folder 37, Wisconsin Historical Society.

32. Rupert Theobald and Patricia Robbins, ed., *The State of Wisconsin 1977 Blue Book*, Madison: Wisconsin Legislative Bureau, 1977, 330. Text available on-line <http://digital.library.wisc.edu/1711.dl/WI.WIBlueBk1977>.

to 69,323.³³ Some of these workers were represented by a Union, others were not; some of these workers were full-time permanent staff, others were part-time or Limited Term Employment (LTE).³⁴ These thousands of workers represented hundreds of different job titles working in a wide variety of different agencies and divisions, they represented vastly different pay grades, and had a wide variety of responsibilities and tasks in their day-to-day work.

As already described above under Organization, a worker's organizational status often had a direct correlation with the size and quality of their workspace. One of the big discrepancies in terms of space planning was between permanent staff and LTE workers. While permanent staff (full or part time) were entitled to an assigned workspace, LTE workers were considered outside the state space standards; they were not assigned official space, and there was no minimum for their workspace size. Therefore, agencies that used a large number of LTE workers had to "squeeze" them in wherever they could fit them. For example, in 1979 when the first space standards came out that specifically stated that LTE workers would not receive assigned space at all, an administrator from the Department of Health and Social Services complained about the lack of assigned space for non-permanent staff (their agency housed as many as 250 non-permanent staff at any point in time) stating that "to ignore these employees in the planning and allocation of space will inhibit their ability to function effectively, and overcrowd offices to the

33. Rupert Theobald and Patricia Robbins, ed., *The State of Wisconsin 1983-1984 Blue Book*, Madison: Wisconsin Legislative Bureau, 1983-1984, 315. Text available on-line at <http://digital.library.wisc.edu/1711.dl/WI.WIBluebk1983>; Lawrence Barish, and Rupert Theobald, ed., *The State of Wisconsin 1991-1992 Blue Book*, Madison: Wisconsin Legislative Bureau, 1991-1992, 303. Text available on-line at <http://digital.library.wisc.edu/1711.dl/WI.WIBluebk1991>.

34. "DOA Employee Handbook," 1984, Wisconsin Historical Society, 25-26.

point where permanent staff lose their operating efficiency as well.”³⁵ This suggests how the efforts to squeeze the many non-permanent staff into the office had a negative impact on both the LTE workers and their permanent counterparts.

The state used an enormous number of LTE workers throughout this period. In 1981, a study on Limited Term Employee usage in the state found that while the official number of LTE appointments was 21,050, many of those appointments were only 300 hours total and also since some LTE workers had multiple LTE appointments, the study recommended shifting the LTE calculation to a full-time equivalent, which in 1981 was 3,043.³⁶ Reading the editions of the Wisconsin *Blue Book* for 1979-1980 and 1981-1982, I observed a massive drop in the reported numbers of LTEs from 16,789 to 4,319 (suggesting the state reduced its LTE usage significantly), but the study on Limited Term Employee usage suggests that what really happened is that the state changed the way in which they calculated LTEs so that it reflected full-time equivalents (FTE) rather than individual LTE appointments.³⁷ This is notable because it shows both the reliance of the state on LTE appointments, as well as their efforts to minimize their usage in their official employment documentation. Instead of recognizing each LTE as an

35. Correspondence from Nate Harris to Willie Nunnery, April 24, 1979, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 34, Folder 3, Wisconsin Historical Society.

36. Report of Investigation of Limited Term Employment Practices, March 30, 1981, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 40 Folder 9, Wisconsin Historical Society.

37. Rupert Theobald and Patricia Robbins, *State of Wisconsin 1981-1982 Blue Book*, Madison: Wisconsin Legislative Bureau, 1981-1982, 724. Available on-line at <http://digicoll.library.wisc.edu/cgi-bin/WI/WI-idx?id=WI.WIBLueBk1981>. Rupert Theobald and Patricia Robbins, *State of Wisconsin 1979-1980 Blue Book*, Madison: Wisconsin Legislative Reference Bureau, 1979-1980, 789. Available on-line at: <http://digicoll.library.wisc.edu/cgi-bin/WI/WI-idx?id=WI.WIBLueBk1979>.

individual worker, it treated them as small fractions of a full-time equivalent reducing the overall number of LTEs in the official record. Such calculations were especially misleading in terms of office space, in that each individual worker required workspace of some fashion, no matter how small or temporary their job. Even as late as 1990, the official space standards indicate that LTE workers could be assigned space “below the recommended standard” that was in use for permanent staff.³⁸ In this way, LTE workers were subject to very different treatment than that of their permanent counterparts, even if they were doing the same work (as they often were).

It is impossible to talk about status in this period without acknowledging and grappling with the complex landscape of race, gender and ability in Wisconsin State Government because of the significant changes in the diversity of state workers in this period as well as the efforts by the state to further increase the diversity of state office workers (particularly in terms of race, gender, and ability). Through the 1970s and 1980s diversity was a critical issue in terms of hiring and retention. Starting in the mid-1970s, the DOA began examining the issue of increasing diversity in state government. In their objectives for 1975-1977, DOA administration stated that “the hiring, development and advancement of women and minorities in all pay ranges and all parts of DOA is the single most important, internal DOA priority.”³⁹ In 1976, Governor Lucey signed an executive order which mandated that every agency create an affirmative action plan in keeping with Title VII of the Civil Rights Act to encourage the increased hiring of women and

38. Wisconsin State Administration Building, 101 E Wilson Street Space Guidelines, January 30, 1990, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 87 Folder 37, Wisconsin Historical Society.

39. “Looking Forwards...: Objectives 1975-1977”, November 1975, Wisconsin Department of Administration Secretary Subject File Part 1, Box 21 Folder 24, Wisconsin Historical Society.

minorities in all aspects of state government.⁴⁰ In 1977, the DOA newsletter bragged about some exciting “firsts” including: the hiring of the first professional female architect; the hiring of the first minority superintendent in Buildings and Grounds (trainee); the promotion of the first female custodial supervisor from custodial staff; the reclassification of a female into a professional purchasing agent position; and the hiring of the first minority engineering technician. The report documented an overall increase of minority workers from 5.8% to 7.7%, and an increase in the number of women in professional positions from 15.6% to 24%.⁴¹

Despite such efforts, actual recruitment, retention and promotion of women and minorities remained slow through the late 1970s and early 1980s. In fact, from 1978 through 1982, the reported percentage of minority workers employed in Wisconsin State Government actually decreased from 5.46% in 1978 to just 4.4% in 1982.⁴² A series of reports from 1979 shows that of the 877 new hires in the Department of Administration for the 1978-1979 fiscal year, only 43 (4.9%) were people of color, 33 (3.8%) were “persons with handicaps” and only 259 (29.5%) were women. In the same year, only one (.9%) of the 107 upward reclassifications in the DOA was a person with a disability, eight (7.5%) of the 107 were people of color, and 54 (50.5%) of the 107 were women. Another report showed the percentage of female, minority, and “handicapped” workers broken down by groups of agencies and in different job types. Across the board, women were in the majority in para-professional and clerical positions, while some of the

40. Patrick Lucey, Executive Order No 9, March 26, 1976, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 21 Folder 25, Wisconsin Historical Society.

41. “DOA Moves Towards Affirmative Action Goals,” *DOA Today*, October 1977, Wisconsin Historical Society, 1.

42. Rupert Theobald and Patricia Robbins, *The State of Wisconsin Blue Book 1979-1980*, 789 and *The State of Wisconsin Blue Book 1981-1982*, 727.

social service agencies also showed high numbers of women in their technical jobs and their professional jobs (60% and 55% respectively). Meanwhile the employment of ethnic or racial minorities was low across the board with only positions in protective services breaking into double digits (10% in education-social services and 15% in general executive). The use of workers with disabilities was even lower with none breaking out of single digits in any agencies.⁴³

In a report reflecting on this period, Kenneth Lindner, the DOA Secretary, described how despite “concentrated staff time and monetary resources” used toward increasing diversity in professional positions, the DOA continued to have difficulty in recruitment for these kinds of positions. The report argues that they are unable to compete monetarily with other employers, and the budget constraints “greatly reduce our ability to develop trainee positions.”⁴⁴ In 1985, the DOA had 33% women, 6.3% minorities, and 13.3% persons with disabilities, and had internal goals to raise those to 35%, 9.4% and 14% respectively. Relative to other units, DOA was second in terms of number of minorities on staff in their agency, third for persons with disabilities, and sixth for women “working in non-traditional jobs.”⁴⁵ Despite these stated goals, internal reports about affirmative action efforts reveal a decrease in the hiring of minorities in the mid-1980s, as

43.”Parity for Wisconsin State Service,” May 1979, Wisconsin Department of Administration Secretary Subject File Part 1, Box 32 Folder 17, Wisconsin Historical Society.

44.Kenneth Lindner, “Affirmative Action Plan,” October 1980, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 38 Folder 2, Wisconsin Historical Society.

45.”All DOA Employees [sic] to Benefit from Approved AA/EEO Plan,” *DOA Today*, August 1985, 1-2.

well as a concern about the decrease in professional women in state government.⁴⁶ As late as 1992, an internal monitoring report on affirmative action efforts in the Department of Administration showed that in 24 hiring transactions, the DOA made Affirmative Action hires only 5 times.⁴⁷

In other words, despite their efforts, Wisconsin State Government continued to struggle with affirmative action throughout this period. Although they made some strides in the 1970s, much of that growth in affirmative action had stagnated in the 1980s. Meanwhile, there is some evidence that there was ongoing discrimination and dissatisfaction with affirmative action policies by some workers. In a study on the Division of Buildings and Grounds completed in early 1981 one person in upper management complained that the Personnel Department was “carrying out affirmative action policies and hiring practices too aggressively” and one worker indicated “that primarily white males are now being treated unequally and that they do not feel that minorities and women should be accorded equal rights because of past discriminatory policies and practices.” Similarly, some complex managers argued that many of their current problems stem from the affirmative actions policies in Personnel resulting in the tendency to hire employees that were “at the bottom of the employment barrel.” Implicit in that comment is the belief that affirmative action choices in hiring resulted in lower quality workers. The report describes how the supervisors in the division were largely “middle aged [white] men who grew up in small rural communities where they had little, if any, exposure to blacks or other minority

46.Memo from Jeffrey Russell to James Klauser, April 23, 1987, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 59 folder 20, Wisconsin Historical Society.

47.”Department of Administration Monitoring Report,” May 1992, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 91 Folder 2, Wisconsin Historical Society.

group members.” Indeed, a few of the building managers “pointed out that their supervisors are not the best qualified and questioned whether the supervisors should have their present positions because ‘they are prejudiced as hell.’” Some of the complex managers indicated that the mostly white supervisors had a tendency to “stereotype the behavior of minority employes [sic] and make judgements without weighing evidence.”⁴⁸

The Division of Buildings and Grounds was a part of the organizational culture of Wisconsin State Government and the Department of Administration, and it is highly unlikely that these attitudes towards minority workers and towards affirmative action hiring policies were isolated to this one unit. In fact an article from *DOA Today* in 1985 about affirmative action policy cryptically indicates that affirmative action “goals” did not represent “quotas” for hiring, “nor does it require supervisors to hire persons who are not qualified to do their job.”⁴⁹ In other words, the belief that women and people of color were less qualified for jobs than white male counterparts was not a viewpoint unique to the Division of Buildings and Grounds. There were likely many workers throughout various agencies, divisions and departments who carried similar attitudes and prejudices. The human relations study also reveals a tendency for the upper level white-collar positions to be filled by white men, and the lower positions, for example the Building Maintenance Helpers, by people of color. In this way, the organizational hierarchy was often inscribed by race and gender such that top positions (official/administrators) were held by

48. Anthony Brown, “Human Relations Study for the Division of Buildings and Grounds,” January 1981, Wisconsin Department of Administration Secretary Subject Files Part 1, Box 38, Folder 11, Wisconsin Historical Society.

49. “All DOA Employes [sic] to Benefit from Approved AA/EEO Plan,” *DOA Today*, August 1985, 1-2, Wisconsin Historical Society.

white men, while women and people of color were typically in lower organizational positions. This is not to say that there were no women or people of color in upper positions; however, the various memos and reports concerning affirmative action indicate that even the top administrators in State Government were aware that there were significant inequities in terms of hiring and promotion for women and people of color.⁵⁰

These inequities were embodied in the organizational structure and further inscribed in the office space, with the top (primarily white and male) workers receiving the larger office spaces, with women and people of color receiving smaller workspaces as a result of their lower organizational status. In her article “Seeing the Invisible,” vernacular architectural historian Diane Harris describes how “the walls, floors, ceilings, windows, fences, sidewalks, and surfaces that surround our daily lives contribute enormously to the cultural constructions of racial and class identity and to racisms embedded within our lives.”⁵¹ In the context of Wisconsin State Office Buildings, the physical environment of the office reinforced the organizational hierarchy, as well as the underlying disparities along race, class and gender lines. I mention class because it is another “invisible” aspect of work—not measured in a direct way by affirmative action numbers for example—and yet implied by the very notion of status and pay as associated with certain positions which were associated with education and training that were themselves reflections of status, upbringing and opportunity. These distinctions of class and status in state government were also reinforced by internal classifications (represented and non-represented,

50. "Parity for Wisconsin State Service Expressed as Percentage of Total Employees [sic] in Category," May 1979, Wisconsin Department of Administration Secretary's Subject File Part I, Box 32 Folder 17, Wisconsin Historical Society.

51. Dianne Harris, “Seeing the Invisible: Reexamining Race and Vernacular Architecture,” *Perspectives in Vernacular Architecture* 13, no. 2 (2006), 100.

permanent and LTE for example) which created differential status in the organization, and different benefits and rules governing work. Among all of these different types of workers, LTE were definitely the most poorly treated with no benefits of any kind associated with their employment (health insurance, retirement, etc.), and no minimum in terms of workspace size.

Thus, although the open plan had initially been conceived as a means of creating greater equity and reducing hierarchy, in practice it ultimately reinforced such distinctions. While a number of private offices were eliminated, that ultimately served to make private offices that much more valuable as an organizational amenity. By the early 1990s, only those workers at the very top of the organization were entitled to a private office. Meanwhile, status continued to be directly measurable in the size of one's workspace, even as space standards decreased over the course of the 1980s (as described above under organization). The disparities in workspace across race, class, and gender lines reflected deeper disparities in terms of hiring and promotion.

These issues of social difference and discrimination also influenced workers' feeling of safety and well-being in the office itself. In 1991, one worker sent a letter to Governor Tommy Thompson describing an incident in the Hill Farms Building in which she felt her personal safety in the office was compromised as a result of finding an anonymous note on her desk with "racial and sexual overtones." In the letter she specifically states that she will no longer be willing to work after hours (evenings and weekends) because she felt the office was no longer safe for her at those times.⁵² It is significant that this particular worker implies that she felt safe in the office

52. Letter from Lisa Helmeid to Governor Thompson, January 17, 1991, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 85 Folder 18, Wisconsin Historical Society.

during the day while she was surrounded by her colleagues, but the same place became “unsafe” to her at night and on weekends when the office was deserted. The personally threatening letter changed this worker’s feelings about her workplace, and although she never mentions the open plan in her letter, there is some evidence that the design of the open plan might have exacerbated her concerns for her safety particularly after hours when the partitions that create privacy for workers to focus during the workday became blind alleys where danger might lurk unseen just around the corner. For example, there was a significant increase in thefts in Wisconsin State Office Buildings after the open plan was implemented as the new space standard, and workers were cautioned not to leave personal items in their workspace overnight lest they be stolen.⁵³ Most importantly, this also suggests the importance of the social context of the workplace. During working hours, the open plan was perceived as a relatively safe environment, but after hours some workers felt their personal safety and well-being was compromised. In this case, feeling safe and welcome is a kind of privilege that some workers may have felt at all hours, and others may have felt only when there was critical mass of workers present in the office to allow them to feel safe from personal harm.

53. For example: “Take Security Precautions in GEF #2,” May 1980 *DOA Today*, Wisconsin Historical Society, 4; and Letter from Neal Steinhoff to C.D. Besadny, November 7 1989, Wisconsin Department of Administration Secretary’s Subject File Part I, Box 73 Folder 38, Wisconsin Historical Society.

Environment: Physical Safety and Comfort in the Open Plan Office

The environmental aspects of the work system model concern the physical aspects of the workplace including air quality (heating and cooling, ventilation, and air pollution), noise and other sensory elements that may influence the worker's experience of their work and the workplace. The issue of environmental quality in state office buildings revealed significant tension between saving money and ensuring worker health and well-being, particularly as ideas about health and safety in the office changed significantly over the course of the 1970s and 1980s.

In the 1970s, the energy crisis influenced the quality of the office environment, particularly in terms of ambient temperature. As early as 1973, Governor Lucey put out a press release requiring state agencies to "take immediate action wherever possible to reduce their consumption of energy fuels." As part of the new mandate to reduce energy cost, state facilities were required to maintain a 68 degree temperature during the workday and a 65 degree temperature at night and on weekends throughout the winter months.⁵⁴ These discussions reveal how significant the issue of energy usage in state offices was. By the time the state was in the process of constructing GEF 2 and 3 in the late 1970s, there was a federal mandate in place to maintain an office temperature of 65 degrees during the winter months.⁵⁵ Both GEF 2 and 3 were constructed to minimize energy usage throughout, and the use of the open plan concept further supported this

54.Press Release, January 15, 1973, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 11 Folder 17, Wisconsin Historical Society.

55.Letter from Kenneth Lindner to Joseph Noll, Dec 7, 1979, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 34 Folder 6, Wisconsin Historical Society.

goal by allowing for more open space, and fewer enclosed spaces which require separate ventilation as well as heating and cooling. Although the federal temperature guidelines were rescinded by President Reagan in 1981, energy conservation remained an issue throughout the 1980s in Wisconsin State offices.⁵⁶ For example, even in the 1980s, many state offices were remodeled to improve energy conservation.⁵⁷ The open plan became one of a number of strategies to reduce energy costs by reducing the number of private offices requiring heating and cooling.

Managing ambient air temperature was made even more complicated by the increase in technology throughout the office (as I will explore later in Technology). Over the course of the 1980s, more computer equipment, copiers and printers became fixtures in the workplace, and all of those machines generated significant heat. In evaluating the "load criteria" for managing office temperatures and air circulation, the Division of Buildings and Grounds had to factor in the growth of these machines in their calculations, and any increase in equipment required a re-evaluation.⁵⁸ Other areas of the office actually required the installation of additional cooling capacity to maintain certain equipment that would malfunction if the space was too warm.⁵⁹

56.White House Press Release, Feb 17, 1981, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 39 Folder 1, Wisconsin Historical Society.

57.For example: Memo from Stan Vinge to Doris Hanson, November 2, 1983, Wisconsin Department of Administration Secretary's Subject File Part I, Box 45 Folder 19, Wisconsin Historical Society.

58.Jaspal Engineering Services Meeting Notes, March 21, 1988, Wisconsin Department of Administration Secretary's Subject File Part I, Box 69 Folder 7, Wisconsin Historical Society.

59.Memo from Thomas Herman to Sue Adix, March 15, 1983, Wisconsin Department of Administration Secretary's Subject File Part I, Box 43, Folder 21, Wisconsin Historical Society.

Yet for workers, the reality was that the ambient office temperatures were often intolerable. In 1979, workers in GEF 1 created a petition complaining about the uncomfortable air temperatures and workers in the Wilson Street State Office Building in Madison were reportedly wearing their coats in their offices in the wintertime, while others were uncomfortably warm in stuffy offices.⁶⁰ Even after the end of the federally mandated temperature restrictions, workers continued to experience erratic and uncomfortable temperatures. The Public Service Commission complained in 1983 that their open plan offices in the Hill Farms State Office Building had such irregular temperature swings that “we literally have staff covering themselves with shawls and coats to keep warm at times while others are concurrently sweltering on the sunlit side of the building”...”staff members are frustrated by what they perceive to be bureaucratic indifference to their reasonable concerns.”⁶¹ Thus, although the open plan concept was intended to improve the quality of air circulation and office temperatures, in practice it did little to improve air temperature quality.

In tandem with concerns about heating and cooling in Wisconsin State Offices, were growing concerns about air quality, particularly related to smoking in the office. Through the 1970s, discussions about smoking in the office tended to emphasize the rights of smokers to be able to smoke, but in 1984, in compliance with the new Clean Indoor Air Act, Wisconsin state

60. Correspondence from Dean Fuller to Neil Steinhoff, November 6, 1979, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 34 Folder 6, Wisconsin Historical Society. Correspondence from Donald Percy to Kenneth Lindner, Wisconsin Department of Administration Secretary's Subject File Part 1 Box 34 Folder 3, Wisconsin Historical Society.

61. Correspondence from Nell Flores to Doris Hanson, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 46 Folder 31, Wisconsin Historical Society.

government began restricting smoking to designated smoking areas.⁶² Interestingly, the new rules allowed offices populated entirely by smokers to be exempt from the restriction.⁶³ At the time that the new rules were put in place, there was some question over whether partitioned offices in an open plan design counted as offices. In other words what constituted an “office” for the purposes of the new law? Was a partitioned workspace an office? To get clarification, the DOA Secretary contacted State Attorney General Bronson La Follette on the issue, and he responded that “since smoking is permitted in an ‘office’ occupied exclusively by a smoker or smokers, then the smoke could escape from the partitioned area and affect the workers in the adjoining area.”... “It is not the area established by the partitions which controls, but rather the entire ‘enclosed, indoor area’ within which the area established by the partition is located.”⁶⁴ In other words, partitioned work spaces were not “offices” for the purposes of this new law. Yet smoking still remained a regular activity in the office. Workers in private offices continued to be able to smoke in their own workspace if they chose (or avoid smoking areas), while workers in open plan offices were subject to the constantly shifting smoking culture in this period. For smokers in the open plan, not being allowed to smoke in their workspace was probably a source of frustration, and for non-smokers the continued practice of smoking within state office space

62. Correspondence from Ed Main to Wayne McGown, October 22, 1974, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 18 Folder 13, Wisconsin Historical Society.

63. Announcement, July 1984, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 49 Folder 20, and Memo from Doris Hanson to DOA employees, July 31, 1984, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 61 Folder 37, Wisconsin Historical Society.

64. Correspondence from Bronson La Follette to Doris Hanson, June 25, 1984, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 49 Folder 20, Wisconsin Historical Society.

would have made the workplace inhospitable. One worker commented that the state should provide a “non-smoking break room” since the only one available is “too smokey for some.”⁶⁵ While some non-smoking break rooms were created, they were sometimes treated as expendable space. In 1988 the Public Service Commission requested that a recently created (but reportedly “infrequently” used) non-smoking lounge be made into a conference room due to a lack of meeting space. In other words, despite the new policies controlling smoking, non-smokers were sometimes treated as the minority.⁶⁶

As late as 1991, the smoking policy which allowed for smoking in designated areas remained more or less intact. In the Wilson Street State Office Building in Madison, smoking was allowed in the restrooms on every other floor. One office (the Bureau of Air Management) complained that the smoke from the men’s restroom adjacent to their workspace drifted out of the louvered door of the bathroom creating a significant air quality issue for the workers in that office.⁶⁷ It was not until 1992 that state office buildings began to be designated as officially smoke free.⁶⁸ This illustrates how ideas about smoking changed significantly in this period, from

65.Evaluations from a seminar, December 9, 1985, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 52 Folder 9, Wisconsin Historical Society.

66.Correspondence from Joyce Narveson to James Burkholder, June 28, 1988, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 69 Folder 25, Wisconsin Historical Society.

67.Correspondence from Donald Theiler to Gerald Whiteburn, James Klauser, and Carroll Besadny, August 28, 1991, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 85 Folder 18, Wisconsin Historical Society.

68.Correspondence from James Klauser to Herbert Grover, May 6, 1992, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 91 Folder 9, and Memo from James Klauser to GEF II employees, March 18, 1992, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 94 Folder 33, Wisconsin Historical Society.

smokers having a right to smoke in the office, to non-smokers having a right to clean air in their workplace.

New technology could also be a source of unpleasant odors in the office. In 1980, one telecopier in a state office reportedly had a particularly noxious smell as a result of an electronic charge which oxidized the paper. Although the smell was unpleasant, it was apparently not harmful; however, the manufacturer acknowledged that the scent was certainly 'not something you want to smell' and recommended the state invest in special paper that is "scented like baby powder" and available through the manufacturer to cover up the smell.⁶⁹ This just further suggests the complicated layers of smells, pollution, and particulates that were potentially circulating around the office at any given time.

The broader concern about air quality and ventilation was often in tension with the issue of energy conservation. Although designed in the open plan style, GEF 3 had significant air circulation issues, particularly in private offices and enclosed areas like conference rooms which were reportedly “stuffy.” In the late 1980s, then DOA Secretary James Klauser stated:

air quality in new state office buildings continue to be a concern for all of us involved in management. A compromise between energy conservation and air quality it is too often weighted to energy conservation. We are watching this balance carefully and recently have had to rescind certain changes that were made in the name of energy and conservation.⁷⁰

In 1990 an Air Quality Task Force found that GEF 2 ventilation only allowed 5 cubic feet of air per minute per person, which was apparently below the standard for good air quality. In their

69.Memo from Robin Riley to Art McClure, December 4, 1980, Wisconsin Department of Administration Secretary's Subject File Part I, Box 36 Folder 7, Wisconsin Historical Society.

70.Correspondence between Herbert Grover and James Klauser, May 6, 1987 and May 13 1987, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 64, Folder 7, Wisconsin Historical Society.

report, they acknowledge that GEF 2 was completed during an era in which energy conservation was the primary concern, and suggest that the state work to increase ventilation, reduce crowding, further reduce private offices and internal partitions to allow better overall air circulation, and even steam clean furniture partitions to reduce dust.⁷¹ Thus, there was continued emphasis on the benefit of the open plan over the use of private offices to increase ventilation. In fact, air quality was important in terms of both worker comfort and health. Throughout the 1980s office buildings became a recognized source of various health problems. In Wisconsin State Government there were concerns about asbestos, and a state office building in Milwaukee was actually shut down very suddenly in 1984 to deal with an asbestos problem.⁷² In 1989, there was a report about several employees having “sick building syndrome” in the GEF 1 building.⁷³ The office space itself was thus seen as a potential source of health problems. Particulates in the air, air circulation, smoking, ventilation, and HVAC were all related to the overall construction and design of the office space. While the open plan in some ways allowed for more circulation of particulates and air pollution (such as smoking), it was also continually treated as a solution to air circulation problems.

71.Memo from Air Quality Task Force to James Klauser, June 1990, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 80 Folder 4, Wisconsin Historical Society.

72.Memo from Doris Hanson to Milwaukee State Office Building Employees, October 30, 1984, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 48 Folder 1, Wisconsin Historical Society.

73.Correspondence from Jake Mitchel to Neal Steinhoff, Feb 27, 1989, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 73 Folder 37, Wisconsin Historical Society.

There were also concerns about office noise. The combination of the new “open space landscape” concept along with the growth of equipment made the office environment increasingly noisy, especially through the 1980s. In addition to the lack of meeting space in open plan offices which created less privacy and more social disruptions in the form of impromptu meetings and conversation, the “clattering” of various office machines (particularly copiers and printers) made it increasingly difficult for workers in Wisconsin State Office Buildings to concentrate on their work.⁷⁴ This is also a good example of the complex interactions of change in the work system itself; the State of Wisconsin was investing a lot of resources in the acquisition of new technologies with the very deliberate goal of increasing productivity and reducing costs, but they often failed to consider how those new technologies would behave in the new open plan office, particularly with respect to the increased noise. Thus, in the mid 1980s, the state scrambled to make recommendations to accommodate the increased noise from machines including using acoustical covers, locating the machinery on outside walls as much as possible, and strategically using partitions to reduce overall noise.⁷⁵ The “Goals for an Ideal Office Building” from 1989 described: the role of furniture panels in absorbing sound; the importance of creating sufficient separation between workers (even suggesting placing workers back-to-back to reduce sound transmission); and also the benefit of “sound masking” or “white noise” to reduce acoustic disruptions. According to the report, the isolation from sounds was not just about

74. Correspondence from Joyce Narveson to James Burkholder, June 28, 1988, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 69 Folder 25, Wisconsin Historical Society; Correspondence from Lon Sprecher to Tom Alt, March 9, 1984, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 52 Folder 10, Wisconsin Historical Society.

75. Correspondence from Sue Adix to Tom Alt, March 25, 1985, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 52 Folder 17, Wisconsin Historical Society.

allowing workers to concentrate and focus on their work, but also about perceived privacy, which the report argued was an important aspect of a workspace. The report stated that “generated sounds” should be “absorbed” (by panels, carpeting, and ceiling materials) or “masked” as much as possible in order to give workers a sense of privacy in their individual workspace.⁷⁶ The official space guidelines created for the new State Office Building in 1990 specified locating “high noise producing activities” away from the open landscape in order to reduce the impact of office noise. They also recommended adding “white noise” where necessary.⁷⁷ Thus, while the open plan in some ways created more noise in the office, systems furniture with its fabric covered panels was viewed as a tool for controlling noise in the office by absorbing sound.

Although the open plan included design elements (acoustical panels, white noise machines, etc.) to manage noise, workers experienced a workspace full of noise distractions that made it difficult to concentrate, and also reduced their sense of privacy within their workspace. It is significant that noise in the office was more than just an irritant, it also was thought to directly relate to workers sense of control and privacy within their workspace. Despite efforts to manage noise in the open plan, the loss of meeting spaces (due to the increasing shortages of office space in general) along with the growth of noisy office machines made the open plan workspace feel less private.

76.”Goals for an Ideal Office Building,” March 1989, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 73 Folder 37, Wisconsin Historical Society.

77. Wisconsin State Administration Building, 101 E Wilson Street Space Guidelines, Jan 30, 1990, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 87 Folder 37, Wisconsin Historical Society.

Technology: Systems Furniture and Computers

In the Work System Model, this section examines the role of new technologies in the process and structure of work. Of all of the changes in this period, perhaps the biggest and most influential on all aspects of work was the increasing use of new technology, particularly the growing use of computers which in turn had a major impact on the open plan, especially the design and use of systems furniture.

In the early 1970s, as Wisconsin State Government was trying to “increase productivity” and reduce costs, the investment in computers and word processing systems became another strategy to address the organizational pressure to meet the mandated cuts. Although the equipment was a significant investment, it was considered justifiable in the long term because it could make the office more efficient. In their report on management improvements from 1977, the DOA described two critical trends that were improving management across state agencies: centralization of management support services, and the “continued substitution of sophisticated technology for human operations to achieve greater speed, accuracy and capacity to handle complex tasks.”⁷⁸ These two trends that were treated as separate ideas were in fact linked in that the new technology itself enabled and encouraged greater centralization of certain functions. The economies of scale achieved in the small agencies by using technology to centralize certain processes and task, became the impetus to create still greater cost reduction by implementing new technology and centralized clerical work across multiple agencies.

78. "Selected Management Improvements from Wisconsin's Executive Agencies, 1973-1976," January 1977, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 28 Folder 13, Wisconsin Historical Society.

In the late 1970s, data processing and word processing were distinct functions handled by separate office machines. For data processing government entities and private industry alike relied on large mainframe-style computers that often handled large data processing tasks for multiple divisions or departments. In Wisconsin State Government, data processing was centralized in three regional data processing centers that serviced all of the state agencies. Data processing jobs were “scheduled” to maximize the usage of the computing systems by all agencies.⁷⁹ A similar trend in centralization was also taking over word processing. Over the course of the 1970s, word processing shifted from a way of organizing work (a centralized unit for typing and dictation) into a distinctive piece of office equipment.⁸⁰ In Wisconsin State Government, various agencies began experimenting with dedicated word processing units throughout the late 1970s. The DOA for example created a centralized unit for their agency that was comprised of seven automatic IBM typewriters that could handle 15 characters per second.⁸¹ By 1979, as they were preparing for their move into their new offices in GEF 2, the DOA purchased a new “Wang” word processing system which featured a CPU (Central Processing Unit) and a CRT (Cathode Ray Tube), and could handle multiple simultaneous inputs (typing) and outputs (printing).⁸²

79.State of Wisconsin, “Data Processing Issues and Organization,” December 1976, Wisconsin Historical Society.

80.Thomas Haigh, “Remembering the Origins of Word Processing and Office Automation,” *IEEE Annals of the History of Computing* 28, no. 4 (2006), 6.

81.Correspondence from Tom Landgraf to Doris Hanson, Sept 15, 1978, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 30 Folder 16, Wisconsin Historical Society.

82.”Wang Equipment a Big Plus,” *DOA Today*, September 1979, 2, Wisconsin Historical Society.

Meanwhile, many agencies began purchasing their own computing equipment to supplement and in some cases replace their use of these centralized services. Through the 1980s and 1990s personal computers and “microcomputers” became increasingly common in offices throughout the state.⁸³ Initially (in the early 1980s) a department might just have one or two computers in their office, which were generally used for limited amounts of time by multiple users. The computer was just one piece of equipment among many that was used for discreet tasks including creating spreadsheets, typing letters and reports, and creating charts and graphs. For example, in 1982 the Division of Administrative Services requested the purchase of 3 computers for their office to manage certain tasks that did not fit with either the centralized data processing or word processing units:

These activities involve reports which integrate text and numbers, use formulas to distribute and allocate cost, monitor work in process through the use of logging systems, etc. Systems and programming costs to place these applications on a large main frame computer would be difficult if not impossible to cost-justify. Yet the applications often involve tedious hours of work using paper and pencil and/or calculators.

Of the proposed computers for the Division of Administrative Services, one would be used full-time by a single user, the second would be shared primarily between just two users, and the third would be shared by several internal departments.⁸⁴ Thus throughout the 1980s, individual computer usage could vary widely with some workers spending their entire day on a computer and others using computers just occasionally for specific office tasks.

83. Correspondence from Veronica Luster to various departmental managers, April 5, 1982, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 41 Folder 18, Wisconsin Historical Society.

84. Proposal to acquire Microcomputers for the Division of Administrative Services, March, 19 1982, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 40 Folder 27, Wisconsin Historical Society.

Over the course of the 1980s, computers became quite common throughout the workplace, with more and more workers spending a large part of their day on a computer. In 1984, a memo from the State Budget Office documented the increasing need for computers and compared “sharing a computer among workers to sharing a ball point pen twenty years ago.” According to the author, “the PC is now not only state of the art, but is fast becoming a basic necessity.”⁸⁵ While some units were intensively using computers in the early 1980s, others were still “sharing” computers into the late 1980s. The Division of Administrative Services stated in a departmental memo from 1989 that the “sharing of microcomputers and their peripherals by several people within divisions remains the object of the department.”⁸⁶ Despite these differences there was increasing saturation of computer usage in all aspects of work. A report from 1990 found over 35,000 microcomputers or workstations in Wisconsin State Government offices, and projected “an increase in PC penetration in those agencies with low workstation to employee ratios.” One estimate even suggested an on-going acquisition rate of computers from 5,000-10,000 annually for the foreseeable future.⁸⁷

The rapid growth of computers and other office technology in Wisconsin state offices required furniture that could physically accommodate the new technology (space for the components, etc.) as well as integrate the power needs of these new machines and manage the

85.Memo from Lon Sprecher to Tom Alt, March 9, 1984, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 52 Folder 10, Wisconsin Historical Society.

86.Department Policies, Division of Administrative Services, September 15, 1989, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 73 Folder 22, Wisconsin Historical Society.

87.Information Technology Advisory Board, “Information Technology Management in Wisconsin,” November 1990, 25, Wisconsin Historical Society.

growing network of cables required for communication. In 1987, the Department of Transportation requested new furniture for one of their offices that had been converted to “open landscape” in 1973, but was no longer suitable for the “new electronic equipment” being used.⁸⁸ In 1988, when the Wisconsin Lottery requested a new layout for their office, they specified that each of their workers (with the exception of field reps) would have a computer, and specifically requested systems furniture that would have both power capabilities and a “cable track” to support computer usage.⁸⁹ In this way, the furniture was viewed as a vital support for technology usage in the office.

The use of computers throughout the office was becoming increasingly necessary, and systems furniture evolved significantly from simple panels and desks into an integrated furniture, power and cable management system to accommodate the increasing demands of computer hardware in offices. In 1989, this issue of wire management became critical in a bid for Badger State Industries to find a manufacturer for a new systems furniture line. While the original bid was awarded to Indiana Chair Frame, a second company, Krueger International of Wisconsin, challenged the award arguing that the Krueger International product was significantly superior to

88. Agency Request for Building Commission Action, August 1987, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 60 Folder 1, Wisconsin Historical Society.

89. "Office Layout for the Offices of the Wisconsin Lottery" (RFP), 1988, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 77 Folder 24, Wisconsin Historical Society.

the other primarily on the grounds of its ability to handle both power and cable management. In his appeal, the president of Krueger argues that:

Krueger's product provides an integrated and extensive computer cable management system while ICF provides little, if any computer cable management. Due to the increased use of computer technology, cable management is required to eliminate unsightly cords and prevent them from being a potential safety (trip) hazard. Krueger's product would meet this requirement for state employee's offices while ICF's product does not.⁹⁰

The Krueger product utilized a so-called "raceway" system on both the top and bottom of the panels to allow for cords, wires and cables to be integrated into the furniture itself.⁹¹ Although the appeal was initially rejected by the Department of Health and Social Services, the decision was overturned by the Department of Administration and the bid process was redone.⁹²

In their 1989 report, "Goals for the Ideal Office Building," the Division of Buildings and Grounds attempted to identify the essential qualities of an ideal office building. Among the specifications for the ideal office building the report stated that systems furniture panels "should be prewired to minimize the wiring exposed by computer cabling." Further the furniture should have "both horizontal and vertical raceways," and provide for telephone, computers and electrical outlets for office personnel and maintenance."⁹³ Less than a year later, the DOA

90. Correspondence from Richard Resch to the Secretary of DHSS, April 18, 1989, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 73 Folder 25, Wisconsin Historical Society.

91. "Analysis and Response to the Krueger Industries appeal of H&SS rejection of the protest to the intention to award bid to Indiana Chair Frame," Wisconsin Department of Administration Secretary's Subject File Part 1, Box 73 Folder 25, Wisconsin Historical Society.

92. Correspondence from James Klauser to Patricia Goodrich, July 14, 1989, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 73 Folder 25, Wisconsin Historical Society.

93. "Goals for the Ideal Office Building," March 1989, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 73 Folder 37, Wisconsin Historical Society.

produced their own guidelines for a new administrative office building being constructed in Madison and with respect to the systems furniture specified that “partitions with cable runs (utility spline) will be used” in order to “accommodate the most moves without significant remodeling.”⁹⁴ In other words, the partitions of the office were no longer simple dividers of workspaces, they were utilities that contained the essential power and cabling needs for the entire office. Connected to the organizational efforts to reduce costs in the long-term, systems furniture with internal wiring was portrayed as a means of increasing the flexibility of the building so that it could effectively meet the technological and organizational needs of the future while also reducing future remodeling costs. Indeed, systems furniture itself was treated as a kind of technology that was directly connected (both literally and figuratively) to the computers and other office equipment in use at the time. The increasing need for cabling and power capability also made older furniture obsolete requiring offices to invest in newer furniture that could meet the power and cabling needs more effectively.⁹⁵

Yet as systems furniture became more complex with its incorporated wiring, it also became less easy to move and change. Early iterations of open plan and systems furniture with simple unwired panels could be easily rearranged, even by workers themselves. For example in 1985 when offices in GEF 2 were redesigned using existing furniture, workers were told not to make any alterations to their work areas as all of the systems furniture components had been accounted

94. Wisconsin State Administration Building 101 E. Wilson Street Space Guidelines, January 30, 1990, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 87 Folder 37, Wisconsin Historical Society.

95. "Goals for the Ideal Office Building," March 1989, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 73 Folder 37, Wisconsin Historical Society.

for in the new plan.⁹⁶ The fact that they had to tell workers not to make any changes implies that it was very easy for workers to make changes; partitions and other system furniture components could be easily moved (even by workers) without the aid of Facilities Management. As systems furniture became more complex in its design, particularly incorporating wiring and power, it became more difficult to make changes. In 1989, a memo on policies for computer usage indicated that one of the challenges for computer usage in state offices was that it was often “costly and impractical” to move cubicle partitions to allow for sharing.⁹⁷ In contrast with the simple partitions of office landscaping that could be easily moved around by anyone, the increasing complexity of the furniture required specialized tools and knowledge as well significant planning to make changes. Thus one of the chief benefits of the open plan concept, its flexibility, was significantly eroded by the inclusion of power and cabling.

Tasks: Interacting with Computers

In the Work System Model, the issue of physical ergonomics typically falls under technology; however, in this research it seems to fit best under “tasks.” This is because in the period, the issue of workers needing ergonomic furniture was directly linked to the amount of time they spent typing and working on a computer. According to the International Ergonomics Association, “Ergonomics (or human factors) is the scientific discipline concerned with the understanding of

96.Memo from Tom Alt to Division Administrators, March 7, 1985, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 52 Folder 11, Wisconsin Historical Society.

97.Department Policies, September 15, 1989, Wisconsin Department of Administration Secretary’s Subject File Part 1, Box 73 Folder 22, Wisconsin Historical Society.

the interactions among humans and other elements of a system.” In this chapter, I will be focusing primarily on one type of ergonomics, often called “physical ergonomics” which examines the fit between the human body (anatomical, physiological, biomechanical, etc.) and the design of work (including workspace, technology, etc.).⁹⁸

In Wisconsin State Government the increasing centralization of certain clerical functions (particularly word processing and data processing) along with the spread of computers and other technology in the office increased clerical work for workers in nearly every job category. Starting in the early 1980s, there was growing awareness of the influence of computer usage on the physical and mental health of workers. As a result, in 1982 the DOA approved a study conducted by Dr. Steven Sauter of the University of Wisconsin Department of Preventative Medicine on VDT users working for the state in order to better understand the physical and psychological impact of VDT usage on workers.⁹⁹ Following this research, Sauter produced a report for the Department of Administration called “Improving VDT Work: Causes and Control of Health Concerns in VDT Use.” The report documents the ways in which the design of the work itself was problematic in terms of creating stress, but focuses primarily on the physical aspects of VDT work and ergonomic deficiencies in the workplace. The report describes a number of factors in workplace design particularly chairs, table heights, and lighting that negatively impact VDT usage, and can be improved to reduce physical stress. Although the

98. International Ergonomics Association, “What is Ergonomics?,” http://www.iea.cc/01_what/What%20is%20Ergonomics.html (accessed July 16, 2013).

99. Wisconsin Data Processing Advisory Council Meeting Minutes, February 25, 1982, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 41 Folder 41, Wisconsin Historical Society.

report makes some recommendations regarding the technology itself (for example favoring detachable keyboards), it mostly emphasizes strategies for working around the constraints created by the technology. The report also features images of workers using computers in ways that would result in injury or physical stress. These images emphasize the misfit between the physical design of the workstation, and the needs of the workers; for example, a monitor that is positioned too low on a desk causing a worker to hunch over in order to see the screen, or a monitor that is too high forcing an uncomfortable upward head tilt.¹⁰⁰ In other words, the design and use of the computer equipment was often treated as a “given” that could not be changed, rather the furniture was treated as both the source of the problem, and the primary solution. As one administrator from the Department of Industry, Labor and Human Relations stated, we “recognize that inappropriate workstation furniture is the primary cause or problems related to VDT use.”¹⁰¹ Thus, this new idea of ergonomics placed a lot of the blame on office furniture rather than the use of the VDTs, and therefore also argued that ergonomic furniture design would alleviate VDT worker’s physical stress.

In this way, the design and use of the computer was treated as an inevitable aspect of work that could not be changed. Workers were going to continue using computers in growing numbers, and since computers themselves could not be changed, nor the work that required computers, the furniture became the primary source of blame and the bandaid for many of the problems. In the state of Wisconsin, a new policy was created in 1985 that mandated the use of ergonomic

100. Steven Sauter, et al., *Improving VDT Work: Causes and Control of Health Concerns in VDT Use* (Madison: Department of Preventative Medicine, University of Wisconsin, 1985).

101. Memo from Helene Nelson to Tom Alt, May 31, 1985, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 52 Folder 17, Wisconsin Historical Society.

furniture for all “target users,” those workers who use a computer for 90 continuous minutes each work day, or a cumulative 5 hours of VDT usage per day. The policy also indicates that absolute rules or guidelines are impossible, and that any worker who reports physical stress as a result of VDT usage was entitled to some kind of action to alleviate that stress. Each state agency was charged with identifying intensive VDT users and ensuring that they had sufficient ergonomic support. In the mid-1980s Badger State Industries (BSI) was the primary office furniture manufacturer used by the state, but they did not yet have furniture that could support intensive VDT users. The state offered furniture from other manufacturers that could be purchased for use by intensive users. Samples of the furniture were available, and the policy encouraged the involvement of the individual workers in selecting furniture as individual needs could vary significantly from person to person. With respect to space and layout, the policy stated that VDT users’ workspaces should be “designed for the nature of the work” allowing sufficient space for their materials (spreadsheets, documents, etc.). For “high priority jobs it is appropriate to replace existing systems furniture with the proper size table and pedestals;” however, “appropriately designed VDT furniture is often more space consuming than other types of furniture” so “larger space may be needed to allow for this.”¹⁰²

The decision to provide ergonomic furniture for workers was closely tied to cost. Even the policy itself acknowledged that ergonomic chairs were especially important and that all workers should have “chairs that can be fitted to their height, offer some back support, and are reasonably comfortable to sit on,” but “a blanket replacement of all existing chairs is neither warranted from

102. Draft of Ergonomic Policy Statement, 1985, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 52 Folder 17, Wisconsin Historical Society.

an ergonomic viewpoint nor fiscally responsible.” Attention to these individual details of chair design should be considered in proportion to “the length of time that a worker must remain seated.” Thus although the state recognized that chair design was important, they could not justify the cost of replacing all chairs for all workers. Individual agencies echoed this same concern with the fiscal impact of the new ergonomic policy.¹⁰³ One administrator in the DOA wrote that the department “recognizes the need to incorporate these recommendations into its operating procedures in a reasonable time frame while maintaining fiscal responsibility.” She went on to say that the proposed changes should “occur in a controlled evolutionary style rather than in a revolutionary situation.”¹⁰⁴ In other words, the various agencies and their administrators were under pressure to honor the new policy and ensure the safety of workers, while also meeting the continuous demand to reduce costs. The pressure to conform to the policy for ergonomically designed workstations was in some ways at odds with the organizational pressure to maintain “fiscal responsibility.” Even some of the recommendations that did not require a change to the design of the space (for example giving workers breaks) were challenged by some agencies who felt that mandated breaks would result in a significant loss of productivity, and a need to hire LTEs to cover the lost work.¹⁰⁵ This suggests how the policy itself was at odds with the on-going demand in the 1980s to reduce costs by any means necessary. Worker’s health and

103.Draft of Ergonomic Policy Statement, 1985, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 52 Folder 17, Wisconsin Historical Society..

104.Correspondence from Sue Adix to Tom Alt, May 24, 1985, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 52 Folder 17, Wisconsin Historical Society.

105.Correspondence from Eileen Merchart to Tom Alt, May 15, 1985, Wisconsin Department of Administration Secretary’s Subject Files Part 1, Box 52, Folder 17, Wisconsin Historical Society.

well-being was thus sometimes treated as a frivolous luxury, and not a necessary component of a safe workplace for all workers.

Even as computers were becoming a standard fixture throughout the office, “ergonomics” remained something that was specialized and non-standard. For example, in 1989 when the DOA created its “Goals for an Ideal Office Building,” the document specifies the importance of systems furniture being wired (as mentioned above) to handle computers, but does not indicate the importance of an ergonomically designed workspace to support computer usage.¹⁰⁶ This appears to be a significant omission. Although the guidelines acknowledge the growing use of computer technology throughout the office (hence the need for wiring capability mentioned above), they do not engage with the importance of a workspace that can reduce physical stress on workers who use computers intensively. In other words, despite the ergonomics policy established in 1985, ergonomics was still treated as a special exception for some workers, and not a standard around which all workspaces and equipment should be designed in the late 1980s. In 1990, the space standards for the new Administrative Building being constructed on East Wilson Street in Madison also did not offer any recommendations with respect to ergonomics. While the importance of providing partitions with cable management is indicated (again implying the spread of computer and other equipment in the office), the issue of designing ergonomic workspaces is left out entirely.¹⁰⁷ Instead of conceptualizing office design as a preventative solution to manage physical stress related to VDT usage, it was treated as an after

106. “Goals for an Ideal Office Building,” March 1989, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 73 Folder 37, Wisconsin Historical Society.

107. Wisconsin State Administration Building, 101 E. Wilson Street, Space Guidelines, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 87 Folder 37, Wisconsin Historical Society.

the fact adjustment intended to compensate for existing physical stress. The onus was thus often on the employee to recognize the source of the stress (the physical design of their workspace) and request accommodation, rather than on the organization to ensure a safe physical environment for all of its workers. In this way, the organizational culture and structure, particularly the emphasis on reducing costs by any means necessary, drove the decision making concerning the implementation of ergonomic policy. This illustrates the complex interaction of change across the Work System Model, and how organizations like The State of Wisconsin navigated these complicated changes in the content, structure and culture of work in real time. Although the organization was concerned enough about the well-being of workers to consider developing an ergonomic policy in the first place, ultimately the State of Wisconsin opted for a policy that favored the need of the organization to reduce costs over the need for workers to have a safe workplace.

The Experience of Work

The implementation of the open plan was just one change among a whole host of changes in the content, structure, and culture of work in Wisconsin State Government during this period. Rather than focusing exclusively on the implementation of the open plan, this chapter used the Work System Model to describe and articulate the complex interplay of changes in the culture and structure of work in Wisconsin State Government in the 1970s and 1980s. Although this is not an exhaustive catalog of the changes that occurred, it nonetheless conveys the ways in which change in one aspect of work created “ripples” of change in others. For example, the economic

crisis of the 1970s resulted in the state investing heavily in new technology to improve efficiency and reduce the overall workforce, but the new technology also created new organizational structures (for example dedicated word processing and data processing units), and new demands on the office infrastructure in terms of power, networking, noise control and Heating, Ventilation and Air Conditioning (HVAC) which in turn influenced the construction of offices and the design and use of office furniture. On-going changes in the design of computer equipment required further changes to the design of office furniture to accommodate the hardware, and the increased use of computers in all aspects of office work created new physical stresses on workers which required further adjustments to the design of the workspace to alleviate physical stress. In other words, what might seem to be a fairly simple and straightforward change in office work (adopting new technology) could have significant and ongoing consequences on a number of other aspects of the work system. As a structure for research and analysis of work, the Work System Model is a powerful tool for describing the expected and unexpected consequences of change.

All of these changes are important because they fundamentally transformed the experience and meaning of work for workers during this time. Although it was predicted that new technology would eliminate white-collar drudgery, the reality in the 1970s and 1980s was quite different. In Wisconsin State Government the increasing centralization of certain clerical functions (particularly word processing and data processing) along with the spread of computers and other technology in the office increased clerical activities for workers in nearly every job

category.¹⁰⁸ Meanwhile the workers in dedicated clerical units in particular experienced highly repetitive and tedious jobs in which they felt overworked and under-appreciated.

One example of this in Wisconsin State government were the workers of the Word Processing Center in the DOA who experienced significant stress, overwork and frustration with their jobs in the 1980s. In 1984, various memos concerning the need for increased overtime reveal how strained the entire Word Processing Center was at that time. The combination of high turnover in the period and increased work meant that the workers in the Word Processing Center were being asked to work evenings and weekends to keep up with the constant flow of work.¹⁰⁹ Meanwhile the new technology allowed for the close monitoring of the unit's output, which was documented in a monthly report. Despite the pressure and strain on the workers of the Word Processing Center, a clerical study found that several divisions in the DOA were frustrated by the errors and delays in memos and reports sent to the Word Processing Center, and had reduced their use of the unit as a result.¹¹⁰

With these many complaints and frustrations, it is perhaps not surprising that the workers in the Word Processing Center reportedly suffered from very low morale. In 1985 the DOA put together a special workshop led by a professor from Edgewood College to help improve morale

108. Clerical Survey responses from various divisions, August 1984, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 48, Folder 7, Wisconsin Historical Society.

109. Memo from Neal Steinhoff to the Clerical Survey Team, August 6, 1984, Wisconsin Department of Administration Secretary's Subject File Part I, Box 48 Folder 7, Wisconsin Historical Society.

110. Memo from Ralph Hollman to the Clerical Needs Team, Aug 16, 1984, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 48 Folder 7, Wisconsin Historical Society.

in the unit and thereby increase productivity. The comments on the evaluations give hints of the deep dissatisfaction and feeling of alienation among workers in the unit. One comment implies that prior to the workshop, many of the workers did not know each other's names, and states that as a direct result of the workshop people "say hello more often." Yet even with the workshop, workers still complained that certain aspects of the job were fundamentally "dehumanizing and disturbing." The evaluations also reveal how workers felt they were perceived by other units. As one respondent stated, word processors are generally viewed as being "at the bottom of the heap mentally." Thus, while the workshop helped in some respects, workers in the unit still felt frustrated with their job, undervalued, and stressed.¹¹¹

The experience of the workers in the Word Processing Center illustrate how the cumulative effects of new technology, job insecurity, overwork, centralization of tasks, physical stress, social isolation, and the loss of control over the work process could create significant stress and unhappiness for workers. In fact, the experiences of the workers in the Word Processing Center were the bleeding edge of changes to white-collar work that directly affected a large number of workers throughout this period. Meanwhile, these same changes that influenced the experience of workers also fundamentally changed the design, meaning, and use of the open plan. Over the course of the 1970s and 1980s, the open plan became a means of cutting costs and a visible symbol of the insecurity of work at the time and the general indifference of the organization for the physical well being of its workers. Although initially implemented in Wisconsin State

111. Evaluations from workshop with cover letter, dated December 9, 1985, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 52 Folder 9, Wisconsin Historical Society.

Government in a spirit of progressive idealism, by the late 1980s the open plan had become a symbol of the many problems of contemporary work, and a perceived source of or contributor to those problems.

Yet even if the open plan was less than ideal for workers, it did create some opportunities for Wisconsin State Government as an organization. The use of the open plan and systems furniture enabled the state to reduce its dependence on leased office space, and fit more people into increasingly less space. While workers saw their workspaces shrinking around them, for the organization this was a valuable mechanism for meeting the constant pressure throughout this period of cutting costs. The relative ease of changing the design of the open plan also facilitated the management of organizational change. Indeed throughout this period, the constant reorganizations of different agencies required a change in office space to accommodate both the creation of new divisions and the dissolution of old divisions. Thus the use of the open plan in Wisconsin State Government had benefits to the organization as a whole, even if workers in Wisconsin State Government came to perceive the open plan more negatively over time.

However even with these benefits, it is clear that the state did not fully anticipate the challenges the open plan and systems furniture would present to themselves or to their workers. One huge issue was that the ability to make changes along with the numerous reorganizations meant that there was an increased demand for the services of the Facilities Management office which could barely keep up with the constant requests for space evaluations and re-designs.¹¹² Meanwhile, although intended to improve air circulation and reduce energy costs, the problems

112.Memo from Dave Ward to Art McClure, April 21, 1983, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 45 Folder 12, Wisconsin Historical Society.

of managing heating and cooling in the open plan also proved more challenging than anticipated, particularly when combined with the increasing use of various types of technology that increased demands in the various HVAC systems. Also, the lack of meeting space in the open plan compounded the problem of "noise" and forced the state to commandeer employee lounges in some offices and turn them into conference rooms in order to accommodate the demand for meeting space. In other words, although the open plan was useful in certain ways to the organization, it created new challenges as well.

Thus, even as the open plan was implemented with so many hopes for a positive transformation of the workplace, the numerous changes in work fundamentally changed the way in which the open plan was implemented and used. Over time it became closely associated with decreasing space costs, shrinking workspace, and the growing instability and inequity of white-collar work. This case study reveals just some of these changes, and how those changes related to the design, meaning and use of the open plan. This study also illustrates how an office design can not be separated from the context of its use. By using a systems approach, I illustrate the complex interplay of change in the context of work, and how those changes related to the use of the open plan. In the next chapter, I examine patterns of representation in various popular media since the early 1990s in order to understand the negative image of cubicles and how that image is tied to a negative image of white-collar work.

Chapter 4:

Exploring Cubicleness:

The Office Cubicle and the Representation of Office Work in Popular Media

As real workers saw their offices converted from conventional plans to open plans in the 1970s and 1980s, so too did their fictional counterparts in film and other media. Starting in the mid-1970s, open plan offices slowly began to appear in popular films like the 1976 movie *All the President's Men* and the 1980 movie *9 to 5*. By the mid-1980s, the open plan was a fairly standard backdrop in the "office work" movies of the period, for example *Wall Street* (1987), *Working Girl* (1988), *Big* (1988), *The Secret of My Success* (1987), and *Nothing in Common* (1986) all used bustling open plan offices as a stimulating backdrop to the larger workplace narrative. Perhaps more importantly, all of these films featured highly motivated and committed white-collar workers who were willing to do whatever it takes to get ahead. In these films, work is exciting and fun, and a source of personal fulfillment.

This image of office work as a source of personal satisfaction is in direct contrast to the disenchanted office workers featured in films, cartoons, books, commercials, and even news stories from the 1990s and 2000s. In tandem with this shift in images of white-collar work and white-collar workers, the open plan too became increasingly drab, and a source of satire. In 1991, Scott Adams, the creator of the syndicated comic strip *Dilbert*, began regularly skewering

the cubicle in his daily strips for example, depicting workers attached to grey cubicle walls by velcro strips to save space (June 17, 1991), and in another comparing the cubicle to a prison cell (July 4, 1991). Through the 1990s, films too began featuring monochromatic and dreary offices, and showed workers literally and figuratively “trapped” in their cubicles. Unlike their highly motivated counterparts of the 1980s, the “heroes” of the 1990s and early 2000s movies were workers who ultimately escaped their dull office jobs in search of personal fulfillment and happiness outside of work.

In this chapter, I unpack the visual and textual imagery associated with cubicles in various popular media including films, television commercials, comic strips, books, and news stories since the 1990s. I argue that “cubicles” as they are depicted in popular culture, have distinct visual and textual patterns that correspond to the perceived experience of working in cubicles, particularly: social isolation, claustrophobia, noise, lack of privacy, and loss of individuality. These themes are echoed in a variety of different media forms, and suggest common social attitudes towards office cubicles among contemporary American white-collar workers. I argue that these representations and references to office work in popular media also reveal a negative image of office work characterized by corporate bureaucracy, boredom, and the feeling among white-collar workers of being “trapped.” By analyzing representations of white-collar work in popular media, I argue that the “cubicle” has become a symptom and visual signifier in popular media of the “dreariness” of white-collar work. This chapter serves as a means of accessing the perspectives and experiences of white-collar workers in order to identify common cultural and social attitudes towards the cubicle and contemporary white-collar work in America.

A Cultural Studies Approach: The Representation of Everyday Life

There is a long history in cultural studies of using representation in popular media to reveal patterns of social and cultural thought in a particular historical moment. Research in cultural studies has often treated popular culture as a bellwether of culturally dominant fantasies, fears, beliefs and ideologies circulating in contemporary society; according to cultural studies scholar Ann Kaplan:

Commercial productions must command an audience sufficient for handsome profit, producers are clever at sensing the fantasies, fears, and desires that preoccupy a majority of people in a given period. Always hoping to be the first to provide desired images, producers keep their pulse on the moment, anticipating fashions before they catch on.¹

In other words, because media producers have a vested interest in producing objects that will appeal to a majority of people, they deliberately draw on the fantasies, fears, and desires circulating in the contemporary culture. Popular media objects are thus embedded with the shared beliefs, ideas, and feelings of the time and are a valuable source for identifying larger social and cultural patterns that are in turn connected to real experience.

The office cubicle is part of a larger system of meaning defined by both the “real” experiences of white-collar work, as well as its visual and textual representation in popular media forms. Indeed, the two parts are interconnected and mutually constitutive so that the “real” experiences inform the representation of the cubicle and the representation of cubicles in turn inform and reinforce the real experience of them. In cultural studies, this interplay is often likened to a circuit in which the producers use the “raw materials” of everyday life to create images, and the receivers in turn use the produced images as “raw materials” within their

1.E. Ann Kaplan, “Sex, Work, and Motherhood: the Impossible Triangle,” *The Journal of Sex Research* 27, no. 3 (1990), 410.

everyday life.² For example, cartoonist Scott Adams, the creator of the *Dilbert* comic strip, often uses the real stories of his readers as the basis for his comic strips about the inanities of contemporary work and his readers then use the comic strip in their everyday life, reading it in the morning newspaper and perhaps even cutting it out and posting it in their workspace.³ Media objects are thus a part of the bricolage of everyday life reflecting ordinary experience through their stories and images, but also becoming a part of everyday life as viewers and readers interact with and incorporate these media objects in their daily routine in different ways.

In this way, "everyday" experience is comprised of both the lived experiences of individuals as they go about their daily routines, and the representation of everyday experience in media which serves to represent and "re-present" the spaces and experiences of everyday life. According to geographers Aitken and Zonn, "everyday life is not immediately 'present' but is *re-presented* simultaneously through the contradictory images that constitute our postmodern world." The collage of images that make up everyday experience create a "convergence between what is *real* in the everyday and how we *image* the everyday."⁴ Rather than treating media representations as pure abstractions, disconnected from real experiences, this research challenges scholarship to consider media representations as a critical part of the experience of everyday life reflecting it as well as influencing how we experience everyday life.

2.Elizabeth Traube, *Dreaming Identities: Class, Gender, and Generation in 1980s Hollywood Movies* (Boulder: Westview Press, 1992), 4.

3.Peter Lewis, "Dilbert's Creator Espouses the Interactive Life Style," *New York Times* (1923-Current File), December 5 1993, F12, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/109135411?accountid=465>.

4.Stuart Aitken, and Leo Zonn, "Re-presenting the Place Pastiche," in *Place, Power, Situation, and Spectacle: a Geography of Film*, ed. Stuart Aitken, and Leo Zonn (Lanham: Rowman & Littlefield Publishers, Inc, 1994), 6-7.

In my research for this chapter, I examined a wide array of media objects including films, comic strips, commercials, novels and news stories. The media objects are all derived from popular media forms and were identified through a variety of different kinds of sources. First of all, I used a combination of database and Internet searches in sites such as: Proquest, Academic Search (EbscoHost), Internet Movie Database (imdb.com), Netflix (netflix.com), YouTube (youtube.com), the *Dilbert* website (dilbert.com), and the *New Yorker* cartoon database (cartoonbank.com). I also relied on a variety of secondary literature sources to identify other media objects for this study such as Emma Bells' book *Reading Management and Organization in Film*, Latham Hunter's Ph.D. dissertation "The Celluloid Cubicle," and Robert Beuka's book *SuburbiaNation*.⁵ Although this is not by any means an exhaustive survey of every reference or image of cubicles since the 1990s, it is a sampling that reveals underlying patterns of representation that correspond to larger social and cultural patterns that in turn reflect perceived common experiences of white-collar work in America. The first part of this chapter will construct a definition of "cubicleness" comprised of both the image of the cubicle (its common physical attributes) as well as the experiential aspects of the cubicle as conveyed in various media objects. The second part of this chapter will trace how the patterns of representation of the cubicle are related to a larger image of contemporary white-collar work.

5. Emma Bell, *Reading Management and Organization in Film* (New York: Palgrave, 2008); Robert Beuka, *SuburbiaNation: Reading Suburban Landscape in Twentieth Century American Fiction and Film* (New York: Palgrave, 2004); Latham Hunter, "The Celluloid Cubicle: Constructions of Masculinity in 1990s Office Movies" (McMaster University, 2005).

Defining “Cubicleanness”

In order to understand the meaning of the cubicle, we must first consider how the word “cubicle” has come to be associated with a shared cultural image and experience of the cubicle, or what I am calling “cubicleanness.” By “cubicleanness” I mean to indicate the common physical attributes of the cubicle (as defined on a purely technical plane) as well as the abstract notion of the cubicle as it is conveyed through the experiential aspects of the cubicle and defined through both the image of the cubicle and the experience of the cubicle in everyday life. In his book *The System of Objects*, Jean Baudrillard argues that we must go beyond the study of the mere function of objects in order to understand:

how objects are experienced, what needs other than functional ones they answer, what mental structures are interwoven with—and contradict—their functional structures, or what cultural, intracultural or transcultural system underpins their directly experienced everydayness.”⁶

Drawing on Baudrillard’s concept I argue that there is a shared understanding of “cubicleanness” that includes the physical qualities of the cubicle office and the experience of the cubicle in everyday life that is reflected in the representation of cubicles in popular media forms since the 1990s. By identifying patterns of representation of the cubicle in both fictional and non-fictional forms we can begin to understand “cubicleanness” as a social and cultural symbol of contemporary white-collar work in the United States.

The term cubicle as it relates to office design predates our contemporary usage; however, it is only since the 1980s that the term has consistently referred to open plan workstations. In the

6. Jean Baudrillard, *The System of Objects*, trans. James Benedict (New York: Verso, 2005), 2.

preceding post-war period, a “cubicle” was generally a small fully enclosed office space often used by lower and middle management. In an article in the *New York Times* from 1950 for example, the private offices in the new U.N. building are referred to as “cubicles.”⁷ Similarly, in the screenplay for the 1960 film *The Apartment*, screenwriters Billy Wilder and I.A.L. Diamond describe the small private offices where the middle managers work in the film as “glass-enclosed cubicles.”⁸ The word “cubicle” was used with increasing frequency in popular media through the 1990s and 2000s; in a word search in the Proquest database (comprising 48 different databases containing hundreds of different media sources), the number of entries containing the word “cubicle” jumps from 2,573 entries in the 1980s to 6,947 entries in the 1990s, and to 25,394 entries from 2000-2010.⁹ Although not every entry refers to an office cubicle (some refer to bathrooms, locker rooms, storage units, hospital rooms, and domestic spaces for example), the majority are in reference to open plan workstations. Similarly, a search of “cubicle” in the cartoon database for *The New Yorker* follows a similar pattern with just one entry from the 1980s, eleven in the 1990s, and 64 since 2000.¹⁰

So what makes a modern-day cubicle a cubicle? A modern office cubicle is typically comprised of several key elements: panel walls, work surfaces, and storage elements. When these disparate features are placed together, the cubicle becomes a spatial object that serves to

7. George Barret, “Too Few Windows, UN Staff Cries,” *New York Times* (1923 - Current File), October 5 1950, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/111570767?accountid=465>.

8. Billy Wilder, and I.A.L. Diamond, *The Apartment* (London: Faber and Faber, 1998), 26-27.

9. Proquest Database, <http://www.proquest.com> accessed July 19, 2011.

10. *The New Yorker* cartoon database, <http://cartoonbank.com> accessed on July 19, 2011.

organize and structure the office at both the individual and organizational levels. As a modular office furniture system comprised of a series of interconnected parts, cubicles are simultaneously the space of an individual worker, and a total office environment dominated by a rigid geometric structure comprised of a large architectural box (the office space), nested with smaller “boxes” usually in a grid formation. In *The System of Objects*, Baudrillard describes how such modular systems serve as an abstraction of order and control in which “everything has to intercommunicate, everything has to be functional—no more secrets, no more mysteries, everything is organized, therefore everything is clear.”¹¹ As a spatial image of the organization, the cubicle office represents an attempt to order and control both the interior space of the office and its workers, who are compartmentalized in identical boxes arranged in tidy rows.

Within the interior space of the individual cubicle, the panel walls and storage elements create a system of individual compartmentalization and order, that in turn contains a multitude of organizing objects and spaces (drawers, pen holders, file holders, overhead storage etc). In the field of Design History, there is significant research on the role of such organizing furniture that serve to create order in the spaces of everyday life. According to design historian Judy Attfield, “containment” in furniture objects is “the embodiment of order *and* clutter” in which the furniture creates a normative sense of order that often only disguises the messiness of everyday experience.¹² In this way, there is an innate tension between furniture (like the cubicle) that attempts to produce order, and the real detritus of everyday life that often resists such organizational efforts. The cubicle thus represents the compulsion of the organization to order

11. Baudrillard, *The System of Objects*, 28.

12. Attfield, *Wild Things*, 152.

and control the people and their objects of work, even as the personalized clutter of an individual worker may belie or challenge the total vision of order and control.

Whether talking about an individual cubicle or a room full of cubicles, the panel walls are the most prominent and visually recognizable features of the modern cubicle office. Cubicle walls have a very distinctive form typically featuring fabric covered panels with plastic, metal or wood trim around the top and sides. Although some representations of offices since the 1990s have used low walls instead of furniture to visually convey “cubicles,” many others use actual system furniture components with their characteristic fabric panels. Whether they are as low as a foot or two above the work surface, or as high as six or seven feet from the ground, they are generally set at right angles, and they serve to separate each workspace. The degree of enclosure can vary significantly from just one panel wall separating individual workstations, to enclosure on all four sides of the workspace, with most representations of cubicles in the 1990s and 2000s featuring enclosure on at least three sides.

In visual media, workers are often depicted sitting within their cubicle workspace, the panel walls creating a sense of partial enclosure (often shielding the rest of the office), yet the “cubicleness” of the panel walls is almost always identifiable by their low height (the top of the panel is generally visible in the frame), the drab color (typically grey, beige, off-white or blue), and the panel seams (where panels are joined together). In other words, there is no mistaking these kinds of panel walls for anything other than a cubicle.

Indeed, the panel walls of a typical cubicle are so recognizable, that even when removed from a conventional office setting, they are still visually legible as cubicle walls. For example, in a pair of cartoons from *The New Yorker*, cubicle walls are transplanted to an old-fashioned

galleon ship separating rowers from one another (March 31, 2003), and to the prehistoric era where cavemen are depicted in separate “cubicles” drawing on the panel walls (January 11, 2010).¹³ In both images, the essential structure of the cubicle (even without the other accoutrements of the office) is conveyed just by having low walls set at right angles separating people from one another. In this way, “cubicleness” is conveyed through the formal arrangement of the panels (low walls set at right angles), as well as through their use in segmenting space and separating people. It is also significant that both images show panel walls “decorated” with paper artifacts and drawings, suggesting that the visual display aspect of the cubicle is a part of their functionality.

The panel walls give a sense of structure to the individual’s workspace and the entire interior space of the office. In visual representations, the cubicle layout is often depicted from an elevated view filling an entire office space with rows upon rows of cubicle workstations all linked together. When viewed as a whole from above, the cubicle office appears to be an expanse of small boxes all linked up together to form an enormous grid. The films *Clockwatchers* (1997), *Jerry Maguire* (1998), *Office Space* (1999), *American Beauty* (1999), and the Pixar animation *The Incredibles* (2004) all feature elevated images of the cubicle office. In Figure 5, a still from the movie *Office Space*, the rigid rectilinear layout with its sharply defined angles and boxy workspaces gives a strong sense of visual order to the space. The elevated view also emphasizes the overall arrangement of the cubicle office which typically features either: a

13.Christopher Weyant, "I don't mind the whip. It's the cubicles I find demoralizing," *The New Yorker*, March 21, 2003, <http://cartoonbank.com>; Tom Cheney, "Cavemen are seen carving into walls in the form of a cubicle," *The New Yorker*, January 11, 2010, <http://cartoonbank.com>.



Figure 5: Still from the film *Office Space* (1999) featuring a "cube farm" with a central spine, shown from an elevated perspective. Mike Judge, *Office Space*, directed by Mike Judge, (1999; Los Angeles: 20th Century Fox, 2002) DVD.

central spine (comprised of linked panel walls) with panel walls jutting out towards a walkway at right angles from either side of the “spine” (*Clockwatchers*, *Office Space* and *The Incredibles*); or small groupings of four or more cubicles in clusters throughout the space (*Jerry Maguire*, *American Beauty*). Visually, the stark lines of the cubicle layout with its rigid grid and dreary neutral panels seems to dominate the office space creating an image of a stifling office environment.

This kind of image of an office full of cubicles is often casually referenced in text forms as a “farm,” “sea,” or “maze” of cubicles. For example, in the satirical novel, *Company*, writer Max Barry repeatedly refers to the cubicle office in the story as a “cubicle farm.”¹⁴ The novels *Company Man* by Joseph Finder and *JPod* by Douglas Coupland similarly refer to an open plan office as a “cubicle farm.”¹⁵ In the comic strip *Dilbert*, the term “cube farming” is used in a series of strips (June 28-July 1, 1999) in which a farmer character (wearing overalls and carrying a gardening hoe) is shown “weeding” cubicles of unwanted people (such as the Pointy Haired Boss) and “replanting” workers in larger containers (a private office). Even the news media sometimes refers to cubicle offices as “cubicle farms,” for example a 2007 article from the *New York Times* about telemarketers and autodialers creating ghost calls refers to the telemarketer

14. Max Barry, *Company* (New York: Vintage Contemporaries, 2007), 14, 19, 22, 227, 231, 281.

15. Joseph Finder, *Company Man* (New York: St. Martin’s Paperbacks, 2006), 3, 91-92. Douglas Coupland, *JPod* (London: Bloomsbury, 2007), 58.

office as a “cubicle farm,” and an article from the *Wall Street Journal* describes the office of a small company called NetEnforcers as a “dim, spare cubicle farm.”¹⁶

The news media more commonly refers to a wide expanse of cubicles as a “maze” or “sea” of cubicles. For example, in an article from the *New York Times* from 1996, an author describes the character Dilbert as a “computer nerd trapped in a maze of cubicles.”¹⁷ Similarly, an article from 1999 refers to the tendency of companies to place their lower ranking “clerical and technical employees” in a “maze of cubicles.”¹⁸ While both of these examples treat the “maze of cubicles” as a generic description of a type of office rather than a description of a specific office, by the 2000s the phrase was often used to describe real offices. An article from 2001 in *Texas Monthly* describes a man’s double life as an attorney in a “rat maze of endless cubicles” by day and an avid collector of rare music by night.¹⁹ An article about the “Orthodox Union,” the so-called “Microsoft of kosher certification,” describes the Orthodox Union offices as a “maze of cubicles,” and an article by Barbara Ehrenreich features a quote from an office worker referring

16. Peter Wayner, “Battling Ghost Calls, That Telemarketing Annoyance,” *New York Times* (1923-Current File), November 15 2007, 12, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/848099795?accountid=465>; Joseph Pereira, “Discounters, Monitors Face Battle on Minimum Pricing,” *Wall Street Journal*, December 4 2008, A1, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/399119924?accountid=465>.

17. Geoffrey James, “It’s Time to Free Dilbert,” *New York Times* (1923-Current File), September 1 1996, F11, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/109604354?accountid=465>.

18. John Holusha, “Commercial Real Estate; This Time, Best Views Go to the Workers,” *New York Times* (1923-Current File), October 6 1999, B11, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/109953465?accountid=465>.

19. “The Collector: Corporate Lawyer By Day, Indie Music Mini-mogul By Night,” *Texas Monthly*, June 2001, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/226973865?accountid=465>.

to his own office as a “maze of cubicles.”²⁰ Other articles refer to offices as a “sea of cubicles” for example, a news article from 2002 describes a new Dell office in Twin Falls, Idaho as a “sea of cubicles” set up in a former grocery store.²¹ Another article from 2002 describes an office as a “blue-gray sea of office cubicles” and an article from the *National Review* about outsourcing, describes a “sea of cubicles” in a call center in India which he says “could have been anywhere in the U.S;” the article even includes an image of a room full of cubicles.²²

Like the elevated views of cubicles in visual media which emphasizes the rigid rectilinear arrangement of the cubicle office, the expressions “cubicle farm,” “maze of cubicles” and “sea of cubicles” similarly imply a space dominated by an expanse of cubicles. The metaphors of “farm,” “maze” and “sea” all suggest an expanse of undifferentiated and identical cubicles filling the office space. These terms dehumanize the office by comparing it to: a prison-like pen for animals, a cruel scientific experiment commonly performed on rats, or a disorienting and undifferentiated expanse of open water. In tandem with the visual imagery of the cubicle office, these metaphors all convey a sharply negative image of the cubicle.

20.Frederick Kaufman, “The Secret Ingredient,” *Harper’s Magazine* 310 (2005), 75; Barbara Ehrenreich, “Your Brain At Work,” *The Progressive* 70, no. 7 (2006): 17.

21.Virginia Hutchins, “Dell Opens Twin Falls, Idaho, Technical-support Phone Center,” *McClatchy Tribune Business News*, February 4 2002, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/464577781?accountid=465>.

22.Sharon Linstedt, “Humorous Kit Can Help Make Life in Office Cubicle More Bearable,” *McClatchy Tribune Business News*, May 27 2002, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/464390135?accountid=465>; Daniel Griswold, “Outsource, Outsource, and Outsource Some More,” *The National Review* 56, no. 8 (2004), 37-38.

The Experience of Working in a Cubicle

In order to fully understand the negative image of “cubicleness” it is important to go beyond the physical qualities of the space as characterized by the layout, structure, and color of the space in order to examine the experiential aspects of the cubicle as depicted in popular media including: social isolation, size, noise, lack of privacy, and lack of individuality. These aspects of the cubicle are repeated again and again in film, commercials, comic strips, novels and in news stories and create an image of cubicle that is defined by more than just the physical aspects of design, but also by the perceived frustrations of working in a cubicle as reflected in these various media forms.

Isolation

First of all, the cubicle is often portrayed as socially isolating. In visual media, the elevated view of the “sea” of cubicles, with workers neatly separated by cubicle walls conveys an image of alienation and social detachment. In the 1997 film *Clockwatchers*, overhead views of the cubicle office throughout the film emphasize the image of social isolation. In an early scene, all of the office workers are depicted in an overhead shot sitting quietly in their cubicles waiting for the day to be over (literally watching the clock). In the image, no one appears to be talking or socializing at the end of the day—instead they are all sitting quietly just waiting to be able to leave. In the 1999 film *Fight Club*, an early scene shows an office space filled with cubicles, and a central area with half a dozen copiers, and a worker in front of each one, each dressed in drab office clothing, drinking coffee out of a disposable coffee cup, and staring into space. The

zombie-like quality of the office workers staring into space reinforces the image of isolation. Similarly, in a cartoon from *The New Yorker* magazine from June 6, 2005, office workers are shown in a sparsely decorated cubicle landscape (depicted in an elevated view) with a sign on the wall that reads “40 days without an on-the-job conversation.”²³ The cubicle landscape reflects the image of solitude and isolation, with each worker boxed off and working independently, seemingly disconnected from the sea of workers around them. In textual forms, the cubicle is sometimes treated as a symbol of social isolation and even depression. For example, in a story by David Foster Wallace published in *Harper’s Magazine* about the experience of depression, the main character is described in the story sitting in her cubicle actively struggling with her depression.²⁴ In this context, the cubicle is used to convey the character’s sense of isolation and hopelessness—as if the cubicle were a spatial manifestation of the depression itself.

Even news stories sometimes treat the cubicle as a symbol of social isolation in the office. In February of 2011, a woman died in an office cubicle and was not found until the next day. The tragic story about a middle-aged woman dying unexpectedly at work was overshadowed by the sinister idea of a dead woman in an office cubicle unnoticed for almost a day. Even the early wire report about the woman’s death, emphasized the cubicle location not only in the body of the article, but also in the title of the report which read “L.A. County Employee Dead in Cubicle a Day Before Being Noticed.”²⁵ In the news media and blogosphere, the mental image of the

23. Matthew Diffie, "Office cubicles with a sign on the wall that reads '40 Days Without an On-the Job Conversation,'" *The New Yorker*, June 6, 2005, <http://cartoonbank.com>.

24. David Foster Wallace, “The Depressed Person,” *Harper’s Magazine* (1998), 63.

25. “LA County Employee Dead in Cubicle a Day Before Being Noticed,” *McClatchy Tribune Business News*, February 15 2011, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/851727899?accountid=465>.

woman dead in her cubicle was portrayed as a rich metaphor for the isolating experience of white collar work, although it was later revealed that she did not die in her own workspace, or even her own office, but a vacant cubicle in an office where she was working temporarily on assignment. In one article from villagevoice.com titled, “Cubicles Will Kill All of Us Eventually,” the story of the woman’s death is described as a symptom of the dystopian alienation and isolation in the modern workplace in which the sad death of a colleague could go unnoticed for an entire day.²⁶ The blog post even included a small image of an office filled with workers in cubicles as an illustration of the dreary quality of the cubicle landscape. Another article about the woman’s death in the *Chicago Tribune* describes the author’s personal horror at the news of a woman dying unnoticed in a cubicle. The author asks the reader to imagine themselves dying at their desk, “what will your eyes seize upon on that desk where you spend so much time?,” and the final paragraph informs the reader that death could come “at your desk, alone, in a vast sea of empty cubicles, to be found by a security guard the next afternoon.”²⁷ In this article, the author connects directly to the reader who is assumed to be a fellow white-collar worker also working away in a cubicle. The cubicle office is thus treated as simultaneously a symptom and a symbol of the alienation and isolation of the modern office.

26. Jen Doll, “Cubicles Will Kill All of US Eventually,” http://blogs.villagevoice.com/runninscared/2011/02/cubicles_will_k.php (accessed July 26, 2011).

27. John Kass, “Death in a Cubicle Puts Focus on Haunting Thoughts,” *Chicago Tribune*, February 27 2011, http://articles.chicagotribune.com/2011-02-27/news/ct-met-kass-0227-20110227_1_open-eyes-cubicle-young-reporter.

Small and Cramped

In addition to being socially isolating, the interior space of the individual cubicle is often portrayed as uncomfortably small. In the 1999 movie *Office Space*, Peter Gibbons' cubicle is shown in a tight birds-eye shot emphasizing the narrow confines of the space as Peter enters his cubicle and sits down to work at the start of the film. In *The Incredibles* from 2004, Mr. Incredible is depicted working in a cubicle workspace that is partially taken up by a giant column in his already small workspace. In the comic strip *Dilbert* the tiny size of the cubicle is a recurring joke that is often portrayed as an often cruel and humiliating loss of status and importance in the workplace. In one strip from April 8, 1995, the Pointy-Haired Boss tells Dilbert that they have installed status adjusters in the cubicle walls so that the size of the cubicle adjusts to reflect the worker's real value. In the final frame of the strip, Dilbert and his colleagues are shown crammed into tiny box-like cubicles. In a strip from September 15, 1996, the evil HR director, Catbert, gives all of the workers "head cubicles" that are essentially boxes placed on top of their heads. Other *Dilbert* strips talk about workers being forced to share their cubicles. A pair of strips from July 31st and August 1st of 1997 depicts Wally being forced to share a cubicle with the copy machine, and Alice arranging an "accident" for her new cubicle mate. All of these strips emphasize the size of the cubicle as a reflection of status and a on-going source of humiliation by being crammed into increasingly smaller spaces and frustration at being forced to share their workspace. Some cartoons in *The New Yorker* depict similar themes. One cartoon from December 18, 2006 shows a man with a box on his head, with a caption that reads

“Cubicles-the next generation.”²⁸ A *New Yorker* cartoon from August 11, 2003 shows cubicles with several people crammed into them, and has the sub-title “cubicle pooling.”²⁹

The image of the cubicle as uncomfortably small was echoed in news reports about “shrinking cubicles” that have run regularly since the late 1990s. For example, a National Public Radio story from August 1, 1997 actually used the *Dilbert* strip mentioned above (about cubicle sharing) as the opening for a story about the decreasing space in offices, and an article in the *Chicago Tribune* from 2002 similarly reported that cubicles were “shrinking” in size.³⁰ In 2004, a news story about diminishing office size began with the line: “Ever feel like your cubicle or office walls are closing in on you. They are.” The article goes on to describe the steady decrease in office size over the late 1990s.³¹ Another article from 2007 led with the provocative title, “Nearly Half of Americans Indicate their Bathroom is Larger Than their Office Cubicle” which described how Americans are feeling “more cramped at work than ever before.”³² Even as recently as 2011, two separate stories appeared about the diminishing size of office cubicles, one

28.Kim Warp, "Cubicles-the next generation," *The New Yorker*, December 18, 2006, <http://cartoonbank.com>.

29.Jack Ziegler, "Offices with three workers in each cubicle," *The New Yorker*, August 11, 2003, <http://cartoonbank.com>.

30.David Molpus, “Employees Say No to Cubicles,” *All Things Considered* (1997); David Greising, “Chicago Tribune David Greising Column,” *Chicago Tribune*, June 12 2002, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/464337523?accountid=465>.

31.Matt Phillips, “US Workers Adapt to Progressively Shrinking Cubicles,” *McClatchy Tribune Business News*, July 11 2004, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/463986006?accountid=465>.

32.“Nearly Half of Americans’ Indicate Their Bathroom is Larger Than Their Office Cubicle,” *PR Newswire*, July 17 2007, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/450960531?accountid=465>.

in the magazine *Fast Company* and another on National Public Radio.³³ All of these stories use the shrinking cubicle to open a discussion of the decreasing status and resources of white-collar workers.

Noisy

Cubicles are also consistently portrayed as noisy, and the average office worker is often depicted as being continuously assaulted with a cacophony of sounds. The open quality of the cubicle with its low walls and its lack of a door symbolize the workers lack of control over the continuous noise of the office. For example, in the opening sequence of *Office Space*, Peter Gibbons arrives at his cubicle and sits down at his desk, and hears the noises of all of the annoying people around him: a woman repeating the same telephone greeting in the same cadence over and over, and the sound of his colleague's radio in a neighboring cubicle. Peter's frustration with the noises is visually apparent by his facial expressions which convey his irritation, while the image of his open cubicle walls with the sounds of his co-workers bleeding into his workspace symbolize his lack of control. In several television commercials from a series of advertisements from 2005-2006 for careerbuilder.com, in which a large group of chimpanzees play obnoxious coworkers (fully dressed in suits and ties) to a single human in a cubicle office, the noises of colleagues are an on-going joke. In one ad, a chimpanzee is shown standing just outside the human's cubicle popping bubble wrap while the human worker tries to politely ask

33. Rachel Arndt, "Redesigning: Cubicles," *Fast Company* 157 (2011): 48. "The Incredible Shrinking Cubicle," *Weekend Edition*, February 13 2011, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/851425492/130E72F858B5E306FC4/1?accountid=465>.

the chimpanzee to take his bubble wrap popping elsewhere. Another ad from the same series shows the chimpanzees engaged in a raucous party in the conference room, while the human again tries to work.

In the comic strip *Dilbert*, the issue of noise from neighboring cubicles is another recurring theme. In a somewhat gruesome strip from August 7, 1994, Dilbert is shown giving a new engineer a tour of the office. In it he describes the cubicle as an “anti-productivity pod” in which “the top is open so none of the background noise is inadvertently muffled,” and “you’re on the main aisle so you will be haunted by footsteps behind you.” In the final frame, the Pointy-Haired Boss is pulling Dilbert aside, while in the background the woman is tying a noose above the cubicle presumably to hang herself. The dark humor of this particular strip emphasizes the perceived frustrations of working in a space so vulnerable to the noises of others. In another strip from November 1, 1996, Alice is depicted sitting in her cubicle listening to the sound of her neighbor Wally clipping his nails. Driven slowly mad by the sound, she tries to control her frustration, while in the final frame, Wally is shown talking to Dilbert and revealing that the sound is actually just a screen saver titled “Wally’s 101 Annoying Cubicle Sounds.” Alice’s frustration about noise outside her cubicle is an on-going source of humor in *Dilbert*; a strip from April 3, 1997 depicts Alice talking to two people engaged in conversation outside of her cubicle and tells one of the offenders, “if you don’t go away, I’ll pound your inconsiderate head so far into your torso that you have to drop your pants to say hello.” The next frame shows the man with the top of his head barely visible inside his trousers, and Wally and Dilbert asking each other about the “strange noise.” Another strip from July 20, 1998 featuring Alice shows her confronting two men having a loud conversation outside her cubicle; she says “I think I speak for

a lot of cubicle dwellers when I say...SHUT UP.” This strip not only conveys the image of Alice being frustrated, but seems to be written for any worker similarly frustrated with their own colleagues—a perfectly post-able comic strip to discourage people from standing outside one’s own workstation talking.

In the news media, the cubicle is often portrayed as a maddeningly noisy place filled with the irritating noises of colleagues and general office noise, making it impossible to concentrate. For example, an article titled “Shhhh” in *HR Magazine* from 1997 cited an American Society of Interior Designers (ASID) study that “verified what most cubicle dwellers will tell you: noise reduction is a major concern to office employees.”³⁴ A news story from 2000 begins with a woman grouching about a colleague who had the annoying habit of using his nail clippers in the neighboring cubicle, making it impossible for her to concentrate.³⁵ An article from 2004 which documented common complaints of “cubicle life,” stated that the most frequent complaint was the noise particularly people talking and the sounds of ringing cell phones, and an article from 2011 similarly stated that noise was one of the most egregious problems of the modern cubicle.³⁶

34.Elaine McShulskis, “Shhhh,” *HR Magazine* 42, no. 2 (1997): 21-22.

35.Nancy Rubin Stuart, “Basic Principals of Civility Cubed,” *New York Times (1923-Current File)*, October 4 2000, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/259343317?accountid=465>.

36.Kim Anderson, “Fredericksburg, VA-area Workers Share Some Common Office Complaints,” *McClatchy Tribune Business News*, July 17 2004, 1, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/463983164?accountid=465>; Charlotte Tucker, “Closed Off,” *McClatchy Tribune Business News*, May 20 2007, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/459321797?accountid=465>.

Just like their fictional counterparts, white collar workers are portrayed in the news media as frazzled people bombarded with the annoying sounds of colleagues, water coolers, printers, and other office irritants.

Lack of Privacy

In tandem with the irritating noises emitted from neighboring cubicles, is the lack of privacy. Obviously acoustic privacy is a central problem—just as the flimsy panel walls allow everyone to hear everyone else’s noises and conversations, they also prevent an individual from having a private conversation. The low walls of the cubicles additionally prevent visual privacy, with workers often shown peeking over the tops and sides of panels, and bosses casually peering into worker’s cubicles. For instance, in the 1998 film *The Truman Show*, the protagonist, who unknowingly lives in an entirely artificial world of a reality television show revolving entirely around him, is depicted early in the film sitting in his office cubicle trying to make a personal phone call to directory assistance in Fiji. As he tries to make the call, the coworkers are shown peering over the top and side of the panel walls clearly eavesdropping and trying to distract Truman and draw the attention of the camera. In the scene, Truman is forced to fake a loud conversation with an imaginary customer to get them to go away, and then softens his voice to almost a whisper to talk to directory assistance. In the scene, Truman even lowers his body almost below the desk to avoid being overheard in attempt to visually and acoustically “hide” in the office despite the exposure of the cubicle layout. The image of a worker trying to avoid being seen or heard in the office is a recurring image in other films as well. In the 1999 film *The Matrix*, an early scene shows the protagonist in his office cubicle trying to avoid being seen or

heard as he attempts to escape being caught by the authorities. Like Truman, Neo is shown speaking very softly into a phone, and at one point even ducking his whole body below the cubicle wall in order to avoid being seen. Similarly, the film *Office Space* shows Peter trying to avoid being asked to come in on the weekend by trying to slip out of the office without encountering the boss. In the scene, Peter is shown hunched over in his cubicle, and frantically peering over the top of his cubicle wall desperate to avoid being seen.

In comic strips and cartoons, workers are often shown peeking over cubicle walls at one another. An early *Dilbert* strip from July 24, 1992 draws a visual comparison between cubicle workers peering over their panel walls and prairie dogs. Similarly, a cartoon from February 8, 1999 in *The New Yorker* shows workers in their cubicles peering over and around their panel walls watching each other, and a cartoon from January 21, 2008 also from *The New Yorker* shows a worker looking over his panel wall at a group of meerkats in the neighboring cubicle looking back at him.³⁷ Another *New Yorker* cartoon from January 17, 2011 shows a worker leaning on a cubicle wall looking in on his neighbor, and the other worker saying: "The entrance to my cubicle is three feet away. Please respect my partition."³⁸ These images all convey both the nosiness of cubicle neighbors who are eavesdropping and watching each other, but also the lack of privacy inherent to the cubicle itself.

37. Tom Cheney, "Office workers peering over their cubicles at each other," *The New Yorker*, February 8, 1999, <http://cartoonbank.com>; Christopher Weyant "A man in his cubicle with meerkats next to him," *The New Yorker*, January 21, 2008, <http://cartoonbank.com>.

38. William Haefeli, "The entrance to my cubicle is three feet away. Please respect my partition." *The New Yorker*, January 17, 2011, <http://cartoonbank.com>.

The lack of privacy in the cubicle is also conveyed in textual forms. In the 2007 book *Then We Came to an End*, one of the characters, Benny, is described sitting in a cubicle in their new job being annoyed by his eavesdropping neighbor. In the novel, he is sitting in his cubicle muttering to himself, when ““his cubicle neighbor, a lank annoying person who never let anything go by without making inquiries, popped his sandy groundhog’s head over the cube wall and said, ‘What’d you say?’” Later, Benny is talking on the phone, and is described “peering over the cube wall” just to make sure his eavesdropping colleague is gone before speaking.³⁹ Like the cartoons that depict workers as meerkats or prairie dogs peering over the panel walls of a cubicle, the novel compares the worker in their cubicle to a groundhog, peeking out of his hole. In the novel *Company Man* by Joseph Finder, the protagonist works in a “cube farm” and sells cubicle furniture. In one scene, he complains that “there really wasn’t any privacy anymore. Everyone knew where you were, when you went out for lunch or to work out, who you were meeting with. If you yelled at someone on the phone, everyone heard it.”⁴⁰ Again, the image of the cubicle is one in which no one has any kind of privacy—not only do people hear your conversations, but they also know your patterns and routines, they know where you are and where you have been.

In the news media too, the lack of privacy in the cubicle is a recurring theme. For example, an article from the *New York Times* from 1997 describes in a slightly ironic tone how cubicles are “great for overhearing gossip fodder. It’s the 90s: no intimate chat on company

39.Joshua Ferris, *Then We Came to the End* (New York: Little, Brown and Company, 2007), 358-59.

40.Finder, *Company Man*, 92.

time.”⁴¹ An article from the *Wall Street Journal* describes in graphic detail the lack of personal privacy in the cubicle office and the creative ways in which workers avoid being overheard: swiveling the office chair to the floor and hiding the receiver under the desk, hiding out in the computer server room, commandeering a boss’s private office, and speaking in code. The article even describes how corporations have tried to give their workers more privacy for example at Alcoa where workers are told to ““respect a person’s workstation as a private place;”” and at PNC bank, workers are told to “refrain from ‘loud talking over the top of your workstation.’” Proctor & Gamble reportedly provides so-called “huddle rooms” where employees can talk freely on company time.⁴² Another news story about a humorous kit for office workers recounts the personal experience of a medical claims processor whose annoying cubicle neighbor would “pop his head over the [cubicle] wall and start jabbering.”⁴³ The visual and acoustic transparency of the cubicle is thus portrayed as a continuous source of frustration for workers who reportedly feel exposed and irritated by the intrusions of their colleagues and the inability to maintain their sense of privacy.

41. Steve Lohr, “Rethinking the Contemporary Workplace: Hey Who Took the Office Doors?,” *New York Times (1923-Current File)*, August 11 1997, 1, 7, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/109774322?accountid=465>.

42. Gwendolyn Bounds, “I Can’t Really Talk, I’m Here in the Office. But Did You Know?,” *Wall Street Journal*, July 10 2002, B1, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/398917315?accountid=465>.

43. Linstedt, “Humorous Kit Can Help Make Life in Office Cubicle More Bearable.”

Loss of Individuality

The cubicle is also used to symbolize a lack of individuality both for the individual worker, as well as for the corporation whose sea of drab cubicles is often depicted as a generic and nondescript corporate environment. In a number of films, the cubicle workspace is depicted as entirely barren of personal decoration. For example in the films *Schizopolis*, *The Matrix*, *Fight Club*, *Office Space*, and *The Incredibles* the protagonist is shown working in their own cubicle which has no personal objects of any kind. In all of these films the undecorated walls of the cubicle seem to suggest both the disconnection of the individual from their job, as well as a sense of anonymity. In other films, the image of the decorated cubicle suggests an individual's effort to stand out despite the pressure to conform. In *Very Bad Things*, Kyle and Michael are shown at the beginning of the film working in a grey cubicle landscape, in neighboring cubicles which are both covered in personal objects (toys, pictures, signs, etc), and reflect the characters' efforts to stand out from the crowd. Similarly, in *American Beauty*, Lester Burnham's cubicle features a few personal items tacked to his cubicle walls, including a sign that reads "look closer." Around him, the other cubicles are very spare and undecorated, which further emphasizes Lester's efforts to assert his individuality despite the dreary and anonymous image of the office.

In *Dilbert*, the issue of individuality in the workplace is another recurring theme. In a strip from January 17, 1994, Dilbert is depicted requesting permission from the "cubicle gestapo" to keep a plastic plant in his cubicle. In a strip from September 28, 1994, the Pointy-Haired Boss tells Dilbert he is not allowed to post cartoons on his cubicle wall because it "hurts morale." A strip from October 28, 2009 shows Alice at her desk surrounded by desk toys, and being told by the Pointy-Haired Boss that she must remove the toys because "it sends the wrong message."

Alice responds saying, “you mean the message that says this company is a creative environment that values the individual?” The Pointy-Haired boss responds saying “Exactly. We’re going for more of an egg carton vibe.” In this way, the lack of individuality is attributed to the culture of the company (as embodied by the Pointy-Haired Boss) which has strict rules about personal decoration in the cubicle. In the news media, stories about personalized office decoration reveal that while some offices allowed workers the freedom to express their individuality in their workspace, many others were very strict about cubicle decorations. A *New York Times* article from 2006 described a Steelcase survey that found a decrease in the number of companies encouraging personalization over the previous decade. The article goes on to describe both extremes, companies that allow personalization (American Greeting), and companies (like Calvin Klein) that had strict rules about decor prohibiting most forms of personalization (photographs, mementos, toys, awards, etc).⁴⁴

Perhaps nothing embodies the image of an impersonal cubicle office better than a new office trend that began in the early 1990s called “hoteling” in which workers were no longer assigned permanent workspaces, but instead would “check-in” each day and be given a workspace on a first-come, first-serve basis. This new office concept was lampooned in *Dilbert* in a series of strips from January 9-10, 1995 in which Dilbert and his colleagues no longer have an assigned cubicle, but were just assigned an available workspace each day. In one of the strips, Wally is depicted pushing a shopping cart in the office (an image of homelessness) and saying that he has “developed a strong interest in graffiti as a way to express my individuality.” In the

44.Elizabeth Olson, “So You Want Ivy Around Your Desk?,” *New York Times* (1923-Current File), November 19 2006, I12, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/93036158?accountid=465>.

news media hoteling was sometimes portrayed as an efficient cost saving office design strategy, and other times as a dreary impersonal office setting.⁴⁵ An article from 1994 titled “Laptops Yield White Collar Nomads” describes how workers in “hotelings” offices “feel evicted and homeless.”⁴⁶ Another article from 1994 about IBM describes the implementation of this new office environment in which each worker was assigned to an “anonymous, empty cubicle”... “with no walls and no adornments.”⁴⁷ The article includes an image of a worker foregrounded in a cubicle, amidst a sea of undecorated cubicles. Not only is the individual cubicle portrayed as anonymous and undecorated, but so too is the entire space of the office. Although an article about IBM, the image has no identifying features or logos visible—this is a “non-place” that has no distinctive visual identity, and even prevents individual workers from claiming and personalizing the space. According to anthropologist Marc Augé, author of *Non-Places: Introduction to an Anthropology of Supermodernity*, “place” is “formed by individual identities, through complicities of language, local references, the unformulated rules of living know-how;” “non-place” strips away individual identities creating a sense of anonymity—the user is “alone,

45. Andree Brooks, “Making Offices Compatible With Flexible Schedules,” *New York Times (1923-Current File)*, October 4 1992, 418, <http://search.proquest.com.ezproxy.library.wisc.edu/hnpnewyorktimes/docview/108873253/13D9F3677077784E575/1?accountid=465>.

46. Lisa Cenasci, “Laptops Yield White Collar Nomads,” *Journal Record*, March 9 1994, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/259302190?accountid=465>.

47. Kirk Johnson, “In New Jersey, IBM Cuts Space, Frills and Private Desks,” *New York Times (1923-Current File)*, March 15 1994, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/259300626?accountid=465>, 1.

but one of many.”⁴⁸ In other words, while places help to create a sense of identity and individuality, non-places strip away individuality.

The representation of the cubicle office as a “soul-sucking pit of conformity and monotony” is repeated again and again in these various media forms.⁴⁹ The dreary image of the anonymous white-collar worker in their cubicle is further emphasized by the term “cubicle-dweller,” a phrase that has become increasingly common in the news media since the 1990s. For example, an article in *The Wall Street Journal* from 1998 is titled “Downtrodden Cubicle-Dwellers Now Can Snicker at DILBX index,” cheekily refers to white collar workers following an index created by Scott Adams that tracks companies Dilbert, the comic strip character, would most like to work in, and an article from 1998 refers to Scott Adams as a “spokesman for downtrodden cubicle-dwellers everywhere.”⁵⁰ In both of these examples the term is employed in a light-hearted manner in order to match the cynical humor associated with the *Dilbert* comic strip, but the correlation of “downtrodden” and “cubicle dweller” together reinforces a dark and unhappy image of cubicles and white-collar work in general. The phrase “cubicle dweller” recurs again and again, for example two articles from the *Wall Street Journal* use the phrase very casually to describe office workers who find ways to nap in the office and have personal

48. Marc Augé, *Non-places: Introduction to an Anthropology of Supermodernity*, trans. John Howe (New York: Verso, 2000), 101.

49. Kiera Butler, “Works Well With Others,” *Mother Jones* 33, no. 1 (2008), 66.

50. Deborah Lose, “Downtrodden Cubicle-dwellers Now Can Snicker At Dilbx Index,” *Wall Street Journal*, February 10 1998, B1, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/398618661?accountid=465>; Leslie Kaufman, and John McCormick, “Year of the Employee,” *Newsweek* 132, no. 3 (1998): 38-41.

conversations despite the transparency of the cubicle walls.⁵¹ A *New York Times* article from 2006 uses the phrase to discuss the use of office email to banter with colleagues and share off-color humor.⁵² In all of these examples, the phrase is used to generalize very broadly about the experience and context of white-collar work.

By correlating white-collar workers to “cave-dwellers,” the phrase implies that the cubicle itself is archaic and old fashioned, but also that the experience of white-collar work itself is in some way antiquated as well. Although the phrase is often used playfully, it is clearly a strongly negative term that is often coupled with the word “downtrodden” suggesting that white-collar workers are an oppressed group. In this way, the cubicle is more than just a generic backdrop of modern office work, it has become a defining feature of the experience and culture of contemporary white-collar work. The repetition of the negative image and experiences associated with the cubicle in such a large number of different media objects, fictional and non-fictional, visual and textual, reveal a pattern of representation in the United States. Thus, “cubicleness” is not just about the image and experience of the cubicle as a type of workplace environment, but about the larger image of white-collar work it has come to represent.

51. Jared Sandberg, “As Bosses Power Nap, Cubicle Dwellers Doze Under Clever Disguise,” *Wall Street Journal*, July 23 2003, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/398899499?accountid=465>; Bounds, “I Can’t Really Talk, I’m Here in the Office. But Did You Know?”

52. Lorne Manly, “Before You Hit Send, Pause, Reflect,” *New York Times (1923-Current File)*, August 9 2006, C14, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/93053968?accountid=465>.

“Cubicleness” and the Negative Image of White-Collar Work

In media objects, the cubicle is often employed as a symbol and symptom of the culture of contemporary white-collar work. By situating the image of the cubicle within the context of the narrative in media objects we can begin to unpack the relationship between the cubicle and the image of office work. The image of the monotonous “sea of cubicles” in films, comic strips, commercials and news stories has become a stand-in for the bureaucracy and boredom of modern office work. Comparing the cubicle to a kind of prison, many of these media objects portray office work (and the office itself) as something to escape. Examining several specific examples derived from different media source types, this section will explore the ways in which the negative image of the cubicle has become connected to a negative image of white-collar work.

Bureaucracy

First of all, the image of the expanse of office cubicles in both visual and textual media has become strongly associated with an image of corporate bureaucracy characterized by: a corporate culture filled with arcane rules, processes and language; and a lack of interest or concern for the well-being of employees, clients, and customers. The cubicle is more than just the dreary backdrop of corporate bureaucracy, it has become the spatial embodiment of it.

In the 2004 Pixar animation *The Incredibles*, the Insuracare corporation where Mr. Incredible works is portrayed as a dull bureaucracy both in terms of its almost impenetrable bureaucratic structure and its utter lack of concern for the well-being of its customers. The cubicle office where Mr. Incredible works visually conveys the idea of an endless and heartless

bureaucracy, with its vast sea of identical grey cubicles. Throughout the first part of the film, Mr. Incredible is shown taking pity on his clients by giving them the means to get through the obtuse bureaucratic red-tape in order to receive coverage. Indeed, in the very first scene in his office, Mr. Incredible is depicted giving an elderly woman specific instructions on receiving coverage including talking to a specific woman on another floor and filling out a “WS-2475” form with the legal department. The quickly whispered instructions convey a sense of the enormous size of the organization with its multiple floors of offices and its thousands of internal forms and processes. Later in the film, Mr. Incredible’s boss, voiced by a character actor in a snide nasal tone, upbraids Mr. Incredible saying that his clients appear to have “inexplicable” knowledge about the inner workings of the corporation: “exploiting” loop holes, “dodging” obstacles, and “penetrating the bureaucracy.” The corporation is thus both visually and textually portrayed as a labyrinthine organization designed to ensure that the company remains profitable at the expense of the well-being of its supposedly insured customers. At one point the boss tells Mr. Incredible that the real people who need help are their stockholders who need to see a profit.

The room full of cubicles conveys a sense of the enormous size of the organization, while the drab grey palette and rigid rectilinear arrangement of the office emphasizes a cold and uncaring corporate culture (despite its altruistic name). In an animation in which the office could have had any design (not at all bound to the limits of a production budget or a camera as in a live action film), it is significant that the office is portrayed as a cubicle farm. In fact, with mid-century Modern style of the furniture and architecture throughout the rest of the film, the office could just as easily been portrayed as an old-fashioned bullpen. The use of a literal cubicle maze in this setting was clearly an intentional decision to convey a familiar image of a dreary

corporate office that would be legible to an adult audience, as well as the film's primary audience: children. While Mr. Incredible is portrayed as a (former) superhero with a big heart who just wants to help people, Insuracare is portrayed as a heartless and monolithic corporation.

The use of cubicles as a symbol of corporate miserliness is also conveyed in a series of commercials for Capital One credit card that were run in the United States in the early and mid-2000s. The advertisements featuring the actor and comedian David Spade use a nondescript cubicle farm as the offices of "another credit card company." In the series of commercials, David Spade plays a call center worker who takes great pleasure in saying "no" to customers wanting to redeem their reward points. In various iterations of the commercial David Spade is depicted in the sterile cubicle office: saying no repeatedly in different ways; teaching new workers to say no; and attempting to catch his colleagues who are saying 'yes.' Throughout the series of commercials, the cubicle office with its dreary grey panel walls and sparse decor (mostly featuring posters that say "no") seems to reflect the cruel corporate culture embodied by the cynical and often snarky humor of David Spade. The image of the cubicle offices with its flimsy grey walls (that are repeatedly knocked down or destroyed through the antics of Mr. Spade) visually signifies the mean and stingy bureaucracy of the "other" credit card company.

The satirical novel *Company* by Max Barry similarly portrays a "sprawling cubicle farm" as the setting of the absurdly bureaucratic Zephyr Holdings, an imaginary company in which even the workers do not know what the company does.⁵³ Throughout the book, the corporation is depicted as an enormous and complicated labyrinth; indeed the protagonist eventually discovers

53. Barry, *Company*, 13-14.

that the company has no external customers at all, but is just a very large social experiment in which the different divisions of the company sell their individual services to other internal divisions. The absurd rules and behaviors of the corporation are revealed to be intentionally ridiculous in order to test the limits of people's willingness to continue to participate in blatantly inane activities. One "project" involves a group of employees being set a "task of critical importance" in which the fundamental goals of the task change in "minor but significant ways" every few hours forcing the group to keep working.⁵⁴ The absurdities of the corporate bureaucracy are not only played out within a cubicle office, they even use the cubicle to create new absurd situations; at one point in the novel the training sales department which had formerly occupied an entire floor is squeezed into a single cubicle on a different floor.⁵⁵ The novel thus treats the cubicle office as both a symbol and symptom of the ridiculous bureaucracy of a corporation.

Boredom and the Tyranny of Tedium

In addition to visually conveying bureaucracy, the cubicle has also become associated with tedious and mind-numbing work and general boredom among white-collar workers. In the 1997 film *Clockwatchers*, the four main characters are temporary workers at a credit company called GCA. From the very beginning of the film, the corporate office with its drab beige cubicles and its drab undecorated office interiors is portrayed as a space of excruciating boredom for everyone in the office. In an early scene, one of the temp workers initiating the newest temp, Iris, into the

54. Barry, *Company*, 121.

55. Barry, *Company*, 194-99.

ways of the office describes how the hardest part of the job is “trying to look busy when there is nothing to do.” This sense of tedious and mind-numbing boredom in the office pervades the film along with the metronomic ticking of a clock that resurfaces again and again in the film often mixed with the sounds of jazzy muzak that plays continuously over the intercom system. While the characters are occasionally shown engaged in various forms of “typical” office work such as: typing forms and memos, making copies, organizing contacts in a rolodex, and stamping envelopes; they are more often depicted sitting in their cubicles not working or pretending to work. In one montage nearly half way through the film, various workers are shown “not working” in different ways: listening to a sea shell; playing “pick-up-sticks” with office markers; painting fingernails with white-out; spinning idly in an office chair; sniffing permanent markers; carving on a desk; and even throwing a paper airplane. An overhead shot of the office reveals everyone engaged in some form of non-work, while the incidental music prominently featuring an accordion seems to emphasize the excruciatingly slow movement of time in the office. In the voice over, Iris, the protagonist, describes how they were all “just killing time,” “waiting for something to happen.” The overhead view of the office cubicles with their beige walls emphasizes the dreary and slow-moving pace of the entire office where nothing ever seems to happen. Indeed, the cubicles are portrayed as a spatial manifestation of perpetual boredom.

Similarly a commercial for Emerald Nuts depicts the modern cubicle office as a dull and mind-numbing place to work. The commercial which premiered during the 2007 Super Bowl, shows white-collar workers literally asleep at their desks. The commercial features singer and actor Robert Goulet, as an imaginary workplace figure that appears in the late afternoon when everyone in the office is asleep whose sole purpose is to make trouble and create chaos in the

office. Goulet is shown dancing through the office: shredding papers; eating phone messages; tap dancing on desks; taping workers to their chairs; pouring coffee on computers; and generally wreaking various forms of havoc. In the commercial, the image of workers literally asleep in their cubicles reinforces the image of white-collar work as boring. The image of workers sleeping at their desks perpetuates the idea that white-collar work itself is tedious. Indeed the one worker who is awake (thanks to a handful of Emerald Nuts) is standing in a copy room not really doing anything. The humor of the commercial relies on the dreariness of the office space (with its drab undecorated cubicle walls), and the sleeping workers to convey an image of dreary white-collar boredom.

The 2007 novel, *Then We Came to the End* by Joshua Ferris similarly depicts office work as excruciatingly dull. Early in the book, the narrator describes the tedious world of advertising, asking the reader “Is this boring you yet? It bored us every day. Our boredom was ongoing, a collective boredom, and it would never die because we would never die.”⁵⁶ Thus from the very first pages of the novel, work (even the supposedly exciting world of advertising) is portrayed as mind-numbingly boring. Addressing the reader directly, the narrator presumes a shared concept of boredom and a sympathetic reader who understands the intrinsic boredom of white-collar work. In this way, the book immediately establishes this image of pervasive boredom, and of

56.Ferris, *Then We Came to the End*, 4.

workers with nothing to do. Indeed the narrator describes the workers as “killing time” in different ways:

we wandered the hallways carrying papers that indicated some mission of business when in reality we were in search of free candy. We refilled our coffee mugs on floors we didn’t belong on. Hank Neary was an avid reader. He arrived early in his brown corduroy coat with a book taken from the library, copied all its pages on the Xerox machine, and sat at his desk reading what looked to passerby like the honest pages of business.⁵⁷

The characters in the novel are thus depicted actively engaged in avoiding work, even as they try to appear busy in order to be able to “pad” their time sheets with work and keep their jobs. The tedious quality of the work is reinforced by the image of the office itself with its “flimsy” panel walls or “particleboard wrapped in cheap orange or beige fiber.”⁵⁸ Indeed, at one point a worker desperate to be able to think goes to a vacant floor (fifty-nine) filled with cubicles (a result of the recent firings) and sits in an “anonymous cubicle.” In the scene the narrator describes the room humming with “a canny stillness”:

as if the blank page had followed him and morphed through a miracle of physics into a pure sound. All fifty-nine was a blank page, separated by cubicle partitions. The floor’s lonesome eeriness surrounded him in its silence and blankness like an all-consuming void, and once he was sucked in, he would lose not only his job, but his mind as well.⁵⁹

The sea of vacant cubicles is thus portrayed as spatial manifestation of boredom and distraction. Even on this floor devoid of people, the character is unable to concentrate or focus, and finds his

57.Ferris, *Then We Came to the End*, 28.

58.Ferris, *Then We Came to the End*, 62.

59.Ferris, *Then We Came to the End*, 265.

mind wandering to other matters “like what he would have for lunch.”⁶⁰ The cubicle is thus portrayed as both the source and the symbol of the character’s sense of detachment from his job and general ennui.

Fantasies of Escape

Underneath this image of white-collar work as bureaucratic and boring, is the idea that office work is something to escape. Again and again, the office cubicle is depicted as a dreary “prison,” and the office worker is portrayed as a character trapped in their tedious white-collar job desperate to escape. Since the early-1990s, a number of films have featured main characters who are “trapped” in their dead-end white-collar jobs. In many of these films the cubicle is depicted as an office prison that the main character must escape in order to find personal fulfillment and happiness. For example, in the 1999 film *The Matrix*, the nondescript cubicle office where Neo works at the beginning with its fluorescent lighting and drab panel walls visually conveys the idea that Neo is trapped. The cubicle office in the film is a spatial representation of the oppressive control of a computer simulated reality that Neo must escape to find his true destiny as the savior of mankind in a dystopian reality. In the early office scene in the film, Neo is depicted literally trying to escape the office, hiding in cubicles and running between panel walls in order to avoid being captured by the authorities.

In the film *American Beauty* also from 1999, the cubicle office is again depicted as a metaphoric prison for the main character, Lester, who is portrayed as a bored and depressed

60. Ferris, *Then We Came to the End*, 265.

white-collar worker. An early image of Lester in his cubicle shows his face reflected in a computer monitor with vertical lines of text on the monitor, which according to director Sam Mendes was intentionally constructed to evoke the bars of a jail cell.⁶¹ In the film, Lester is thus portrayed as a trapped white-collar worker, unhappy in his job and his marriage, and desperate for escape. These themes of white-collar imprisonment and escape are repeated in *Clockwatchers*, *Jerry Maguire*, *Office Space*, *Being John Malkovich*, *Fight Club*, and *The Incredibles*. In all of these films, the protagonist must escape their dull dead-end white-collar job in order to find personal happiness and fulfillment. As fantasies of escape, it is significant that so many of these films use the dreary cubicle to convey the image of an ordinary character trapped in the present circumstances and desperate to break away.

This theme of escape is also echoed in other media forms. For instance, a 2009 commercial for the website careerbuilder.com uses a repetitive cumulative structure in which various indicators of needing a new job are progressively added on top of one another ending with: “if you hate going to work, your co-workers don’t respect you, you wish you were somewhere else, you cry constantly, you dream of punching small animals, and you work with this guy, it’s probably time.” Each phrase is associated with a distinctive image: a woman in her car screaming, a man walking by a cubicle saying “Hey, dummy,” a person riding a dolphin, a man crying at a bus stop, a stuffed koala wearing glasses getting punched in the face, and a coworker wearing underwear and laughing annoyingly. Though clearly a humorous depiction of white-collar work, the commercial plays on a number of stereotypes about office work including a

61. Sam Mendes and Conrad Hall, “Storyboard Presentation,” *American Beauty*, DVD, 1999.

sense of unhappiness, lack of respect from coworkers, a desire to be somewhere else, sadness, anger, and annoyance. The two interior images of offices feature drab cubicles and irritating coworkers, again reinforces the negative image of office work. The commercial suggests that these people feel trapped, and by using careerbuilder.com can find an escape to a better job, and a happier life. Another commercial for Hummer (cars), depicts a group of white-collar workers in an elaborate escape plan involving automatons and dummies, rope ladders, and disguises to go for an exciting adventure in a brand new Hummer. The contrast between the drab office (portrayed by an office landscape of beige cubicles and visibly bored workers) and the visual thrill of the Hummer off-roading in a natural landscape reinforces the image of white-collar work as something to escape. Indeed, at one point in the commercial one of the workers is “caught” by other workers as a fax machine jams and tells his friends to “save themselves.” In this way, the office is visually depicted as a dreary jail and the escape is compared to a prison break-out.

In the news media, the cubicle office is often described as a kind of prison in which the average white-collar worker is trapped (both physically and metaphorically). For example, an article about running errands during the workday describes the reader as “chained to your cubicle for eight hours a day.”⁶² The strong image suggests that the cubicle is a kind of prison in which the reader is forcibly held for a full day. An article from the *Wall Street Journal* about why people talk to themselves uses the title “From Crib to Cubicle,” and within the article describes

62. Kafia Hosh, “Casual Friday Wasting Time,” *McClatchy Tribune Business News*, February 23 2007, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/462695234?accountid=465>.

the time between toddler-hood and old age as “your cubicle-bound life.”⁶³ The term “cubicle-bound” again suggests that the reader is somehow trapped or stuck within a cubicle. Other articles describe people who managed to “escape” their cubicle by doing something creative. For example, Scott Adams, creator of *Dilbert* is often portrayed as someone who escaped his office cubicle prison. An article from the *New York Times*, describes how Adams’ unhappiness at work inspired him to “escape into a flourishing career as the nation’s leading cube basher and the creator of *Dilbert*.”⁶⁴ An article from 2006 about workers using iPods in the office was titled, “Rockin’ in Your Cubicle: More Workers are Using iPods to Escape Boredom and Chaos in the Office.”⁶⁵ Again, the office (and the cubicle in particular) is textually described as something from which workers need to “escape.” The correlation of the cubicle to a kind of prison even in the news media reinforces the negative image of both office cubicles and the broader experience of white-collar work.

63.Jared Sandberg, “From Crib to Cubicle, a Familiar Voice--our Own--reassures,” *Wall Street Journal*, May 25 2008, 1, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/399097822?accountid=465>.

64.Lohr, “Rethinking the Contemporary Workplace: Hey Who Took the Office Doors?”, 7.

65.Peter Hubert, “Rockin’ in Your Cubicle: More Workers Are Using Ipods to Escape Boredom and Chaos in the Office,” *McClatchy Tribune Business News*, February 7 2006, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/461435516?accountid=465>.

Conclusion

Since the early-1990s, office cubicles have become a recurring symbol of the culture and experience of contemporary white-collar work in the United States. Indeed, the image of the dreary sea of office cubicles with its rows of drab and often undecorated cubicle walls has become inextricably linked to a larger image of white-collar work. By tracking the repetition of cubicles in so many different media forms, and examining the patterns of representation that emerge from these disparate media sources, one better understands what the cubicle is and what it means. Far from a meaningless backdrop of modern office work, the image of the cubicle office is embedded with thoughts, beliefs, experiences, fears, and frustrations about contemporary white-collar work.

Often treated as both a metaphoric and literal prison, the cubicle has become a visual symbol of the trapped white-collar worker at the mercy of external economic and social forces beyond their control. The drab panel walls and the “sea” of workspaces perpetuate an image of an oppressive corporate bureaucracy and tedious and unimportant white-collar work. It is significant that these kinds of images of the cubicle are repeated in so many different media forms (visual and textual, fictional and non-fictional) which suggests just how pervasive this kind of imagery has become. The references to the cubicle in the news media are particularly revealing examples because they often presume that its contemporary readers have a shared understanding of what a cubicle is, but also what it is like to work in a cubicle (the example of addressing the reader directly as a fellow “cubicle dweller”). The negative imagery of both the cubicle and white-collar work are so common that even children are expected to be able to read the maze of cubicles in a movie like *The Incredibles* and understand the image as a reflection of

a dreary and unpleasant corporate bureaucracy, and recognize Mr. Incredible as a bored and trapped white-collar worker. Ultimately, the negative image of cubicles that circulate in so many popular media forms reflects and reinforces a negative image of white-collar work in contemporary American culture.

Obviously there is a significant gap between the vision of the architects and designers who first promoted the open plan, and what the cubicle has come to mean in American popular culture since the 1990s. I argue that this "gap" between the very high expectations of the open plan and the negative image of "cubicles" in popular media is tied to deeper changes in the structure and culture of work in the United States. While the "images" of cubicles in popular culture often suggest that the cubicle's design is the source of the problem, I argue that the negative imagery of the cubicle in popular media reflects the deeper changes in white-collar work that transformed the design and use of systems furniture and the open plan, and fundamentally changed the experience of white-collar work.

Chapter 5

From Vision to Reality:

Understanding the Transformation of the Open Plan

As a design concept, the open plan emerged in the economic, social, and political climate of the 1960s—a moment that in the United States was infused with optimism and a genuine belief in the possibility of radical change. As I suggested in Chapter 2, the architects and designers promoting the open plan were not just promoting a new kind of office design, they were articulating a new vision of work itself that was shared by scholars and writers in a number of fields. This new vision integrated the social research on the value of teamwork and social relations in the workplace, the new theories of management that were circulating at the time which emphasized worker autonomy and motivation, and a belief that new technology and new organizational structures would eliminate hierarchy and bureaucracy from office work. In the late 1960s and early 1970s, the open plan became a spatial symbol of this vision, evoking an office culture and office climate that was supportive of its workers and committed to transforming the nature and culture of office work. In Chapter 3, I used a case study of Wisconsin State Government to describe and analyze the numerous changes in the structure, content and culture of office work in the United States during the 1970s and 1980s, including: implementation of the open plan; changing economic climate; changing ideas about workplace safety; increasing diversity of American workplaces in the 1970s and 1980s; and the increasing use of technology in all aspects of office work (particularly computers). Adapting a model from Industrial Systems Engineering, the Work System Model, I illustrated the complex interaction of these myriad

changes across the different aspects of the work system (organization, people, environment, technology, and tasks) and how these various changes interacted specifically with the design, implementation, use, and meaning of the open plan and systems furniture and the experience of office work in Wisconsin State Government. Although implemented initially as a progressive office design to support a developmental management culture and eliminate the signifiers of hierarchy, by the end of the 1980s "cubicles" in Wisconsin State Government offices were perceived by state office workers as noisy and claustrophobic. As I argue in Chapter 4, starting in the early 1990s the cubicle had become a symbol in popular media of the most bureaucratic and tedious aspects of white-collar work, evoking images of: social isolation; boredom and white-collar drudgery; and entrenched organizational hierarchy and bureaucracy.

In this last chapter, I return to the three central claims articulated by the proponents of the original open plan in the 1960s, specifically that the open plan would: 1) improve communication in offices; 2) increase flexibility in order to support organizational, managerial, and technological change; and 3) eliminate hierarchy and bureaucracy. Revisiting these three core beliefs about the original open plan, this last chapter will consider some of the ways which these three critical attributes of the open plan may have failed to manifest in the 1970s and 1980s. Through this last chapter, I suggest that the "failures" of the open plan to achieve these central attributes were actually related to the changes in office work in this period which countermanded some of the design intentions that were embedded in the original open plan concept. In this way, I suggest that the use of a particular office design can not be separated from the broader context of its use. Yet this research only scratches the surface of this larger question of how the open plan evolved and how changes in work influenced the meaning and use of the

open plan. More research would be necessary to fully explain this transformation of the open plan from a progressive design idea to a negative symbol of "dreary" office work. While this analysis of a single example does not fully address the question I set out to answer, it does suggest some new questions that might be worth exploring in the future.

From Supporting Communication to Creating Isolation

Without a doubt, the promise of improved communication in the office was one of the most celebrated claims of the new open plan. In office landscaping, communication was central to the overall design, and served as the basis of the layout. The Quickborner Team viewed communication as the lifeblood of the organization, and by using communication patterns as the basis of design, workers would be productive, effective, and efficient. Even the more generic concept of the open plan, which emphasized the elimination of private offices was heralded as a means of improving communication in the office. In descriptions of the post-war workplace, office walls were described as hindrances to communication as well as the source of unnecessary corporate bureaucracy by encouraging workers to communicate through letters, forms, and secretaries rather than talking directly to each other. Just by removing walls, even executives and managers would be liberated from such tedious bureaucracy, and would be free to interact with each other as necessary.

To encourage interaction and communication, early open plan designs used partitions somewhat sparingly, mostly to create some spatial division and definition, and to provide modest amounts of privacy, but the goal was not to fully enclose every workstation with partitions. In

fact, in office landscaping in particular, designers were encouraged to focus on setting workstations at odd angles to give workers “visual privacy” without compromising the central goal of encouraging communication and interaction.¹ Even Action Office which used partitions in a new way to create “vertical space” for storage, emphasized the importance of balancing enclosure and access, so that workers could have space to focus on their “thinking” work when necessary, but also be available for social interactions when that was appropriate.

While the early installations had featured minimal use of partitions, increasingly partitions became both a means of aiding noise problems in the open plan (a common complaint among workers) and of creating visual and acoustic privacy for workers in the open plan. Over the course of the 1970s and 1980s, partition heights rose to standing height (5 feet or higher), and increased in number so that more workers had enclosure in their workspaces on 3 or even 4 sides. At the time, research in the emerging field of environment and behavior suggested that workers preferred greater enclosure in the open plan (for example the BOSTI study mentioned in Chapter 1). The idea of the modern “cubicle” really developed in this period as organizations favored higher degrees of enclosure creating what would eventually become the “cube farm.” It is significant that the research at the time suggested that higher degrees of enclosure were better for noise control, and better in terms of worker satisfaction and “ease of communication.” Further, although the research had used managers and technical workers it was assumed that the findings were relevant to clerical workers.² BOSTI did not consider that clerical workers might have different needs or preferences than managers or technical workers. In fact, other research

1.Duffy, *Office Landscaping: a New Approach to Office Planning*.

2.Brill, et al., *Using Office Design to Increase Productivity*, 89-102.

from the 1980s suggests that while managers and technical workers may have wanted higher degrees of enclosure, not all clerical workers necessarily did. According to business administration scholar Shoshana Zuboff, some clerical workers may have perceived the implementation of systems furniture workstations (particularly with high partitions) as a means of deliberately preventing opportunities for socializing at work.³ Even so, in adopting these recommendations, it is critical to recognize that the architects, designers, and organizations were trying to support the perceived needs and preferences of workers for higher degrees of enclosure. In other words, the decision to use higher panel walls was likely not intended to inhibit conversation among workers, rather there was a belief based on highly publicized research that higher partition walls would actually improve communication.

The use of higher panel walls coincided with the increasing use of computers in all aspects of office work. As workers spent more time working on computers, they spent less time interacting directly with each other. According to Shoshana Zuboff, as computer mediated workflows became more prevalent in the office, workers had fewer opportunities to interact with each other in the course of their work processes.⁴ The spread of computers also changed workers' physical movement and use of the larger office. Through the early 1980s computers were often a shared piece of equipment requiring workers to leave their own workspace and travel to a shared computing area; as computers became a fixture of every workspace the need to move in the office was decreased. In their study of workers and their workplaces, BOSTI found that

3. Shoshana Zuboff, *In the Age of the Smart Machine* (New York: Basic Books, 1988), 139-52.

4. Zuboff, *In the Age of the Smart Machine*, 139-52.

managers, administrators, as well as professional, technical, and clerical workers spent significant parts of their workday in places other than their own workspace: others workspaces, “task areas,” conference rooms, and even time outside the building. The BOSTI study revealed a growing trend for there to be ‘dedicated’ equipment in workspaces, particularly for clerical workers, although professional and technical workers sometimes had dedicated graphics and drawing equipment in their own workspace. Yet even as the trend was growing, most workers still used computer terminals and other automated equipment outside of their own workspace.⁵

This is significant because it reveals the small influence of a change in technology on everyday behavior. The act of moving through the office, and spending time in an alternative location creates opportunities for exchange and conversation with other workers. It also meant that workers were spending some time moving around the office, and not just sitting at their desks all day long. As computers became a fixture in individual workspaces, the need to leave one’s workspace for work purposes declined as well. In their report, “Race Against Time,” the National Association of Office Workers described how office automation alongside other changes in services had limited clerks need to move around the building, which also limited a worker’s “opportunities to form face-to-face relationships with other workers.”⁶ Although the open plan is often blamed in popular media as the symbol of “isolation” in representations of “cubicles,” perhaps the real source of the problem was the increasing use of computers which

5.Brill, et al., *Using Office Design to Increase Productivity*, 81-82.

6.Working Women, *Race Against Time: Automation of the Office: an Analysis of the Trends in Office Automation and the Impact on the Office Workforce*, 9.

limited opportunities for some types of workers to interact (particularly clerical and other data entry positions). Future research might look more deeply at the role of computers in changing the social aspects of the workplace.

From Supporting Change to Preventing Change

Flexibility was another promoted “benefit” of the open plan office in the 1960s. In their description of contemporary office life, organizations were depicted as rapidly changing to adapt to new technology, new organizational structures, and new ways of working, and the office space needed to be flexible enough to respond to such changes. Office landscaping which used minimal specialized furniture required very little effort to rearrange, and Action Office was specially designed to be easily adjusted by the individual worker (no tools required) as well as by the organization. In their vision of the new workplace, architects and designers imagined an office that could be quickly adjusted (even overnight) to reflect any kind of change. Through the 1970s, when open plan offices were comprised of free-standing partitions and conventional furniture, such changes were very easy to accommodate. Even early systems furniture could be changed with relative ease. For example, in Wisconsin State Government there is evidence that early open plan systems furniture was similarly movable and adjustable. From the 1970s through the mid 1980s offices were continuously redesigned to accommodate various changes including layoffs, reorganizations, and office moves. This kind of workplace flexibility in the 1960s and 1970s was a visual symbol of organizational flexibility. By making the workplace easy to change, the idea was to reflect an organization that could be nimble and adaptable to change as well. In other

words, the image of the flexible workplace in the form of the “open plan” was a symbol of the death of bureaucracy, and a reflection of a more dynamic and fluid organization.

Meanwhile, individual work stations were also very mobile and flexible, at least enough that workers were able to make adjustments to their workspace, sometimes even taking furniture or partitions from other work stations and adding them to their own. This kind of behavior was sometimes considered a problem among facilities managers and designers who wanted to preserve the integrity of the design, and maintain control of the arrangement of the furniture. An article in the magazine *Administrative Management* in 1974 described this problem as one of “blight” in the open office, and encouraged administrators to establish rules preventing workers from destroying the design through their individual adjustments. According to the article, partitions and plants would “creep” due to “indiscriminate changes” made by workers. In fact, the author recommends that after the open plan design is complete, workers should be notified “that no changes are to be made without approval of the designer.”⁷ In this way, the very principle of creating work stations that could be adjusted was often directly countered by rules from the organization preventing such individual changes. According to the BOSTI study from 1984, more than half of workers surveyed indicated that they needed permission to make changes to their workspace. This suggests a fundamental dissonance between the goals of the early proponents of the open plan who articulated the importance of workers making their workspace supportive for their unique needs, and the reality of organizational cultures that imposed stiff restrictions on the use of the space. As systems furniture became more complicated

7. Al Schumann, “Keeping Blight From the Open Office,” *Administrative Management* (1974): 26-28, 80.

in its design it also became more difficult to change by workers as well as by the organization. In their study on office design, BOSTI found that half of the workers surveyed wanted to make adjustments to their workspace, but found making changes was “physically impossible or impractical.”⁸

The increasing use of computers in work stations throughout the office only further complicated the design and use of systems furniture, as well as limiting their “flexibility” for change. Wired panels and cabling in particular transformed systems furniture into a critical office utility that took time, money, and special knowledge (not to mention tools) to make changes. In fact, the responsiveness of a particular systems furniture line to change was often limited by its particular design, and the needs that were being addressed. Some systems furniture companies continued to upgrade their furniture lines in order to help companies that had purchased older furniture adapt to necessary changes.⁹ Meanwhile, the landscape of office furniture options was made even more complicated because technology companies in this period also began producing dedicated furniture for their equipment which could be used in tandem with conventional or systems furniture.¹⁰

Companies updating their offices or redesigning in this period had to navigate a wide field of furniture and design options. Although they were often making decisions on their office design based on present need, they were often aware that change, particularly in technology, was

8.Brill, et al., *Using Office Design to Increase Productivity*, 325.

9.Pile, “The Second Generation.”

10.William Pulgram, and Richard Stonis, *Designing the Automated Office: a Guide for Architects, Interior Designers, Space Planners, and Facility Managers* (New York: Whitney Library of Design, 1984).

so rapid that their choices might already be out of date by the time the furniture was selected, purchased, and installed. In their book *Designing the Automated Office*, authors William Pulgram and Richard Stonis describe the “domino” effect of changes in technology on workstations, and how one change could result in a whole host of other necessary changes.¹¹ While organizations were encouraged to think expansively about future needs, their actual choices were often framed in terms of present furniture availability, technology capability, and cost, and the needs of the organization often trumped the needs of workers. For example, in weighing the critical attributes of different systems furniture lines, one aspect that was often ignored in Wisconsin State Government was the issue of ergonomics. Even though they recognized that furniture could be used to ameliorate some of the negative physical consequences of computer usage, they did not make ergonomic furniture a standard for state office usage. In other words, although the furniture was expected to effectively accommodate the needs of the technology, it was not necessarily expected to accommodate the physical needs of individuals using the technology.

This attitude towards ergonomics was not unique to the State of Wisconsin. In his book *Office Politics*, Vernon Mogensen describes how computer manufacturers were strongly opposed to any kind of VDT regulation, and how the industry created an interest group called CWT (Coalition for Workplace Technology) which lobbied heavily to prevent the development of government mandated regulations on computers. According to Mogensen, computer manufacturers in the early 1980s were concerned that any regulation would increase their production costs and decrease their sales. It was in fact common practice in many organizations

11. Pulgram, and Stonis, *Designing the Automated Office: a Guide for Architects, Interior Designers, Space Planners, and Facility Managers*, 146.

to shift the burden of self-care and prevention of injury onto the employee through recommended exercises and breaks, rather than providing specially designed equipment. Mogensen's study reveals how reluctant computer companies were to change their equipment to be more ergonomically safe, and how their battles against any kind of regulation left VDT workers at the mercy of the policies of their state and their employer.¹² Despite growing awareness of the dangers of intensive computer usage on workers' physical health, ergonomics has remained a special accommodation rather than a standard of design.

Although originally developed to enable flexibility for change, in reality the open plan became less flexible as the demands of technology and the needs of organizations began to influence the design of systems furniture and the implementation of the open plan. Ironically, the increasing use of computers required furniture that could offer greater flexibility to individual workers (to adjust the workspace to make computer usage less damaging to their physical health), but the design and use of systems furniture actually became less flexible and adaptable as it evolved in the 1980s. Even ergonomic furniture, which was designed to be more adaptable to the needs of individual workers, was treated as an accommodation for certain workers rather than a standard for all. Future research might explore the history and usage of ergonomic furniture in other types of organizations, and perhaps include an analysis of different ergonomic features associated with different furniture lines as well as how those ergonomic features were used or not used in the specification of office furniture for particular installations.

12. Vernon Mogensen, *Office Politics: Computers, Labor, and the Fight for Safety and Health* (New Brunswick: Rutgers University Press, 1996).

From Eliminating Hierarchy to Reproducing Hierarchy

Another critical attribute of the open plan as it was conceived and promoted in the 1960s was its elimination of the symbols and culture of hierarchy in contemporary offices. Proponents of the open plan argued that eliminating private offices, and reducing status symbols in the open plan would help to create a new flat organization in which all workers would be treated as executives. Some companies, like McDonalds, did try to embrace the idea of the non-hierarchical office plan by placing all workers (even executives) in the open office, but in many others in the 1970s and 1980s, the open plan just reproduced hierarchy.¹³ In the State of Wisconsin, the open plan reduced the allocation of private offices overall, but the scarcity of private offices actually made them more desirable. Although the early space standards for the open plan emphasized privacy as a "functional" need rather than a reflection of status, by the late 1980s, the status based allocation of private offices had returned for people who were Bureau Director or higher. Meanwhile among other state workers, the allocation of individual space in the open plan was determined by their organizational status. The physical space of the office remained a landscape of differential power and authority, spatially representing organizational status in a manner that was visible to all workers in the organization. In this way, while the original open plan had been conceived as a means of reducing the symbols of hierarchy, the implementation of the open plan often just created a new way of representing hierarchy.

This culture of hierarchy was further complicated by the increasing diversity of the white collar workforce. In Wisconsin State Government, the state enacted a number of programs in the

13.Malcolm, "McDonald's Sold on Its 'think Tank' Office."

1970s to specifically increase diversity among workers, but those workers had an uphill struggle for equity in Wisconsin State Government, particularly in the 1980s when affirmative action efforts stagnated. In offices across the United States women and people of color were increasing in number through the 1970s and 1980s, but were often trapped in low-level positions, with little opportunity to advance. In general, women and minorities were often disproportionately represented among clerical workers, and only modestly visible in managerial and executive positions. In their study on automation and clerical work, The National Association of Office Workers found that in 1970 74% of clerical workers were women, and by 1980 a full 80% of clerical workers were women.¹⁴ Minority workers faced similar disparities in the workplace, and according to the massive study conducted by the U.S. Government, *Work in America*, young black white collar workers in the early 1970s were twice as likely to be unhappy in their jobs as white workers. Further, young black white-collar workers often reported higher levels of discrimination in the workplace than older black workers or black blue-collar workers.¹⁵ In Wisconsin State Government, office workers remained predominantly white and male through the 1980s. These systemic inequalities in the workplace created a physical landscape in the office that mapped organizational status as well as social differences including gender, race, and class.

Another growing “class” of workers in offices in the U.S. were temporary workers, the use of whom increased significantly in the 1980s in many industries, and perpetuated yet another form of internal hierarchy. A *New York Times* article from 1982 describes how temporary

14. Working Women, *Race Against Time: Automation of the Office: an Analysis of the Trends in Office Automation and the Impact on the Office Workforce*, 5.

15. O’Toole, *Work in America*, 52-53.

employment had grown 25 percent since 1977. According to the article, temporary workers were less expensive than permanent workers, and significantly more expendable at a time of so much economic uncertainty.¹⁶ Short term or temporary workers were sometimes used side-by-side with full-time employees, and often were doing the same work as their permanent counterparts, but without many of the benefits associated with being permanent. In his study on a high tech company, sociologist Gideon Kunda found that temporary workers were generally denied the social and economic benefits associated with permanent status. He describes how full-time permanent workers were given turkeys each Christmas, with any leftover turkeys given to “charity,” but temporary workers were not allowed to take a turkey. One permanent worker described how this inequity made her uncomfortable, especially since some of the temporary workers performed the same work as she did.¹⁷ In other words, the use of temporary workers and other types of non-permanent workers created structural inequities and hierarchies related to permanent and non-permanent status.

This was also true in Wisconsin State Government with their use of Limited Term Employees (LTEs) which they used to manage short term staffing needs in different agencies and offices. Unlike their full-time counterparts, LTE workers were not entitled to many of the benefits that full-time employees took for granted for example: health insurance, paid sick leave, vacation, and holidays.¹⁸ In their allocation of workspaces, LTE workers were typically left out

16.Brooks, “‘Temp’ Work Becomes Full-time Way of Life.”

17.Gideon Kunda, *Engineering Culture: Control and Commitment in a High Tech Corporation* (Philadelphia: Temple University Press, 1992), 213.

18.Department of Administration, *Employee Handbook*, 1984, Wisconsin Historical Society, p 26.

of calculations so that agencies and their workers experienced chronic overcrowding.¹⁹ Further, official space standards for the state indicated that LTE workers could be assigned a workspace below the minimum standard, meaning they could be squeezed into any corner of an office.²⁰ This created yet another form of spatial hierarchy in which LTE workers were “squeezed” anywhere, and permanent workers were entitled to minimum space standards.

Thus, although the open plan was meant to reduce the symbols of hierarchy, its use in offices often reinforced hierarchy and created new forms of distinction. Private offices remained an amenity for high level workers, and other workers were assigned workspace based on their organizational status. The physical arrangement of the workplace remained a landscape of power and authority informed by differences in race, class, and gender. While the architects of the 1960s envisioned a new kind organization in which hierarchy was eliminated in large part due to the use of the open plan, the implementation of the open plan in many organizations just created new symbols of hierarchy and status. Exploring this issue more deeply, future research might analyze space standards across different organizations over time in order to understand how hierarchy has or has not been used in office design over time.

19.Memo from Nate Harris to Willie Nunnery, April 24 1979, Wisconsin Department of Administration Secretary's Subject File Part 1, Box 34 Folder 3, Wisconsin Historical Society.

20.Wisconsin State Administration Building, 101 E Wilson Street Space Guidelines, January 30, 1990, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 87 Folder 37, Wisconsin Historical Society.

Mind the Gap: Different Workplaces; Different Work Experiences

Reflecting on the transformation of the open plan, I suggest that the image of the open plan is tied to a broader understanding of the meaning and purpose of office work. In their fantasy of the future of white-collar work, architects and designers of the 1960s envisioned an entirely new kind of office culture that would be more collaborative, less bureaucratic, less hierarchical, and more flexible and adaptable to imminent change. Yet, the reality of the 1970s and 1980s with the downturn of the economy, the loss of jobs, and the influx of computers and other technology in the workplace all made office work more bureaucratic, more hierarchical, less interactive, and ultimately less personally fulfilling than it was forecast to become. In her 2002 book *White-Collar Sweatshop*, journalist Jill Andresky Fraser describes how changing technology, job insecurity, and declining job conditions in late 20th century America led to increasingly poor working conditions for American white-collar workers creating a culture of over-work and stress.²¹ Thus, although the "cubicle" is often blamed in popular media for making workers unhappy, this research suggests that the true source of unhappiness for workers is perhaps the changes in office work itself, not the changes in office design.

Yet, not all offices are "made" equal in terms of price, quality, functionality, or design. While this study used just one case study of offices in Wisconsin State Government, other research might explore the experience of workers in higher end furniture installations and architect-designed interiors. The most celebrated of these "higher-end" offices are those in the high-tech and creative industries. These kinds of workplaces which are regularly featured in the

21. Jill Andresky Fraser, *The White-collar Sweatshop: the Deterioration of Work and Its Rewards in Corporate America* (New York: W.W. Norton & Co, 2001).

press often have wacky decor (featuring unique design themes), a mix of individual workspaces, lounge areas, and conference areas, and feature amenities like foosball tables, basketball courts, free food, coffee bars and even alcohol. Much like the original open plan, these highly supportive workplaces are designed to encourage collaboration, allow flexibility for the needs of individual workers, and reflect a flat organizational culture in which all workers are equally committed to the needs and goals of the organization. Although these bright, colorful, and permissive offices seem to be the antithesis of the "dreary" cube farm, they are only available for the highest paid white-collar workers of the most successful organizations. These organizations that invest top-dollar in their workspaces do so in part to create an environment in which their workers "stay" throughout the work day and beyond. In other words, these beautifully appointed and playful offices are often designed to keep workers in the office for very long hours (there is no reason to leave the workplace for food, exercise, relaxation or even sleep).²² Future research might explore these high-tech and creative office environments more specifically to understand how they are perceived and used by workers.

However, the majority of white-collar workers in the United States do not work in such luxurious surroundings. Indeed, many workers still work with older furniture that may have been installed anywhere from 10-40 years ago. For example, when I visited the John Deere Company corporate offices in 2008, I was surprised to discover that they still had the original systems furniture from 1978 in use in some parts of their office. While in 1978, that furniture had been

22. Sara Kehaulani Goo, "At Google, Hours Are Long, But the Consume is Free," *Washington Post*, January 24 2007, <http://www.washingtonpost.com/wp-dyn/content/article/2007/01/23/AR2007012300334.html>; Kelly Clay, "What it's Really Like to Work At Google," *Lockergnome* (2012).

the height of fashion and was even featured in architectural magazines, by 2008 that same furniture was shabby and ill-equipped to manage contemporary needs (particularly computer equipment).²³ Of course, even the most thoughtfully designed and installed furniture will be tattered and incompatible with contemporary work after several decades of use. Thus the aging of systems furniture over time is an inevitable aspect of its use.

In addition to companies keeping their own furniture for many decades, there is also a lively second-hand market for systems furniture in which older panels and workstations may continue to be re-sold and re-used by other organizations, and perhaps even re-sold and re-used a second or third time. Therefore even the "old" systems furniture discarded by one organization may have a long life beyond its original installation. In fact, at the time of my visit in 2008, John Deere was in the process of redesigning their offices with new Herman Miller furniture and their old 1978 systems furniture was donated to several local charities that re-purposed the older furniture for their own offices.²⁴ Those older workstations that John Deere was removing from their own offices are still in use today, just in a different office and populated with a different group of workers.

This aging of furniture and its reuse over time indicates the importance of thinking about change over time in any analysis of office design. While a traditional architectural history perspective tends to focus always on the "new" in architecture and office design, a vernacular architecture approach that considers the change of an office over time might be an even more

23. William Marlin, "Excellence Attracts Excellence," *Architectural Record* (1979): 85-92.

24. "re-Purpose Program" press release, Herman Miller, available on-line at http://www.hermanmiller.com/content/dam/hermanmiller/documents/environmental_advocacy/rePurpose_John_Deere.pdf accessed on November 29, 2013.

interesting direction to consider. Future research might thus delve further into the life of systems furniture beyond the initial installation, even including a study of the used systems furniture market to understand how old furniture is purchased and re-used by other organizations.

Although high design environments are not the "normal" for most workers in the United States, high design concepts still matter because those "high design" ideas often get filtered down to mainstream use over time. The open plan is in fact a perfect example of exactly this issue in that it began as an architect and designer driven idea that was celebrated in the architectural press and was eventually translated into more ordinary applications (for example in Wisconsin State Government offices). Although some of the central goals and intentions became "lost" in the translation as these organizations took some of the ideas they liked and ignored others that did not fit with their culture or their budget, they nonetheless remain important to the design concept. Recognizing this complicated relationship between the most expensive and elaborate offices and their more mundane counterparts is critical to understanding how design ideas circulate. Future research might examine this relationship more closely to understand how office design ideas are developed and circulated in mainstream office design.

Meanwhile, office work continues to change as new technology and new ways of working create opportunities and challenges for organizations. For instance, the extensive use of laptops, wireless internet, and cloud computing allow workers in some organizations to be "nomadic" in the office, working in different places and spaces as their whim dictates (in some cases even working from home). As part of this trend, some offices have adopted the "hoteling" method in which workers do not even have assigned spaces, but rather "check out" work stations when they need one. Companies first began experimenting with hoteling in the early 1990s; one notable

example was at the advertising agency Chiat/Day which in 1994 eliminated individually assigned permanent workspaces and instead encouraged its workers to work wherever they were comfortable. This particular example was a much documented failure as workers were reportedly frustrated with the lack of dedicated workspace, and after only one year Chiat/Day abandoned the "hoteling" experiment in favor of a more traditional office space.²⁵

It is important too to recognize that this kind of "flexibility" for movement is not a product of the office space, but rather it is the product of a work culture and job design that allows workers to determine their own work schedule and process. Further, policies and organizational culture can change. In 2013, both Yahoo and Hewlett Packard made big headlines for rescinding the privilege of allowing their workers the flexibility to "work from home."²⁶ Of course these high tech workers who were no longer allowed to work from home likely still had significant freedom and flexibility to move around within the office itself during their workday, in contrast other types of workers are still tethered to a workstation with few opportunities to even walk around during the day; for example call center workers are often closely monitored throughout the day to ensure optimum productivity (often called "adherence"). A news story from 2013

25. Phil Patton, "The Virtual Office Becomes Reality," *New York Times (1923-Current File)*, October 28 1993, C1, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/109142554?accountid=465>; Phil Patton, "Technology in the Office: Individual Power and Collective Standards," in *On the Job: Design and the American Office*, ed. Donald Albrecht, and Chrysanthé Broikos (New York: Princeton Architectural Press, 2001), 117; Thomas Hine, "Office Intrigues: the Interior Life of Corporate Culture," in *On the Job: Design and the American Office*, ed. Donald Albrecht, and Chrysanthé Broikos (New York: Princeton Architectural Press, 2001).

26. Will Oremus, "Now Meg Whitman Wants Everyone to Stop Working From Home, Too," *Slate*, October 8 2013, http://www.slate.com/blogs/future_tense/2013/10/08/hp_working_from_home_ban_marissa_mayer_s_yahoo_policy_becomes_industry_narrative.html.

described how one worker at a T-Mobile call center was forced to use vacation time to go to the restroom during her pregnancy because her visits to the toilet exceeded what was permitted by the organization to achieve optimum adherence rates.²⁷

In this way, it is critical to acknowledge the enormous gap in experience among different types of workers in different jobs and different organizations. While some workers in some organizations have the freedom to roam the office at will, other workers in other organizations are penalized for using the restroom too frequently. Of course the T-Mobile example also suggests how even in a single company, different workers may have vastly different work experiences, after all I doubt that T-Mobile executives have to "clock out" to use the restroom. Future research might explore these contrasts among different types of white-collar workers in order to understand the full breadth of work experiences that exist.

Applied to office design, this contrast reveals the deep systemic inequities that exist among different workers and different workplaces. In other words, the most elaborate and expensive offices are reflective of a system that privileges the needs of one group of workers over those of another. For example, the highly paid workers at Apple's elegant headquarters in Cupertino, California are dependent on the corporate reliance on low paid sweatshop workers who produce

27. Abby Ellin, "Pregnant T-mobile Employee Clocked Out to Use Toilet," *ABC News*, May 1 2013, <http://abcnews.go.com/blogs/business/2013/05/pregnant-t-mobile-employee-clocked-out-to-use-toilet/>.

their high end-devices in terrible working conditions.²⁸ Future research might delve more deeply into the hierarchies of design and privilege that exist among different organizations, but also within the same organization among different classes or types of workers.

In Search of the "Ideal" Workplace

At the time of its widespread implementation in American offices, the open plan was just one change among many. Although the "cubicle" is often described as the source of worker's unhappiness, I suggest that perhaps the real source of that unhappiness is within the culture and structure of contemporary white-collar work. My case study of Wisconsin State Government offices hints at just some of the changes that occurred in the 1970s and 1980s for example: tightening budgets, a desire to decrease space and energy costs, and the increasing use of computers in all aspects of work. Yet this study is just one example of how the open plan was implemented in a single organization, therefore it is by no means exhaustive of all types of organizations or all types of office work. Future study should expand beyond the narrow confines of a single case study in order to see what larger patterns of change existed among different types of organizations in this period.

In fact, I do not think that the architects and designers of the late 1960s could ever have predicted all of the ways in which work would change in the 1970s and 1980s. Yet, one lesson

28. Rob Cooper, "Inside Apple's Chinese 'sweatshop' Factory Where Workers Are Paid Just £1.12 Per Hour to Produce iPhones and iPads," *Daily Mail*, January 25 2013, <http://www.dailymail.co.uk/news/article-2103798/Revealed-Inside-Apples-Chinese-sweatshop-factory-workers-paid-just-1-12-hour.html>.

from this research might be the importance of thinking more holistically about how the various aspects of work might influence the ways in which an office design is implemented, and how an office arrangement might be used and perceived by the workers that inhabit that space. In other words, contemporary architects and designers could perhaps benefit from thinking more globally in their design process about the full work experience, as well as the possibility for future change. The systems approach which thinks about the influence of change across various aspects of work might be a useful model for architects and designers to analyze a particular work context as an interdependent system.

Most importantly, I am not suggesting that the open plan was just a neutral (and "innocent") bystander to the activities of white-collar work; in fact, the widespread adoption and use of the open plan represented a critical change in the physical design of American workplaces in the late 20th century and created some new opportunities both for organizations and for workers. For organizations like Wisconsin State Government, the open plan enabled the reduction of space usage (fitting more people into less space) and also reduced energy usage by eliminating a number of enclosed offices. In a period with exceptionally tight budgets, the ability to minimize space costs was a huge benefit to organizations like Wisconsin State Government that were struggling to meet obligatory budget cuts.

The widespread popularity of the open plan also allowed for growth in the use of leased office space, which in turn allowed organizations to accommodate organizational changes as well as sudden growth or attrition of their workforce or market with relative ease.²⁹ In Wisconsin State

29. Irwin Ross, "Bargains Galore in Office Space," *Fortune*, February 17 1986, http://money.cnn.com/magazines/fortune/fortune_archive/1986/02/17/67143/index.htm.

Government, there was a strong interest in eliminating all leased office spaces, so that all of the agencies would be housed in state-owned and operated facilities.³⁰ In part because the open plan and systems furniture allowed for a net reduction in space usage, Wisconsin State Government was able to require further space reductions for its various agencies. Meanwhile, although the inclusion of power and cable management made systems furniture in some ways less flexible and less easy to change (as was documented in Chapter 3), it also made systems furniture even more critical to the daily functioning of the office as computers and other forms of technology filled workspaces, and systems furniture workstations became the primary source of power and network access for workers.

Also, in terms of its design, the number of different systems furniture lines and the many options for finishes, fabrics, storage options and workstation configurations meant that organizations purchasing new furniture often had a vast number of options for designing a workplace that reflected their distinct culture and supported their particular organizational needs. Thus systems furniture became a means for organizations to deliberately reflect their organizational values and culture. In fact, there has been significant research in the field of organizational behavior on the role of the physical office environment in expressing organizational culture and values.³¹ The design of systems furniture also enabled organizations to create a visual sense of order and compartmentalization for the office that could reflect an image

30. Announcement, October 26, 1983, Wisconsin Department of Administration Secretary's Subject Files Part 1, Box 39 Folder 1, Wisconsin Historical Society.

31. Per Olof Berg, and Kristian Kreiner, "Corporate Architecture: Turning Physical Settings Into Symbolic Resources," in *Symbols and Artifacts: Views of the Corporate Landscape*, ed. Paquale Gagliardi (New York: Aldine de Gruyter, 1992).

of corporate power and control. The crisp lines of a new systems furniture installation and the fresh fabric on the partitions convey a sense of visual order which in turn could reflect the "order" and control of the organization as a whole. In other words, the image of the newly designed open plan office with its fully coordinated and articulated furniture parts, designed to the unique specifications of a particular organization, implies a sense of authority and control over the workspace, the work process, and the workers themselves.

For workers, the same panel walls that have been so reviled in so many media also present opportunities for transgressive activities as well as self-expression. Although for some upper-level workers, the implementation of the open plan represented a net loss of privacy (visual and acoustic), for a large number of workers who had worked in an open bullpen previously the panel walls actually created a modest amount of privacy. In some positions, this change along with the growth of computers allowed for greater opportunities for what cultural and spatial theorist Michel de Certeau calls "la perruque" which he describes as the worker's own work disguised as work for his or her employer.³² According to de Certeau, these small acts of resistance serve as strategies for finding pleasure in everyday life. The quasi-privacy of the cubicle along with the addition of computers equipped with a network connection allowed new forms of subversive behavior for example perusing on-line pornography at work or keeping up with the day's basketball scores during "March Madness."³³ Even much smaller acts such as playing a game of

32.de Certeau, *The Practice of Everyday Life*, 25.

33.Trip Gabriel, "New Issue At Work: on-line Sex Sites," *New York Times (1923-Current File)*, June 27 1996, C1, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/109645248?accountid=465>; Swett Clint, "Hoops in Your Cubicle," *McClatchy Tribune Business News*, March 17 2005, <http://ezproxy.library.wisc.edu/login?url=http://search.proquest.com.ezproxy.library.wisc.edu/docview/460364951?accountid=465>.

solitaire, reading the news on-line, or writing a personal email can be viewed as small transgressions of time in the workplace. The panel walls which serve as a mechanism of order and control, thus also create opportunities for some workers to disguise such personal activities as "work." Of course not all workers or jobs have equal opportunities for such forms of resistance; while some workers may have positions that allow significant freedom others may have more tightly controlled jobs that do not permit such activities.

The walls of the "cubicle" also presented an opportunity for workers to personalize their workspace. Although some organizations have strict rules in place about personal decorations (limiting or outright prohibiting personal decoration), others (including the State of Wisconsin) have had a much more permissive view of workplace decoration allowing workers to display pictures and other items. In the era of *Dilbert*, panel walls can also become a space for potential subversion by giving workers the opportunity to post a cartoon or other object that might serve as commentary on the behavior of coworkers, the boss, or the organization. Thus, the same panel walls often described as "dreary" in popular media actually enable these small subversive acts that cumulatively represent a significant transgression of the more controlling (and perhaps dissatisfying) aspects of work. Although these small acts of self-expression and subversion will not turn a tedious job into a fulfilling one, they do serve as critical daily strategies for workers to express themselves and potentially find small pleasures in their workday. By their very nature, such acts are often unseen by the organization (or at least they are essentially ignored), but they represent a critical aspect of contemporary white-collar work. While the physical structure of the

cubicle and the office as a whole can on the one hand be viewed as a mechanism of control and order on the part of the organization, it also creates opportunities for quiet resistance and subversion by workers.

Rather than thinking of the open plan as a single unified space, this suggests that "the open plan office" is a series of intertwined and nested spaces that reinforce, challenge and inform each other. In his seminal text, *The Production of Space*, philosopher Henri Lefebvre describes how social spaces "interpenetrate" and "superimpose" each other such that even "visible boundaries , such as walls or enclosures in general, give rise for their part to an appearance of separation between spaces where in fact what exists is an ambiguous continuity."³⁴ Thus, rather than conceiving of space as something merely contained by the boundaries of walls and other 'visible' elements, Lefebvre describes space as both a process and product of social relations. In the space of an office, the organization's control of the space is asserted in the form of the total workplace, its design, and its underlying process and structure, while each worker's ownership of the space is asserted through their daily use of the space as well as their actions of resistance and subversion. Yet the spaces of control and the spaces of subversion are not mutually exclusive, rather these different spaces in the workplace shape and inform each other. Indeed, the partition walls which both expose workers to the organization and create opportunities for subversion perfectly symbolize this complex interplay of space. The open plan office is a tangle of interacting, nesting, and contradicting spaces that are defined by the physical space and boundaries of the office design as well as by the organizational setting, the social interactions,

34. Henri Lefebvre, *The Production of Space*, trans. Donald Nicholson-Smith (Cambridge, MA: Blackwell, 1991), 87.

and the individual behaviors of workers in the office. As a result, the open plan concept and systems furniture together created a new vision of corporate culture and corporate space, and new ways for workers to inhabit and use the workspace.

What is most surprising is not that the open plan failed to make the changes in the experience of white-collar work that it was forecast to make, rather it is surprising that the open plan retained much of its progressive sheen for as long as it did. Through the 1970s and 1980s the open plan continued to be embraced by workers and organizations as a symbol of a progressive workplace, even though the context of its use was often altogether different. Images of the open plan in films in this period were often very positive featuring attractive bustling offices and highly committed workers who were passionate about their jobs and their careers. In Wisconsin State Government there is also evidence that the open plan and systems furniture were attractive to workers in the late 1970s and early 1980s, to such an extent that agencies accepted space reductions in order to be allowed to purchase the newest furniture. There are other indications that in various agencies the staff were excited to adopt the new open plan, and were a part of the planning teams choosing the open plan, sometimes over the wishes of their agency heads.³⁵

Meanwhile, the ideas promoted in the 1960s of an office that fostered communication, offered flexibility to workers and to the organization, and supported a flat organizational

35. Letter from Donald Percy to Kenneth Lindner, October 19, 1981, Wisconsin Department of Administration Secretary's Subject File Part I, Box 39 Folder 17, Wisconsin Historical Society.

structure are still considered primary goals of contemporary office design.³⁶ In other words, even as office work has changed significantly since the 1960s, the idealized image of work has actually not changed very much at all. Indeed, systems furniture manufacturers still describe the open plan as the spatial counterpart to the modern organizational culture, just as the architects and designers who first promoted the open plan did in the late 1960s.

Although the designs have changed significantly, many of the underlying principles particularly regarding communication and flexibility are still articulated. For example, a white paper written in 2008 for Knoll by their head of workplace research, Mike O'Neill, describes how well designed and planned offices should: be responsive to change, support collaboration, "promote spontaneous interaction," reduce distractions, "balance communication and privacy," and support "learning and mentoring."³⁷ Similarly, a recently published article in the *Wall Street Journal* similarly described the "postcubicle" office in which the office is designed to encourage conversation and teamwork (lower partitions, areas specifically designed for collaboration), the work stations are easily adaptable to the changing needs of workers and organizations, and the workspace is no longer reflective of status or hierarchy.³⁸

Thus even the "new" open plan represents a vision of white-collar work that is remarkably similar to that articulated by the post-capitalist ideas of the late 1960s. The image of white collar

36. Russell, "Form Follows Fad: the Troubled Love Affair of Architectural Style and Management Ideal," 66.

37. M. O'Neill, "Open plan and Enclosed Private Offices," Knoll, 2008, available online at http://www.knoll.com/media/878/738/OpenClosed_Offices_wp.pdf, p 4.

38. Ben Kesling, and James Hagery, "Say Goodbye to the Office Cubicle," *Wall Street Journal (Online)*, <http://search.proquest.com.ezproxy.library.wisc.edu/docview/1322548761?accountid=465> (accessed April 3, 2013).

work characterized by dynamic communication and collaboration among workers, a flexible work environment that can be adapted to the individual needs of workers and the changing needs of organizations, and a “flat” image of the organization in which the conventional symbols of status are removed is still held up as an ideal office culture and environment. In this way, the open plan has retained its progressive idealism even today, despite the “negative” associations of its direct descendant, the “cubicle.” Of course the cubicle is not a mere descendant, twenty years ago the workplaces that we call “cubicles” were considered to be open plan workspaces. As architects, designers, and office furniture manufacturers re-package the open plan as “new” they are actually returning to the initial concept of the open plan as it was promoted in the late 1960s.

This reinvention of the open plan as the ideal workplace suggests a return to a vision of white-collar work itself that is remarkably similar to the utopian image of white-collar work forecast in the late 1960s. Just as the architects and designers of the late 1960s predicted a new image of white-collar work characterized by high levels of communication and interaction, an emphasis on social, cultural, and technological change, and the elimination of hierarchy, the contemporary image of the idealized experience of white-collar work emphasizes the importance of “collaboration” among workers, the ability to adapt to rapid organizational and technological change, and the need to eliminate spatial hierarchy in order to support the culture of a “flat” organization. This does not mean that all contemporary workplaces (or contemporary white-collar work) actually look like this idealized image, but the fantasy of an idealized workplace is nonetheless important in that it reflects shared social and cultural values, expectations about what work should be, and hopes for the possibilities of the future, particularly the desire to experience personal fulfillment through work. Despite the numerous changes in the culture and structure of

white-collar work in the last forty years, a number of American architects, designers, furniture manufacturers, and organizations continue to aspire to achieve a working environment that fosters worker autonomy, encourages collaboration, and eliminates hierarchy and ultimately creates a workforce of engaged, motivated, and creative workers for whom work is a joy rather than a burden.

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